

Cardiovascular Structures on Chest X-ray

Roentgenologic Anatomy of the Heart, Chambers, and Valves

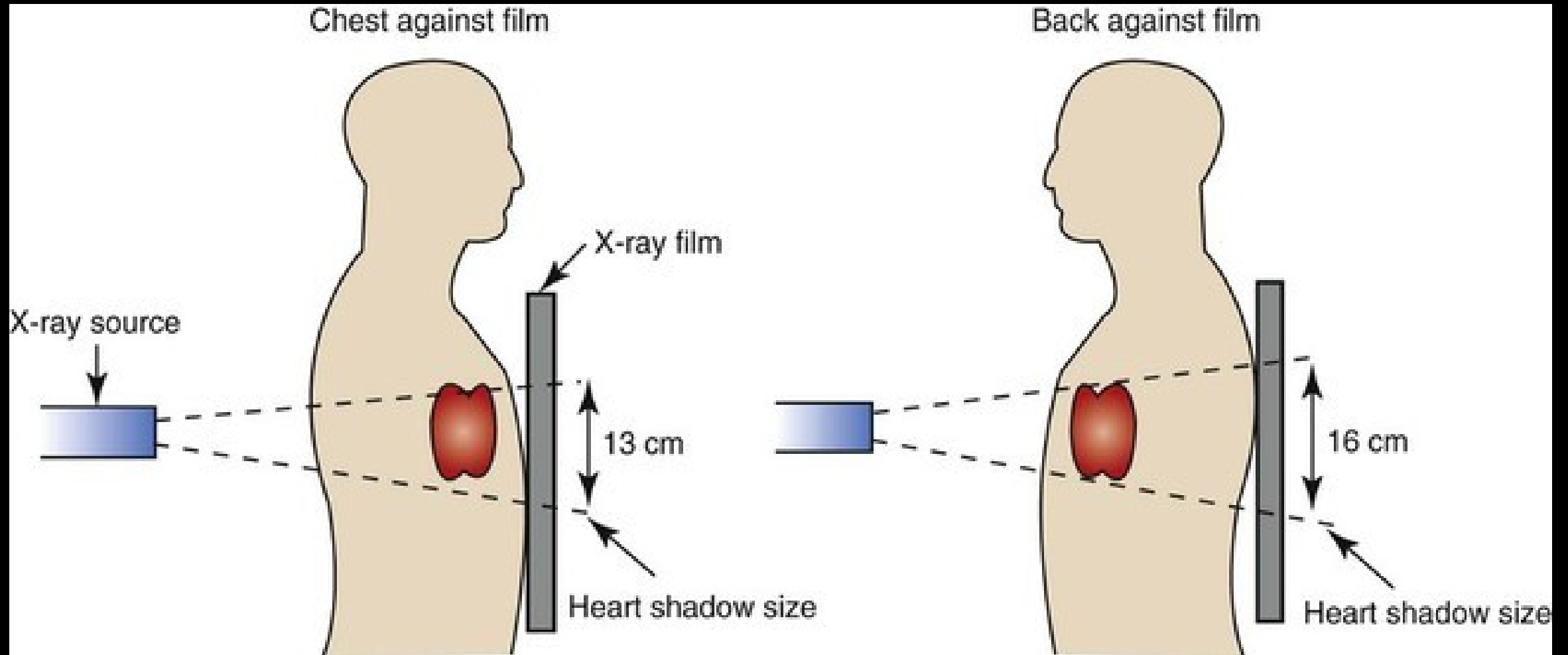
Warath Chantaksinopas, M.D.

Cardiovascular Imaging Unit | Department of Radiology
Faculty of Medicine | Prince of Songkla University

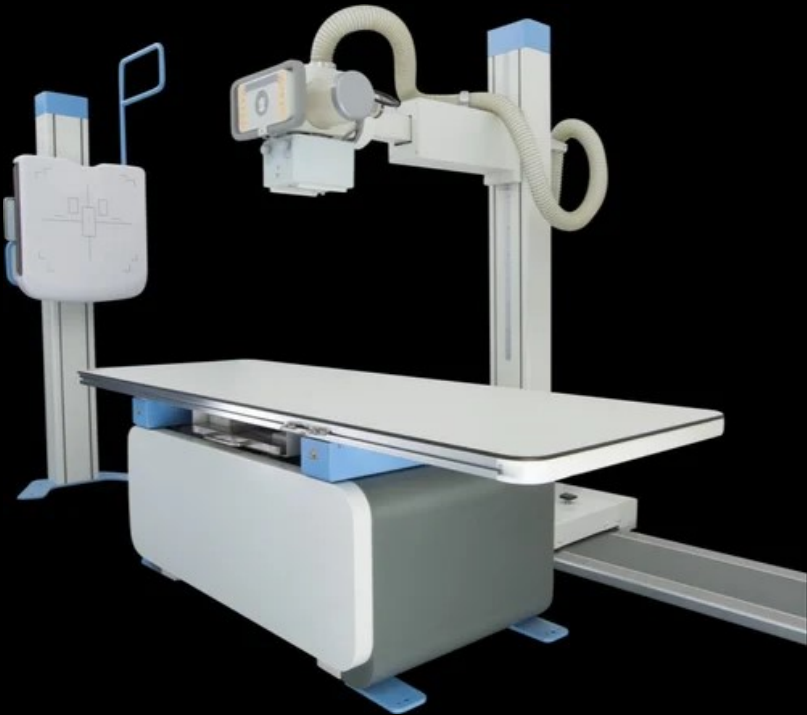
Outline

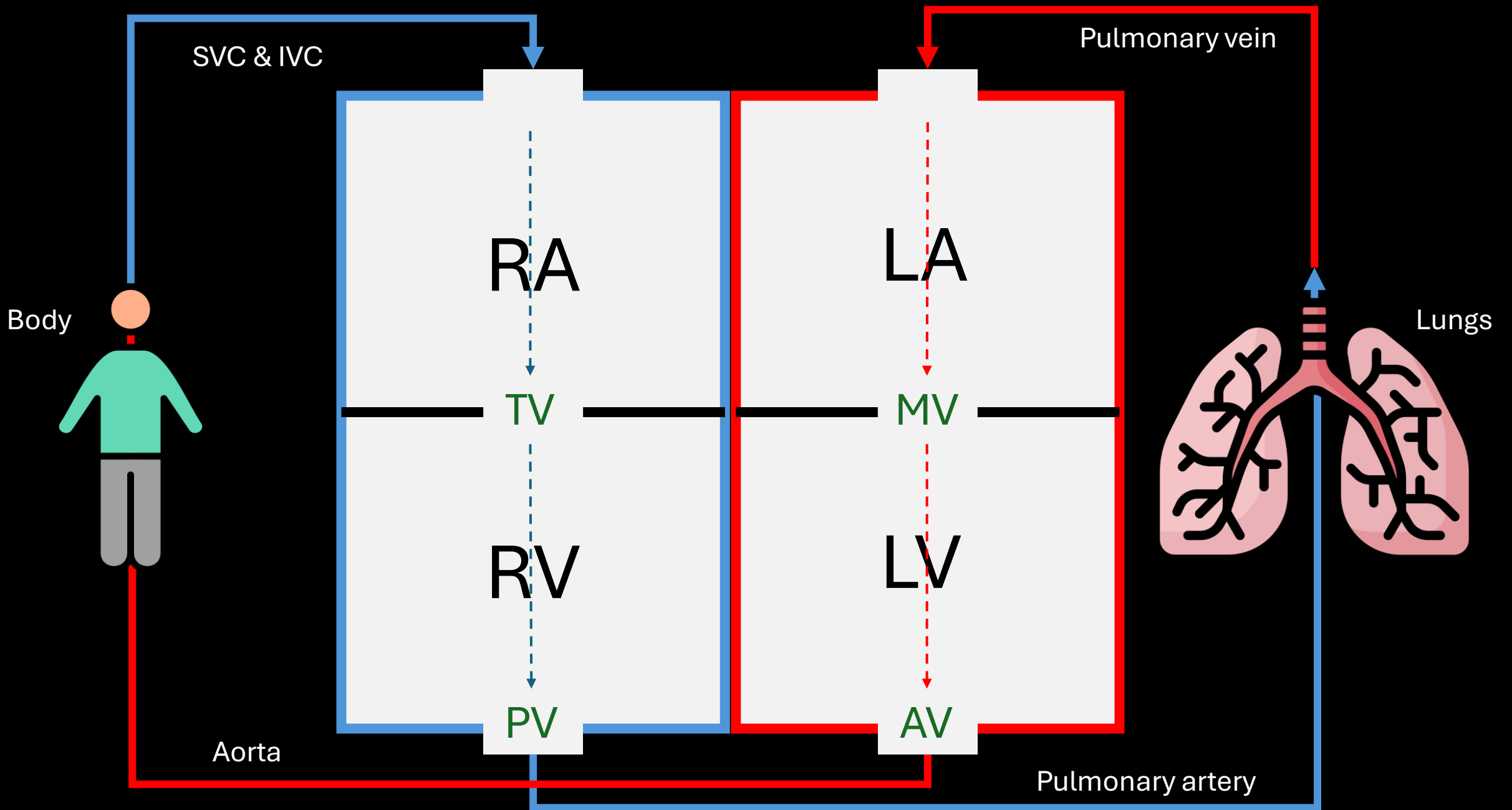
- 1 Patient's positioning for CXR
- 2 Cardiovascular structures on CXR
- 3 Cardiac chamber evaluation and valve positions
- 4 Pulmonary vasculature evaluation
- 5 Common cardiac morphology and Configuration
- 6 Acyanotic and cyanotic CgHD

PA vs AP views



PA vs AP views



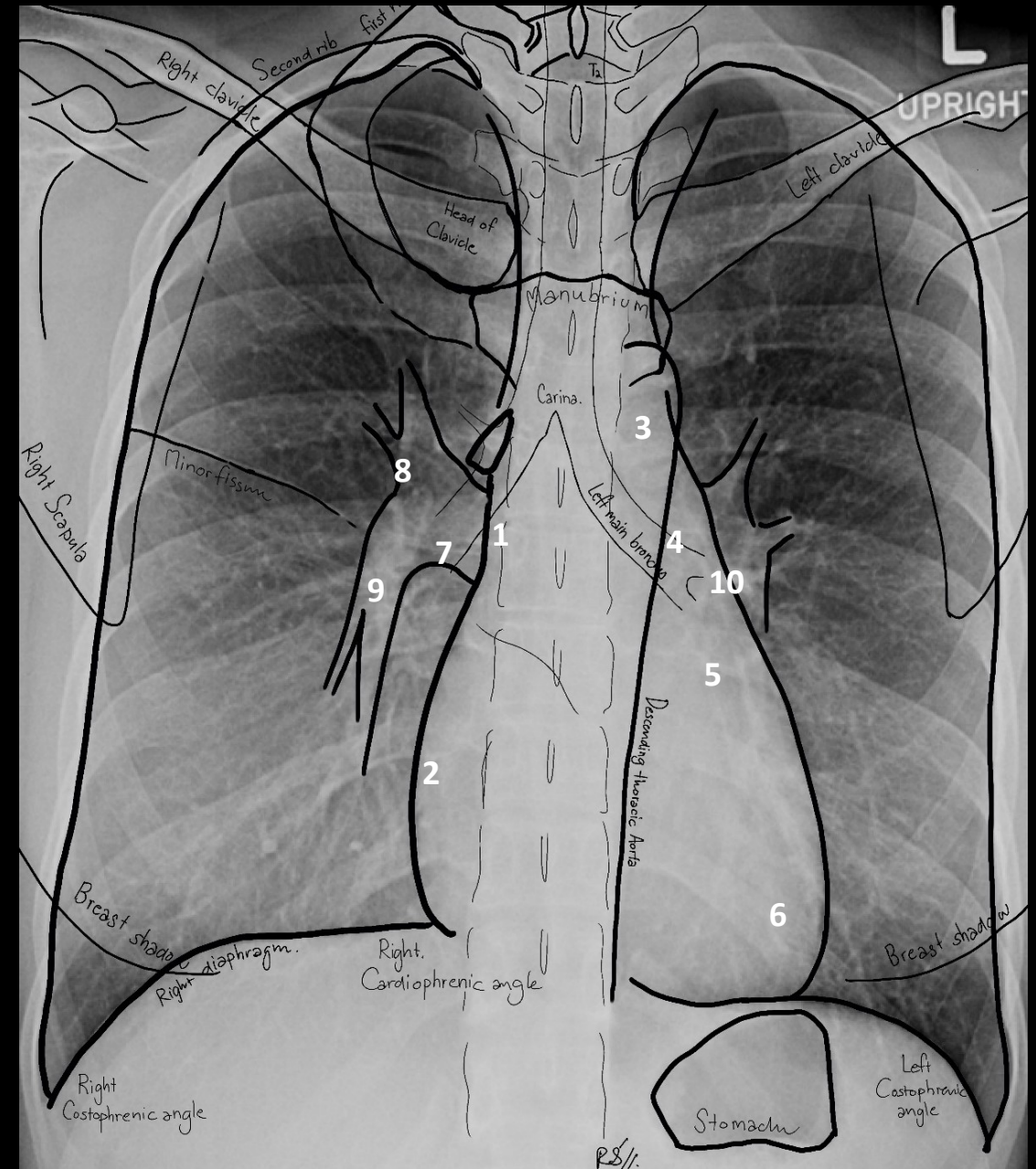
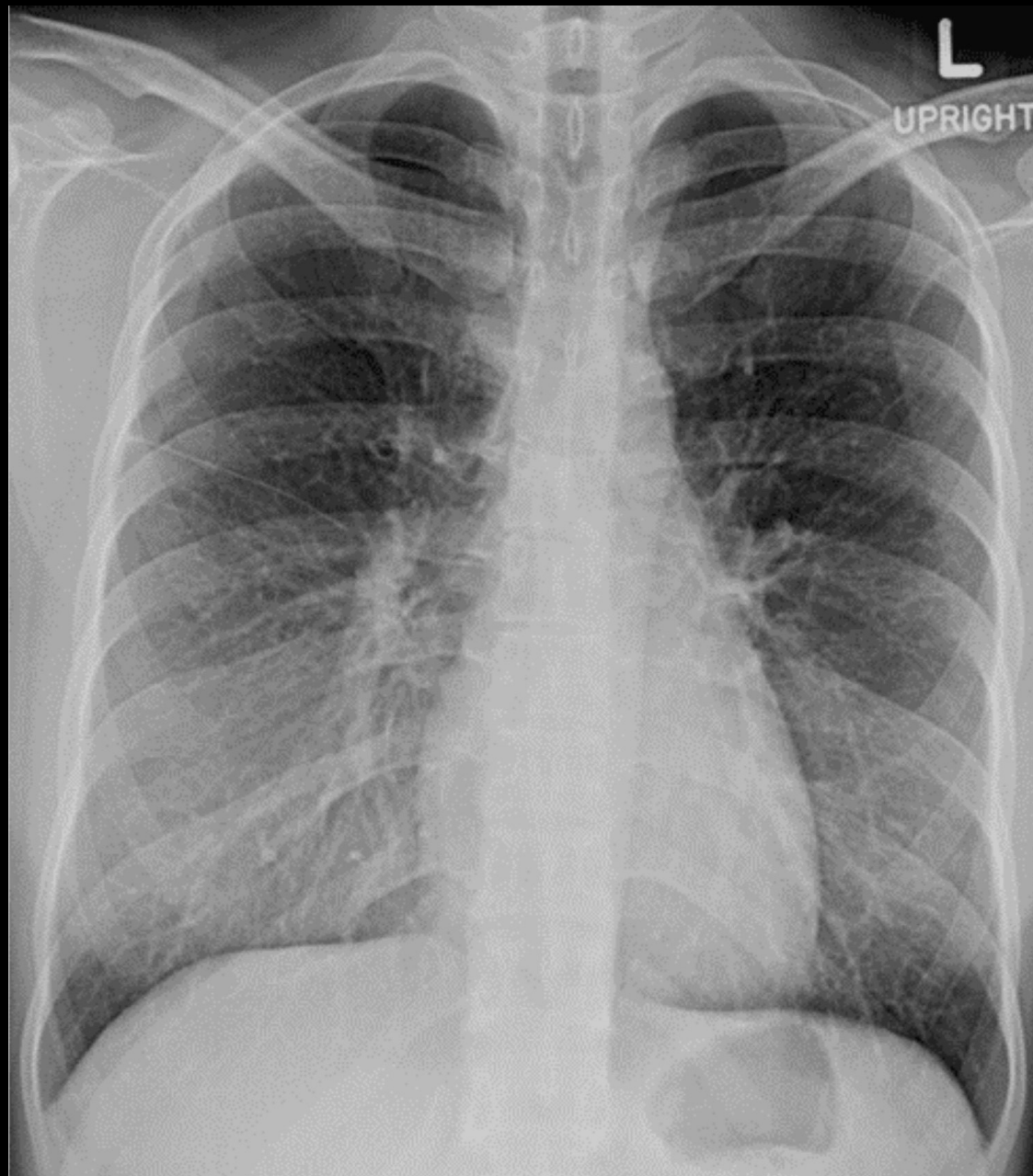


Basic physiology of the Circulatory system

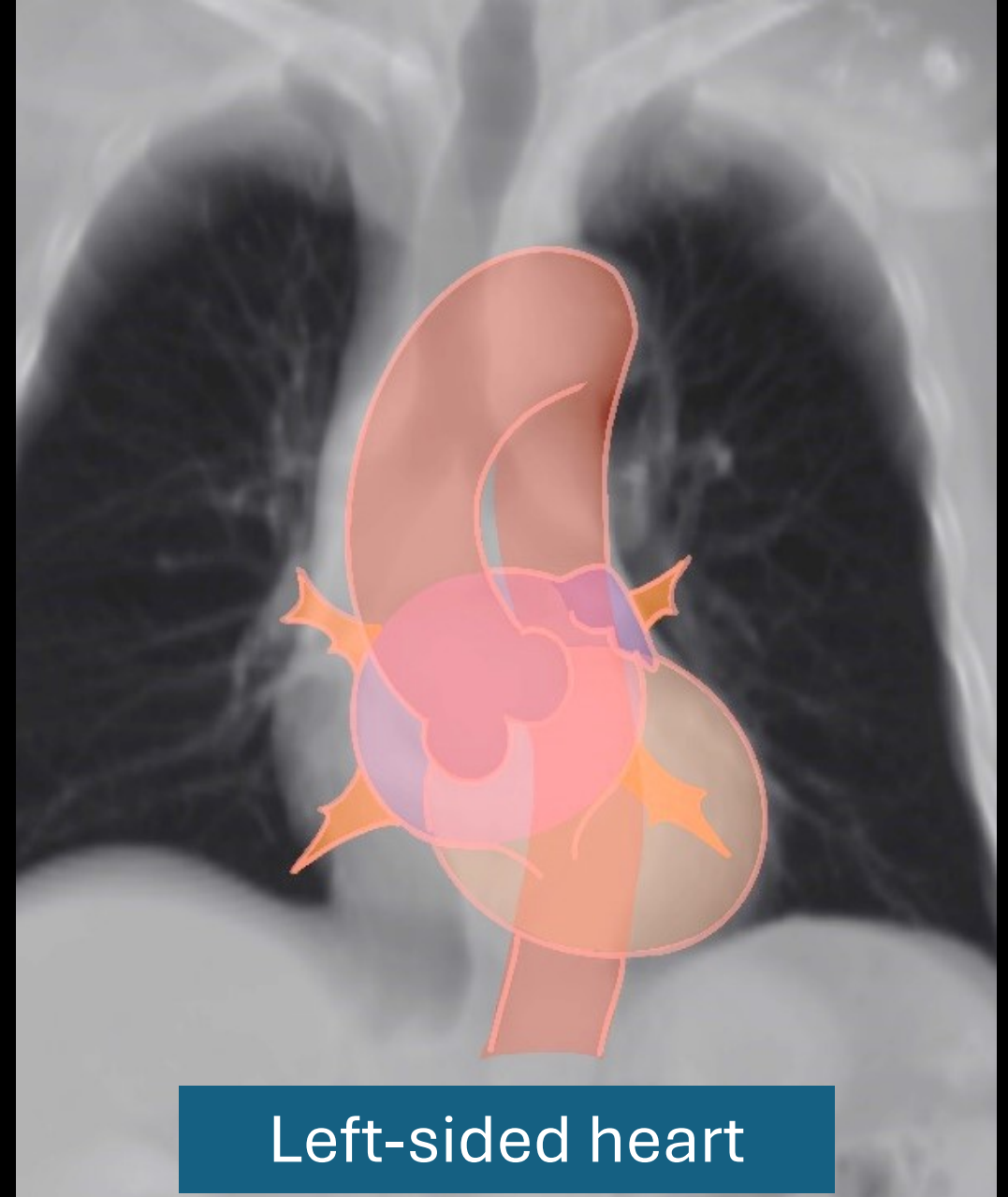
Cardiovascular Anatomy on Chest Radiograph



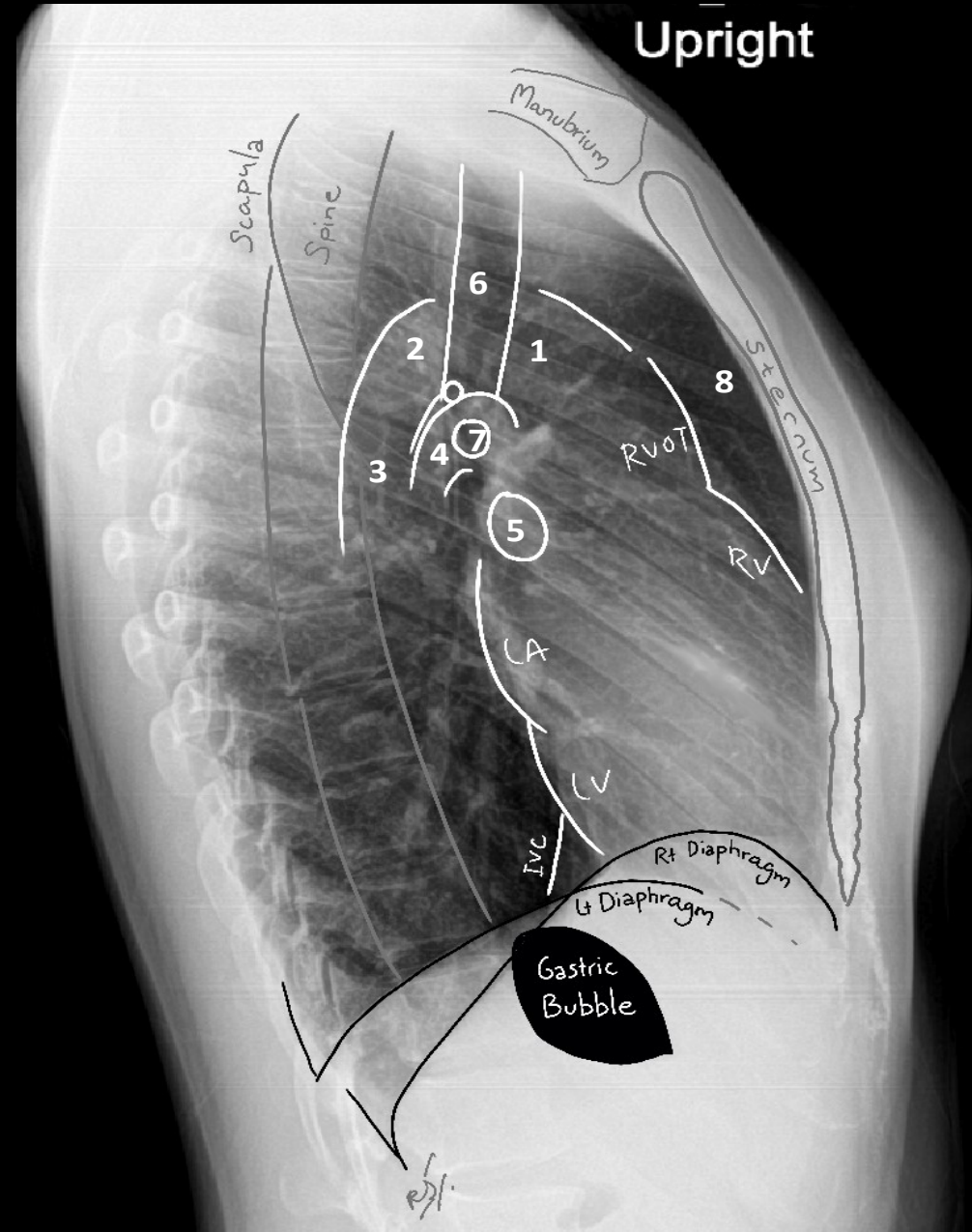
Anatomy of Cardiovascular Structures on CXR



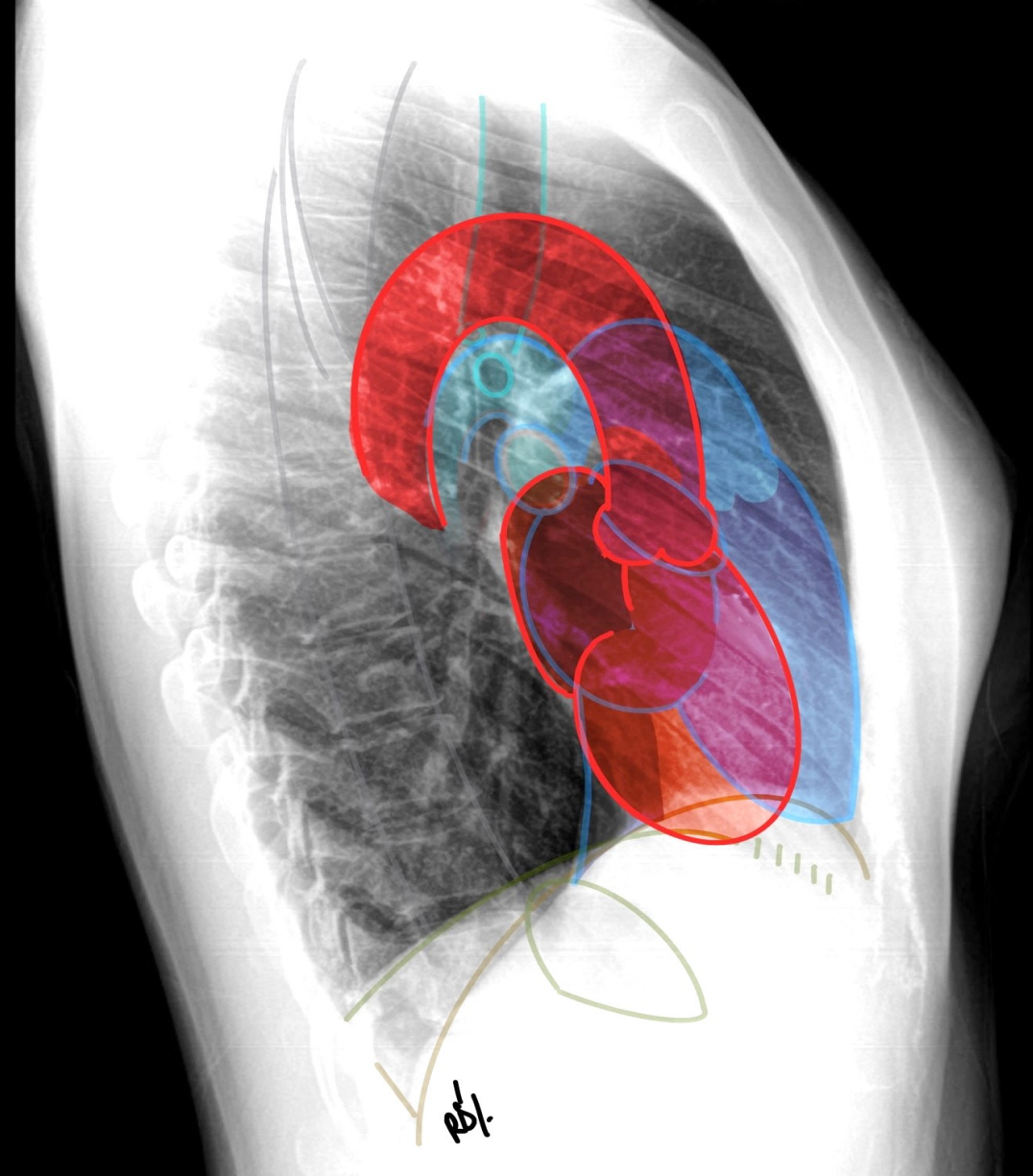
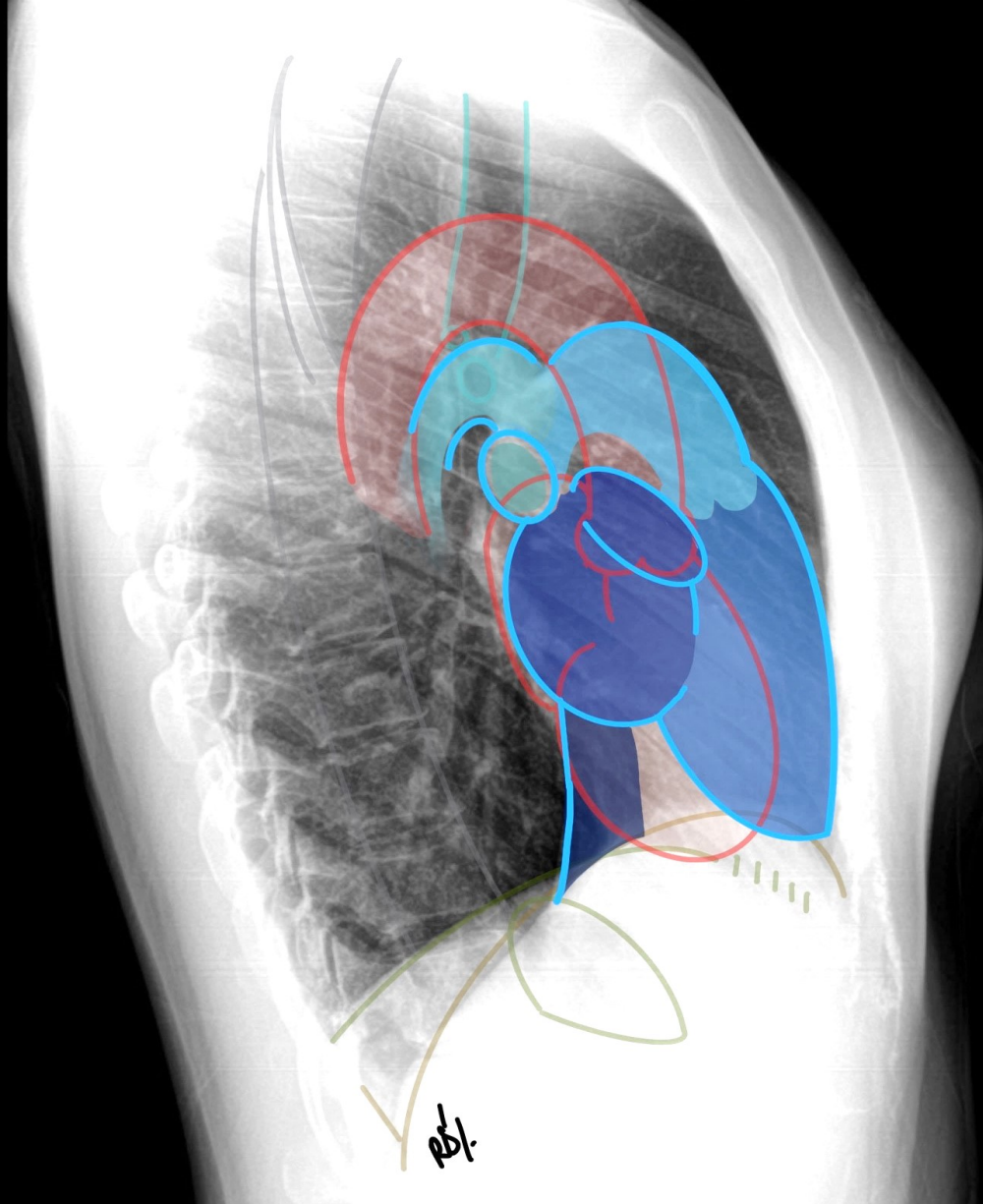
Anatomy of Cardiovascular Structures on CXR



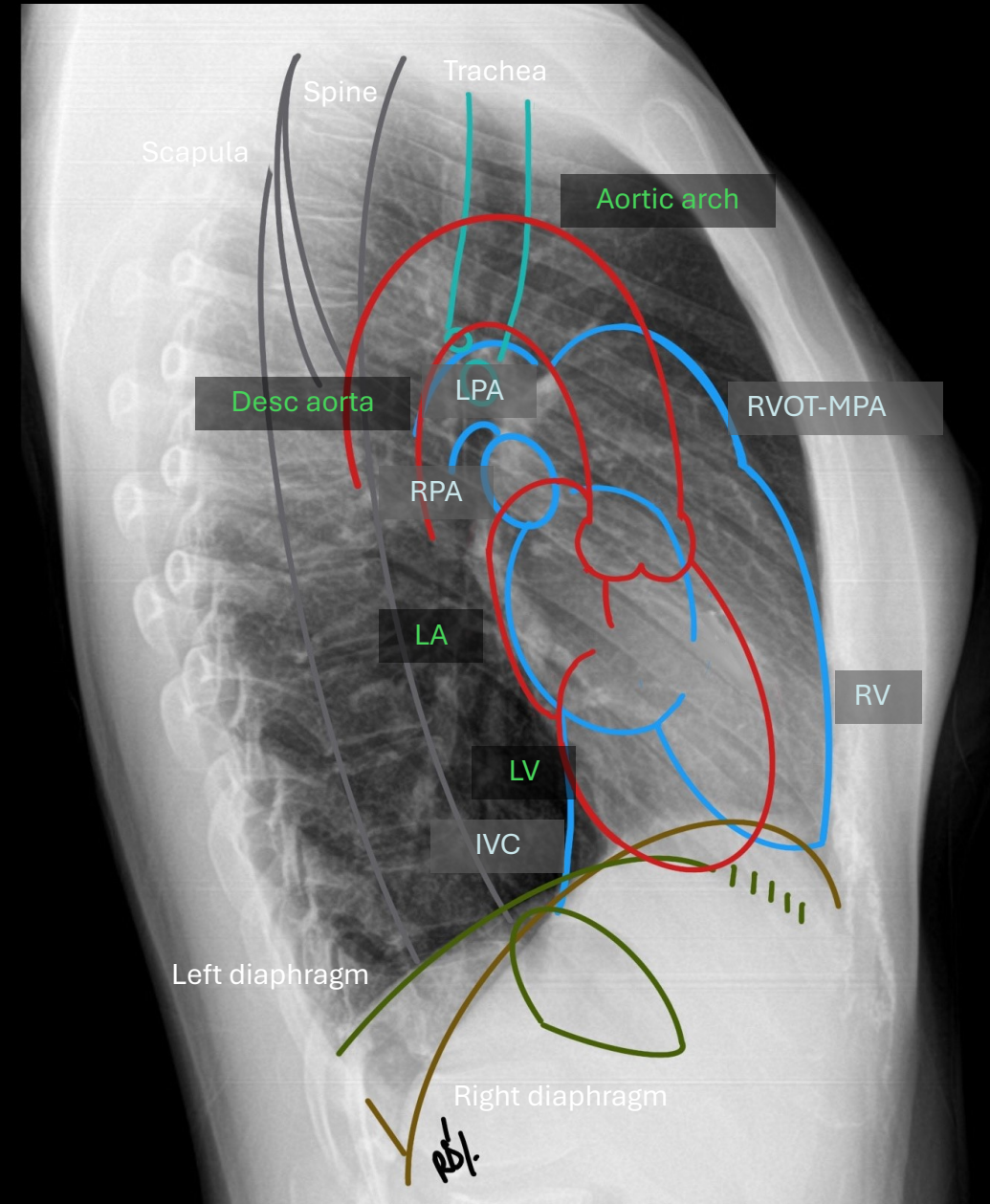
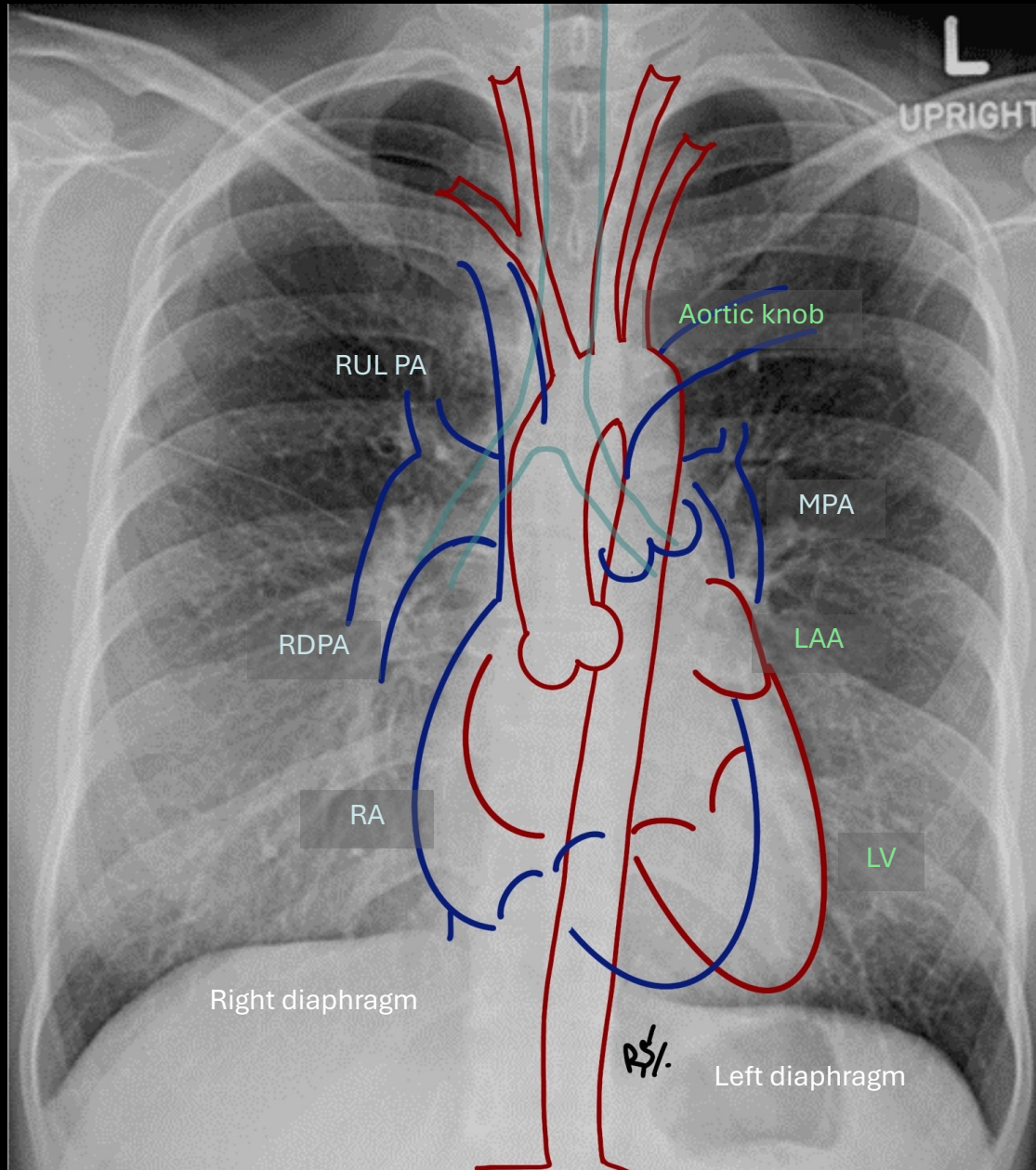
Anatomy of Cardiovascular Structures on CXR



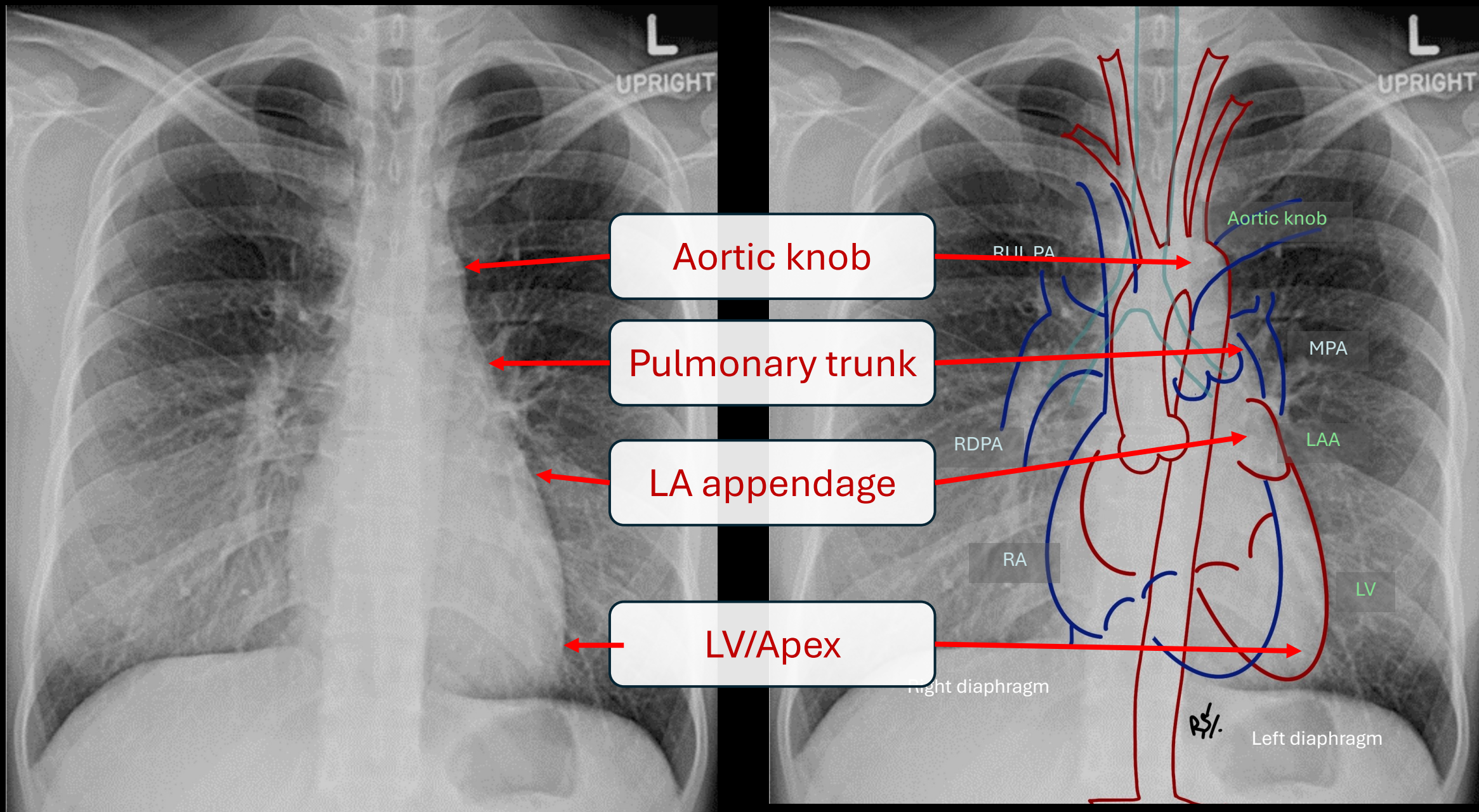
Anatomy of Cardiovascular Structures on CXR

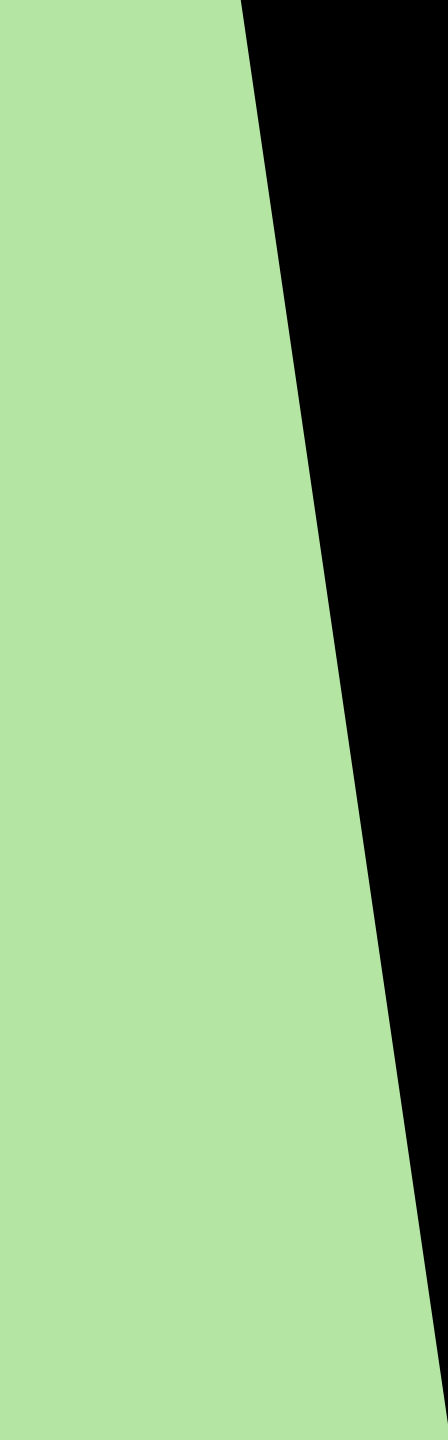


Anatomy of Cardiovascular Structures on CXR



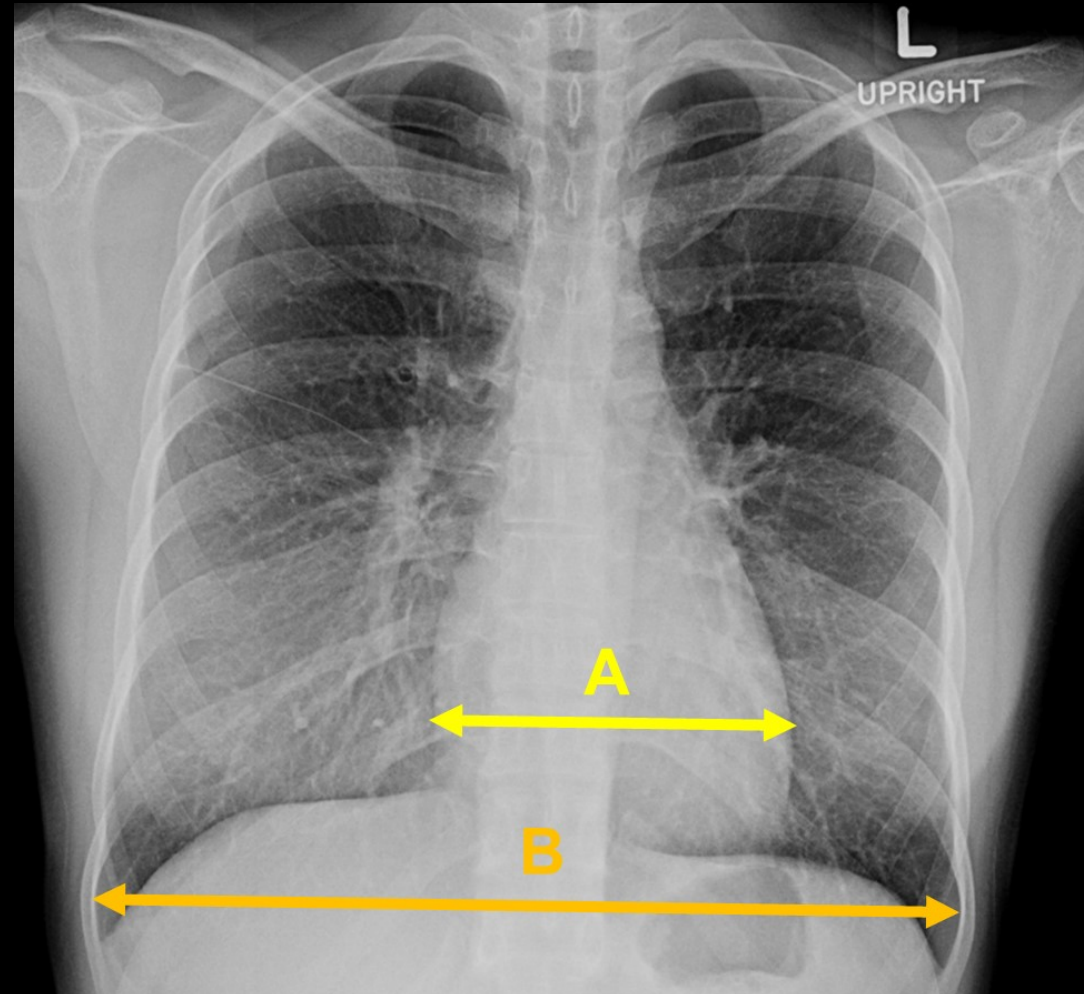
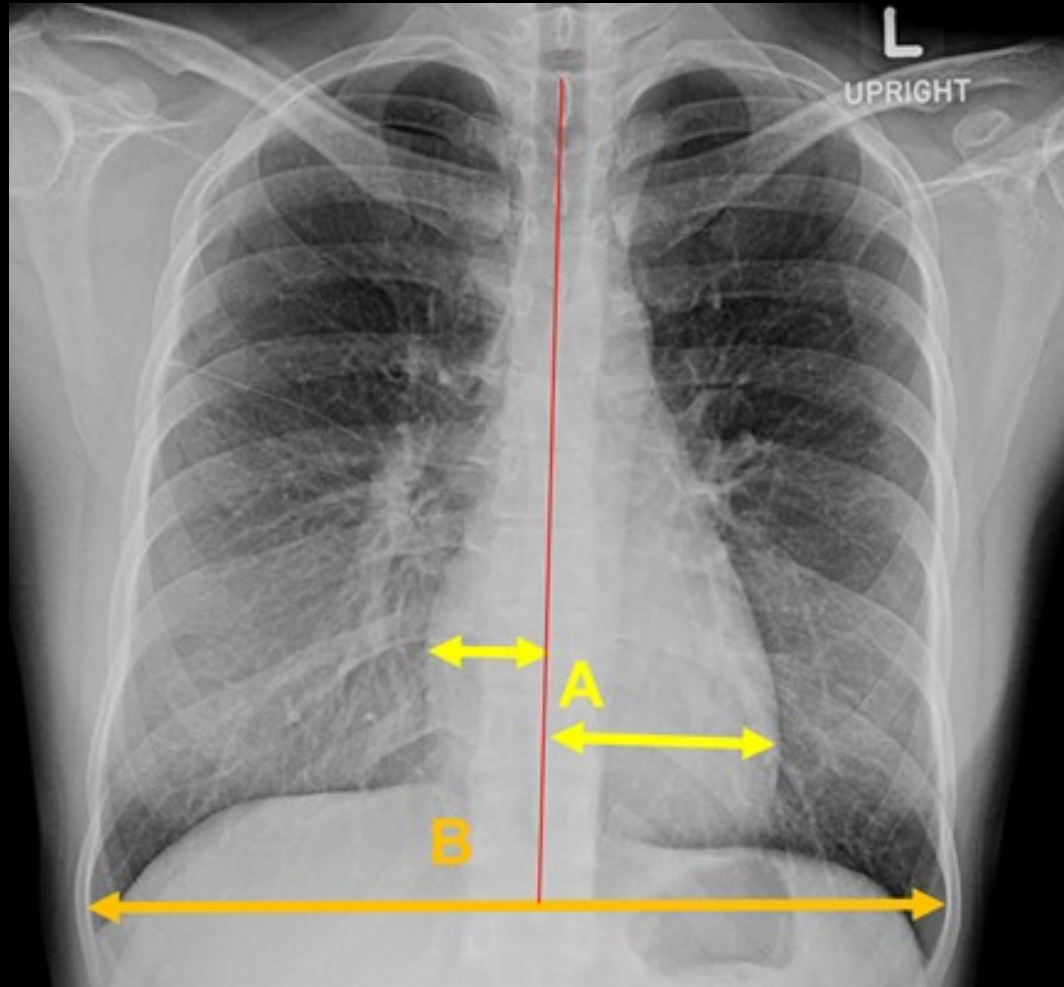
Left-sided Mediastinal Contours





Heart size and
Each chamber size

Heart size: Cardio-thoracic ratio



$$CT\ ratio = \frac{A\ (maximal\ transverse\ diameter\ of\ the\ heart)}{B\ (greatest\ internal\ diameter\ of\ the\ thoracic\ cage)}$$

≤ 0.5

Increased CT ratio

Abnormal Heart

- Cardiomegaly
- Cardiomyopathy
- Heart failure
- Volume overload
- Valvular abnormality
(with dilated chambers)

Normal Heart

- Pericardial effusion
- Pericardial cyst or mass
- Portable AP chest X-ray
- Obesity

RA Enlargement



Pathophysiology

- Right heart volume/pressure load
- 2nd from right ventricular dilatation

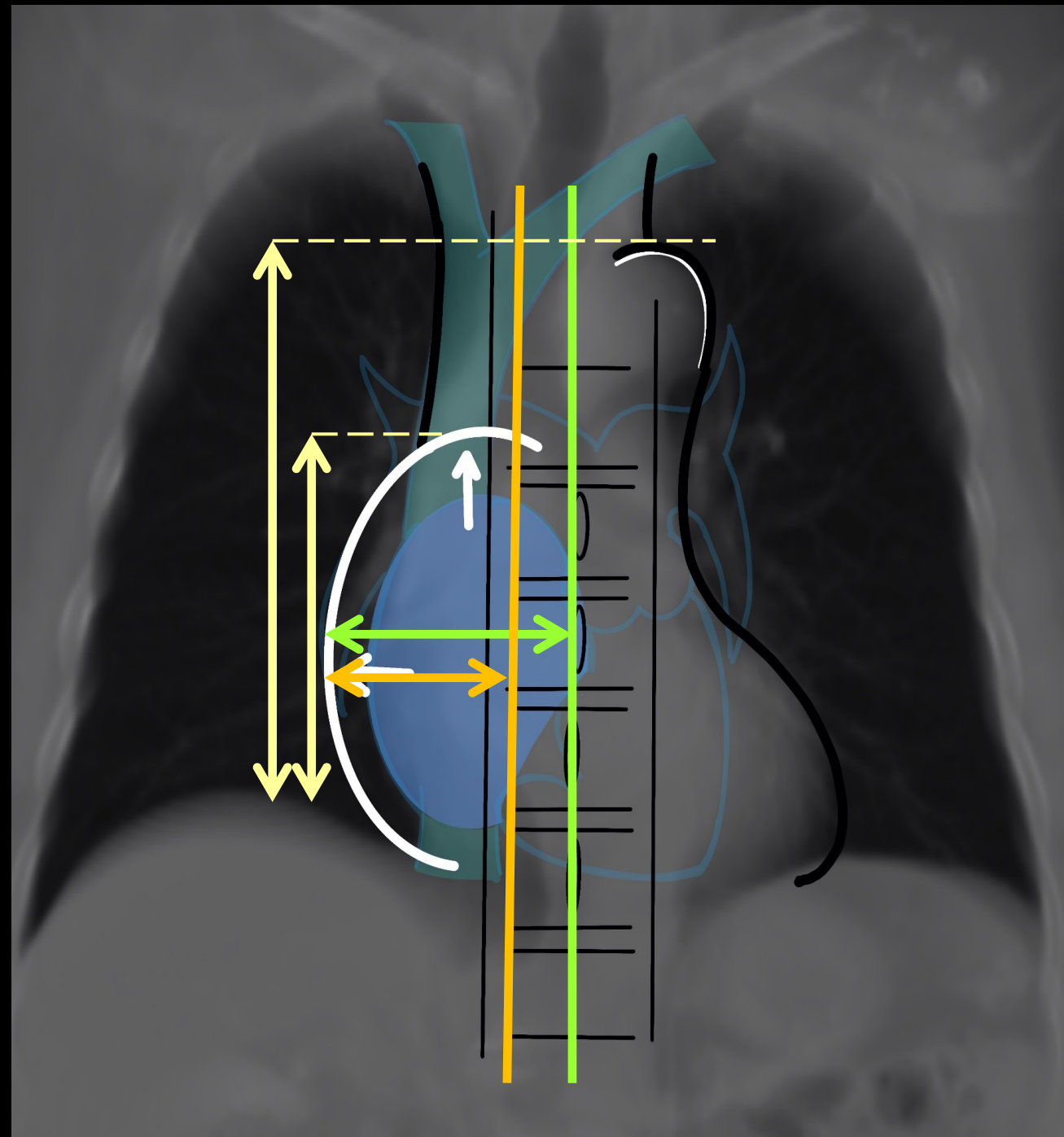
Possible causes

- Congenital/Developmental disorder
 - Left to right shunt e.g. ASD, VSD
 - TV abnormality (e.g. Ebstein's anomaly)
 - Anomalous pulmonary venous return
- TV or PV regurgitation/stenosis
- Constrictive pericarditis
- Restrictive cardiomyopathy
- Congestive heart failure

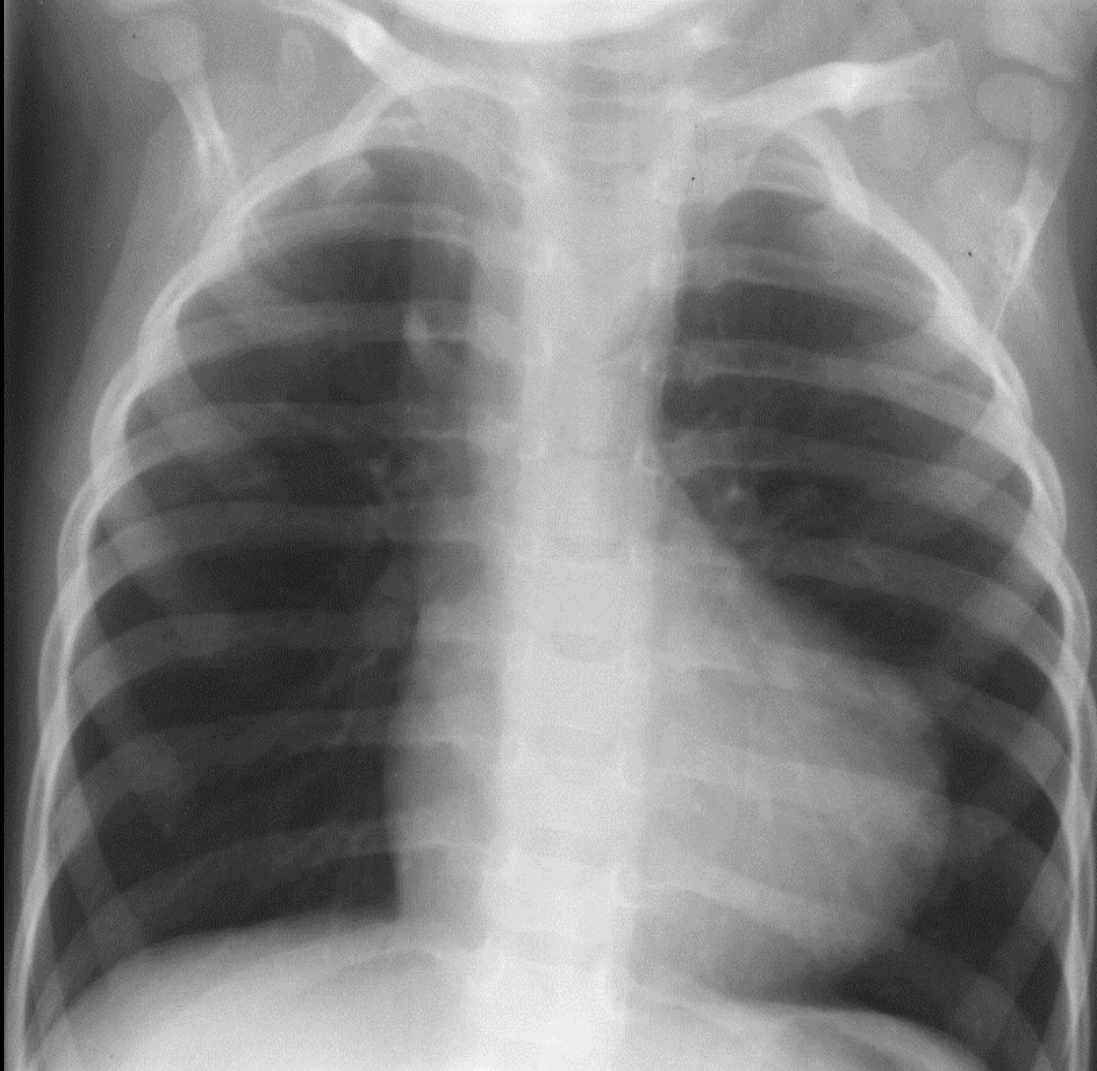
RA Enlargement

Radiologic signs on PA film

1. $\uparrow$ CTR
2. > 5.5 cm from midline
3. > 2.5 cm from right border of thoracic vertebral body
4. $> 50\%$ RA/aortic arch to right diaphragm



RV Enlargement



Pathophysiology

- Right ventricular volume/pressure load

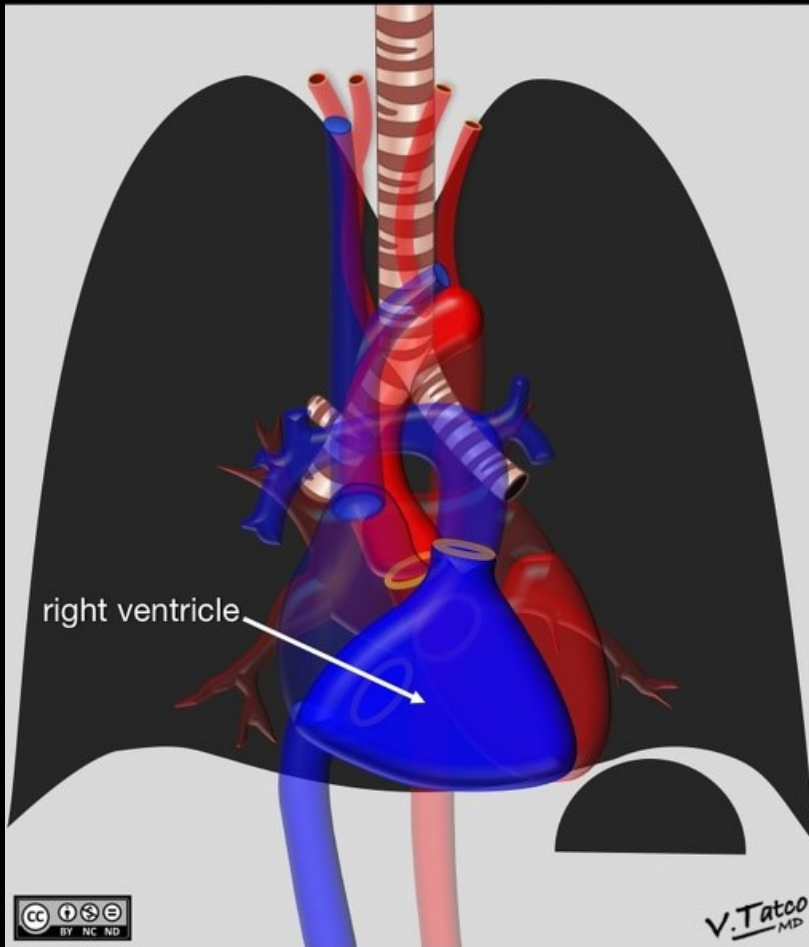
Possible causes

- Pulmonary arterial hypertension
- PV stenosis/regurgitation
- Left to right shunt e.g. ASD, VSD
- Cardiomyopathy
- Anomalous pulmonary venous return
- Tetralogy of Fallot (from RVH and PS)

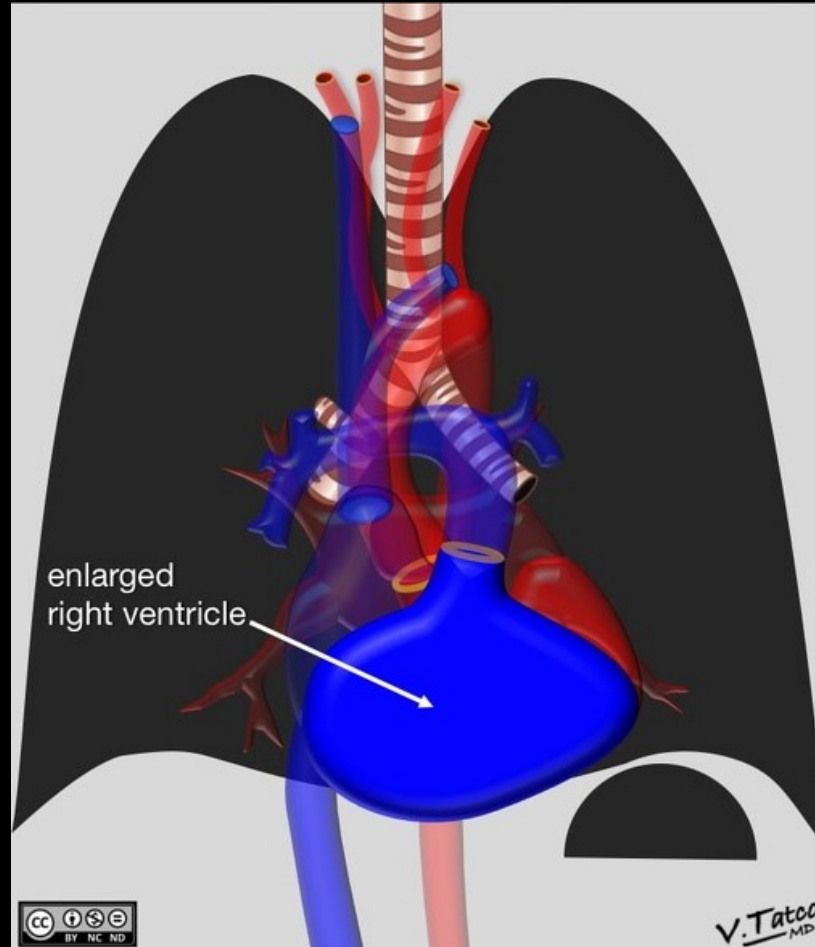
Radiologic signs

- PA view – uplifted cardiac apex, +/- ↑ CTR
- Lat. view – Retrosternal clear space fullness

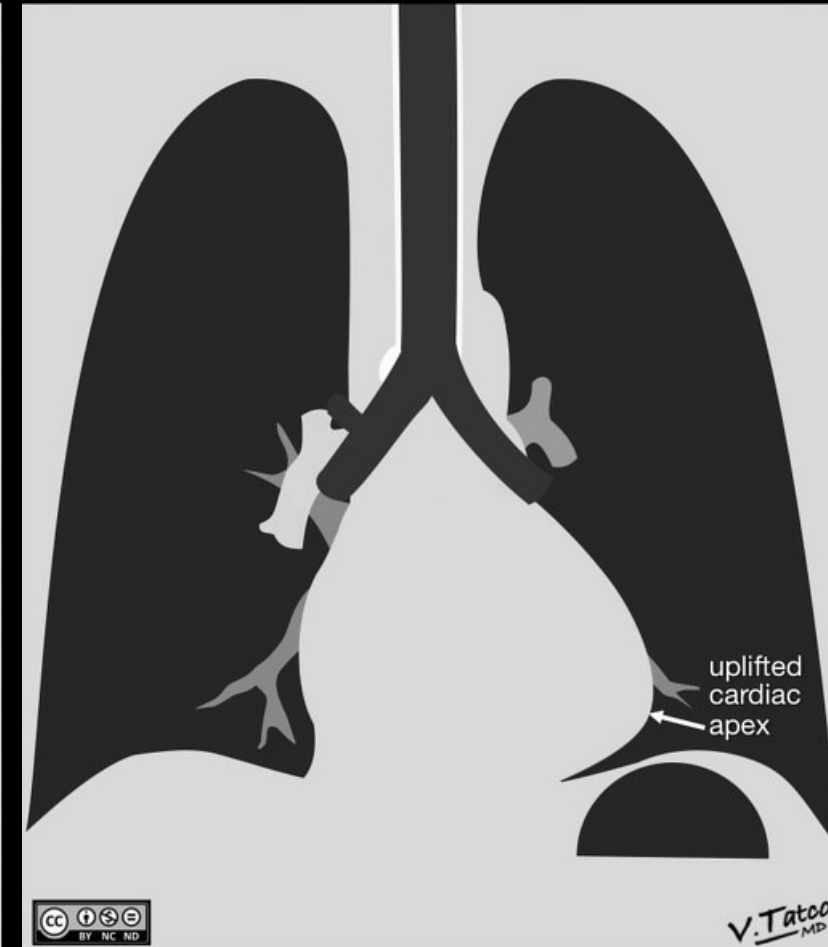
RV Enlargement



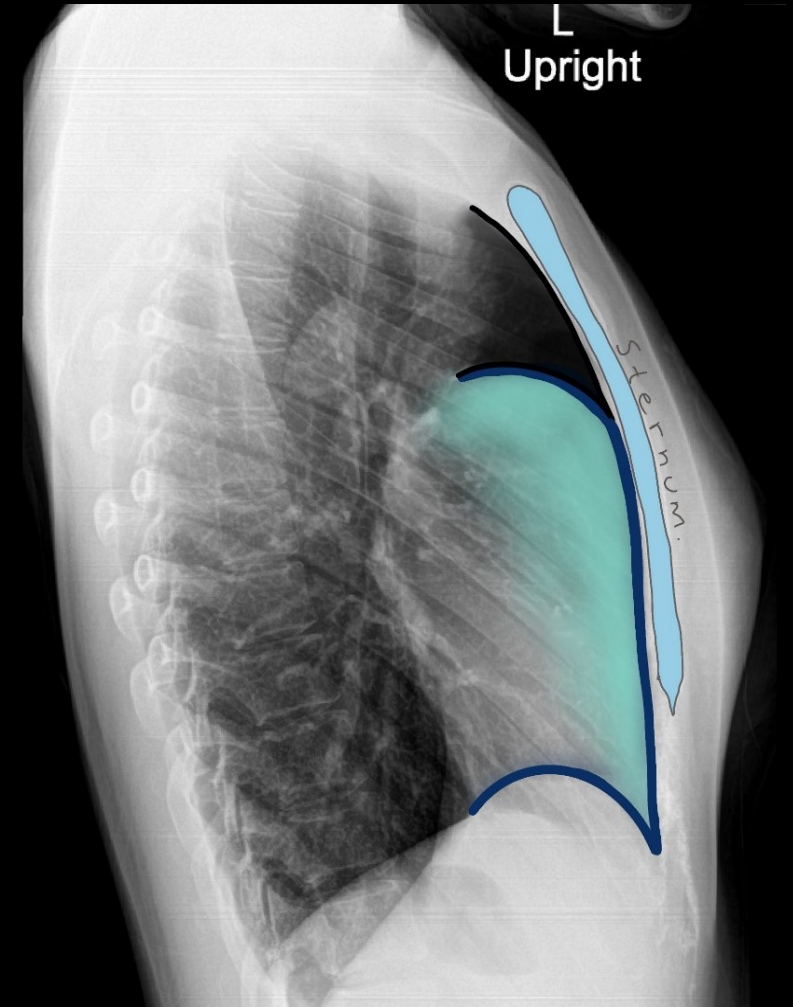
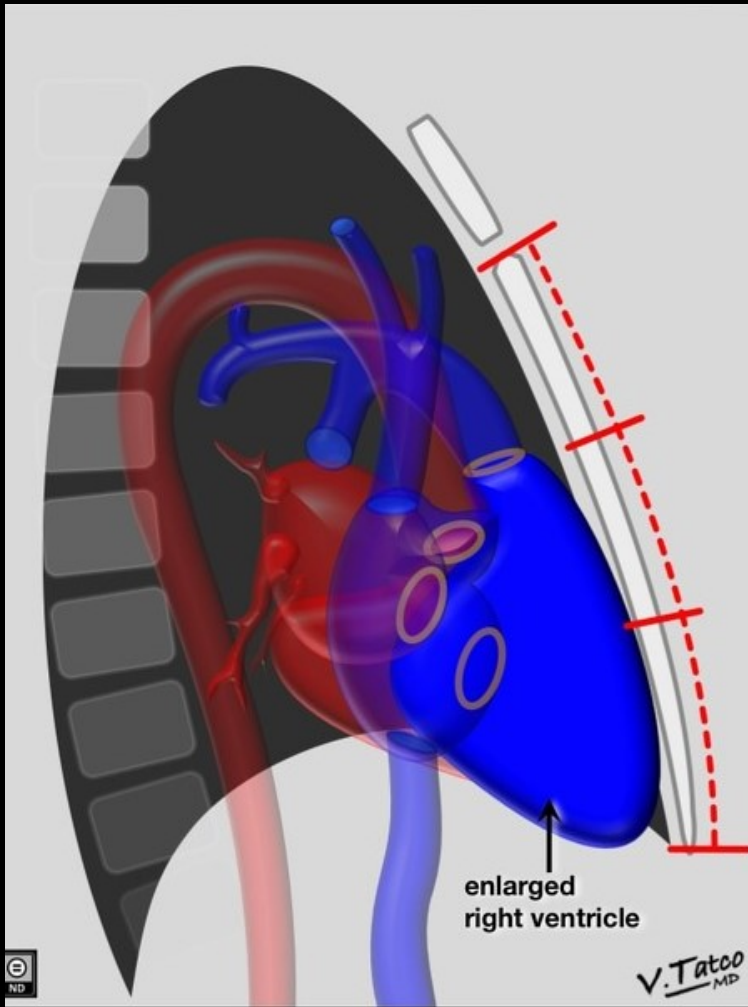
Normal sized RV



Enlarged RV – uplifted cardiac apex +/- increased CTR



RV Enlargement



Enlarged RV – Retrosternal fullness $>1/3$ of sternal length

LA Enlargement



Pathophysiology

- Increased LA filling pressure/tension

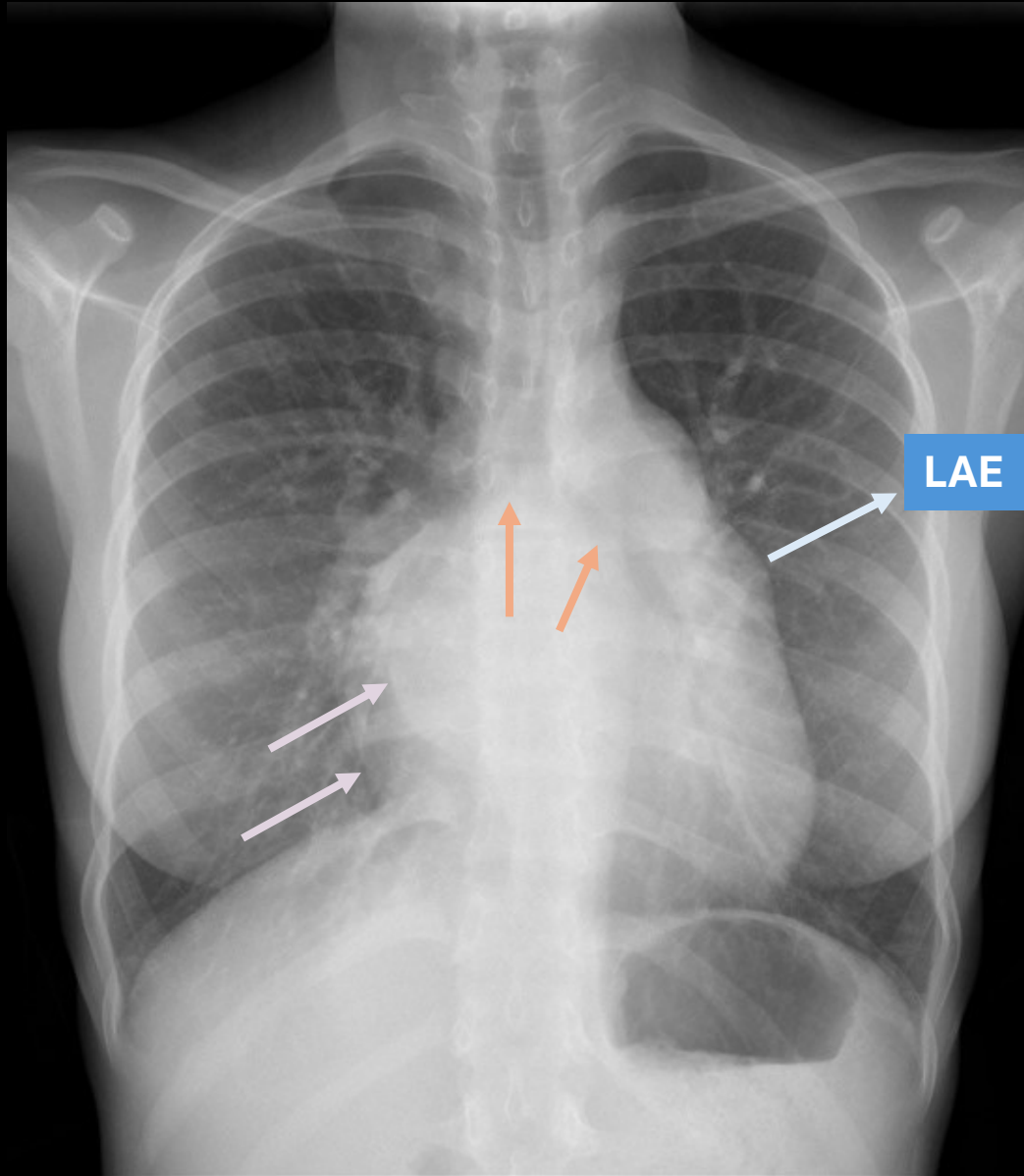
Possible causes

- Left ventricular failure
- Atrial fibrillation (associated with LAE)
- MV stenosis/regurgitation, rheumatic heart
- Left atrial myxoma/mass
- Cardiomyopathy (esp. restrictive CM)
- Left to right shunt; e.g. VSD, PDA

Note

- LA = most posterior chamber of the heart
- Difficult to see on CXR (it does not form cardiac border, except for LA appendage)

LA Enlargement



To LEFT side

- Fullness beneath pulmonary artery shadow
- Straightened or convexity of LA appendage
- May be the earliest sign

To RIGHT side

- Reach the right heart border (and +/- beyond)
- Causing “double density sign”

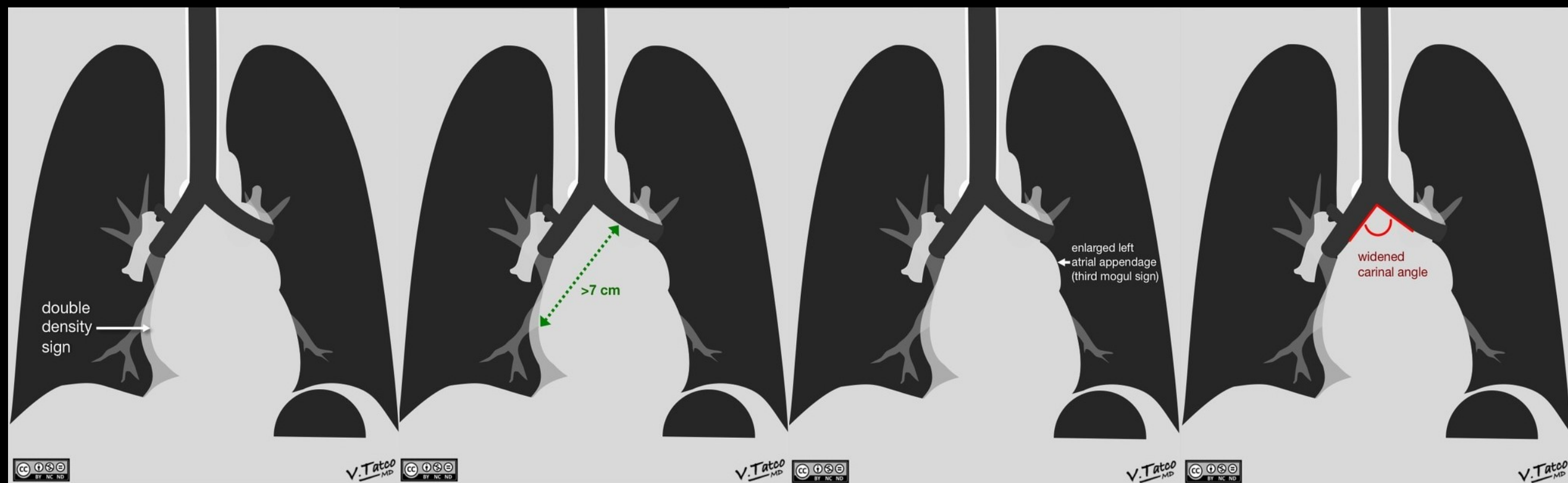
To SUPERIOR

- Lifted left main bronchus
- Widened carinal angle (>90 degrees)

To POSTERIOR

- Posteriorly displaced left main bronchus
- Push esophagus to spine (not seen on CXR)

LA Enlargement



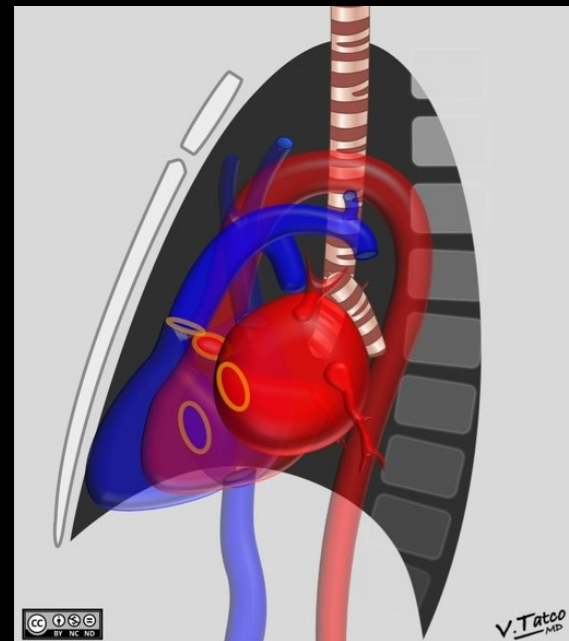
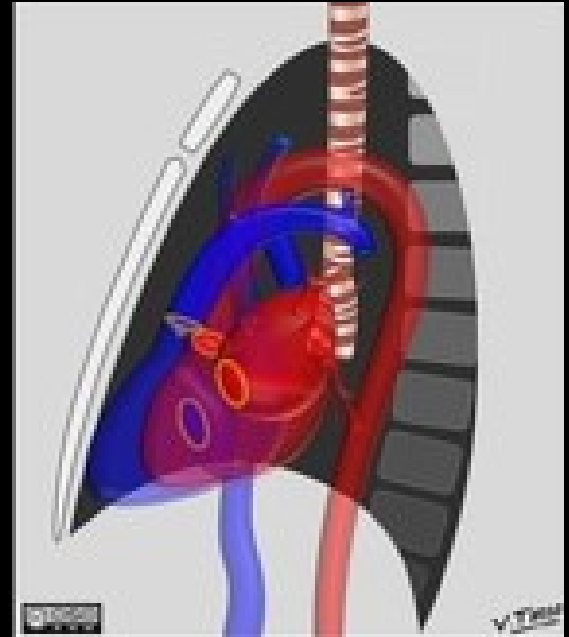
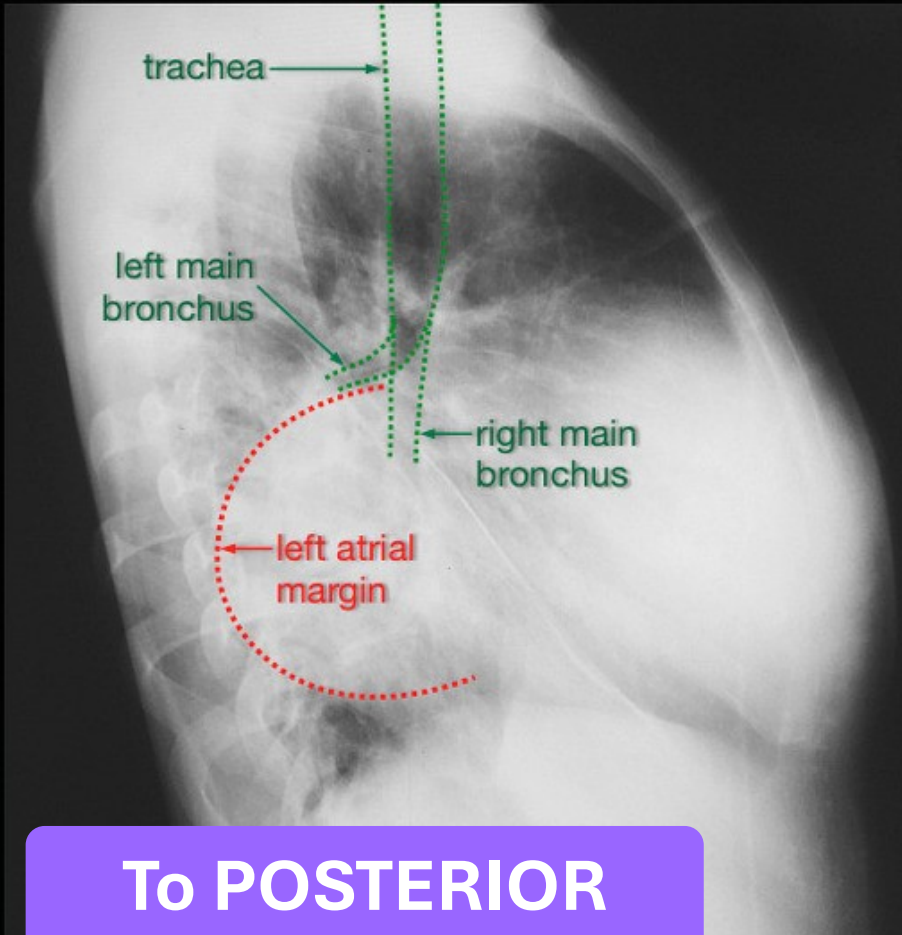
Double density sign

> 7 cm distance

Enlarged LA
appendage (LAA)

Widened carinal
angle > 90 degrees

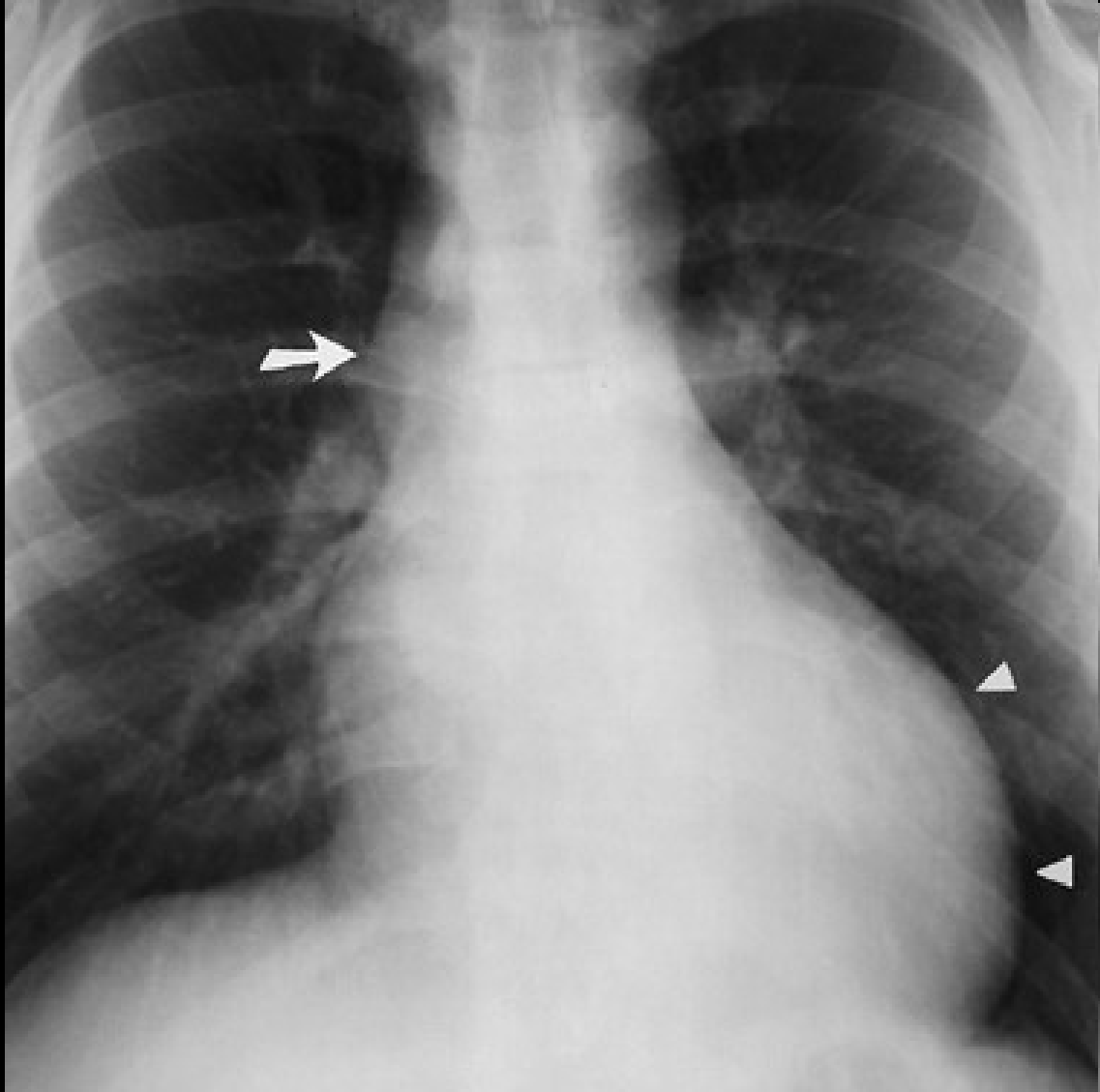
LA Enlargement – lateral film



To POSTERIOR

- Posteriorly displaced left main bronchus (walking man sign)
- Prominent upper posterior cardiac convexity

LV Enlargement



Pathophysiology

- Pressure overload/Volume overload
- Wall abnormality

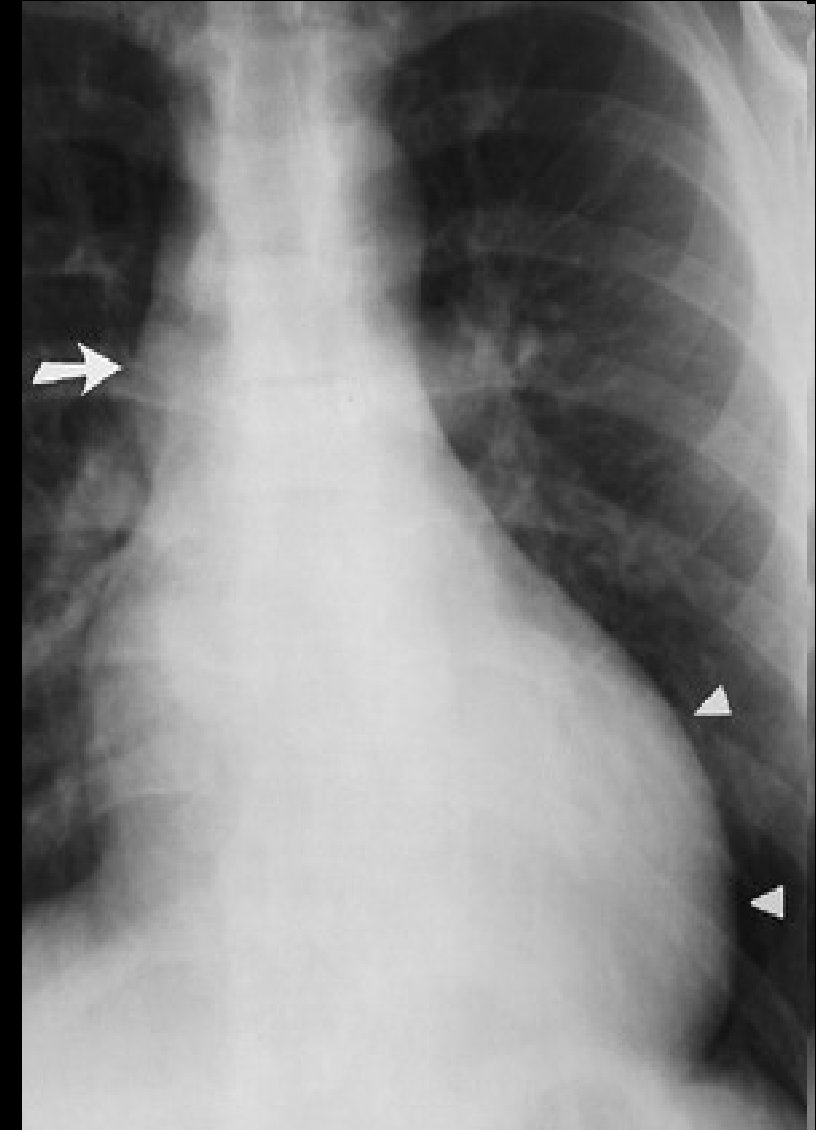
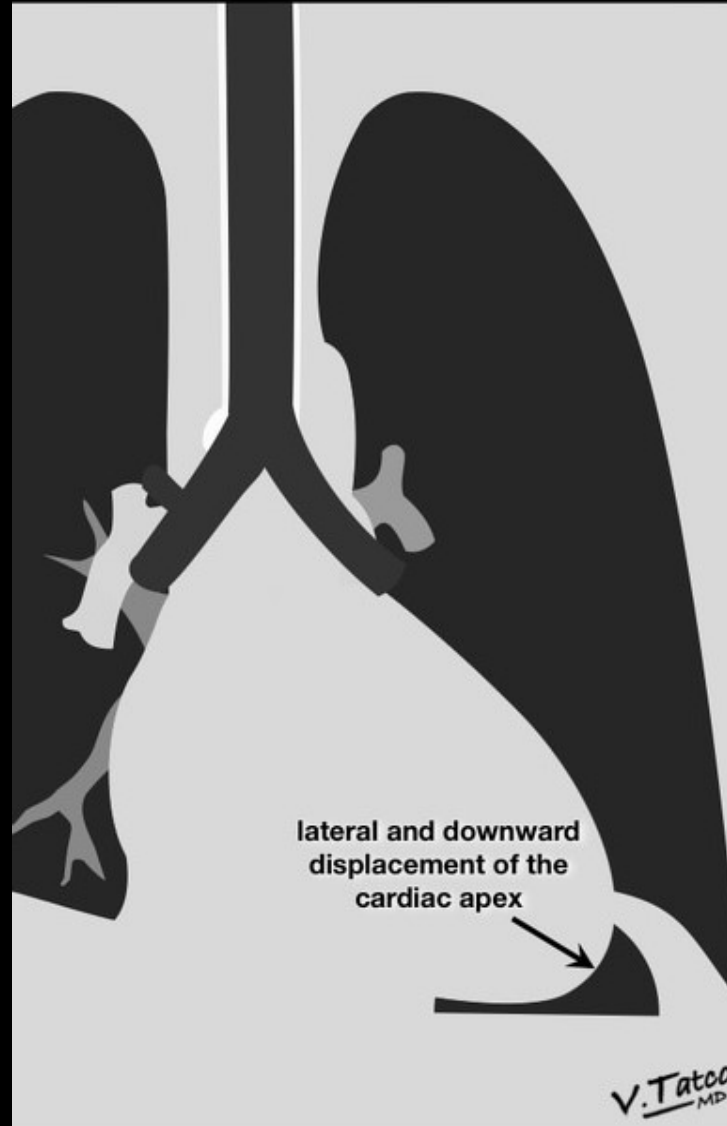
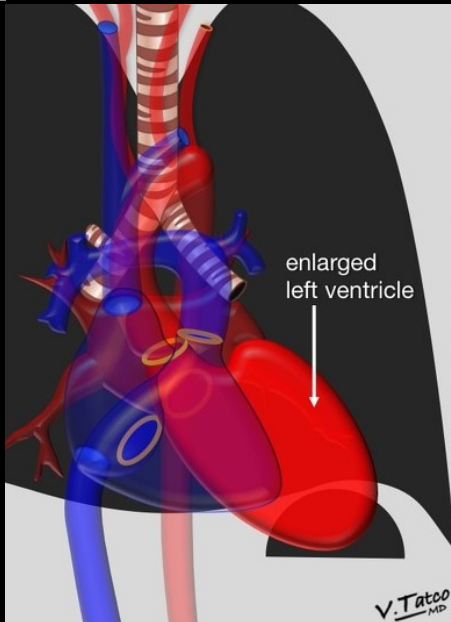
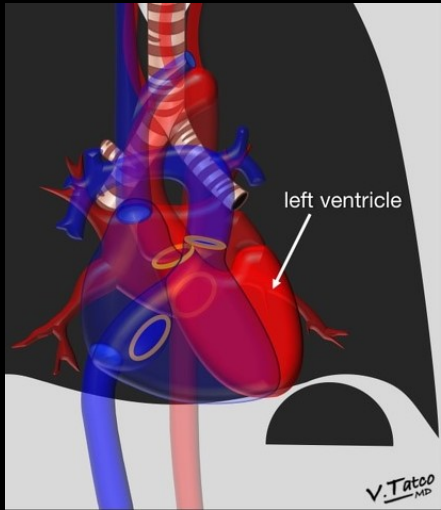
Possible causes

- Hypertension/ AV stenosis -> pressure
- AV or MV regurgitation -> volume
- Left ventricular aneurysm
- Hypertrophic cardiomyopathy
- Dilated cardiomyopathy and etc.

Radiologic signs

- Increased CTR, downward shift of apex

LV Enlargement



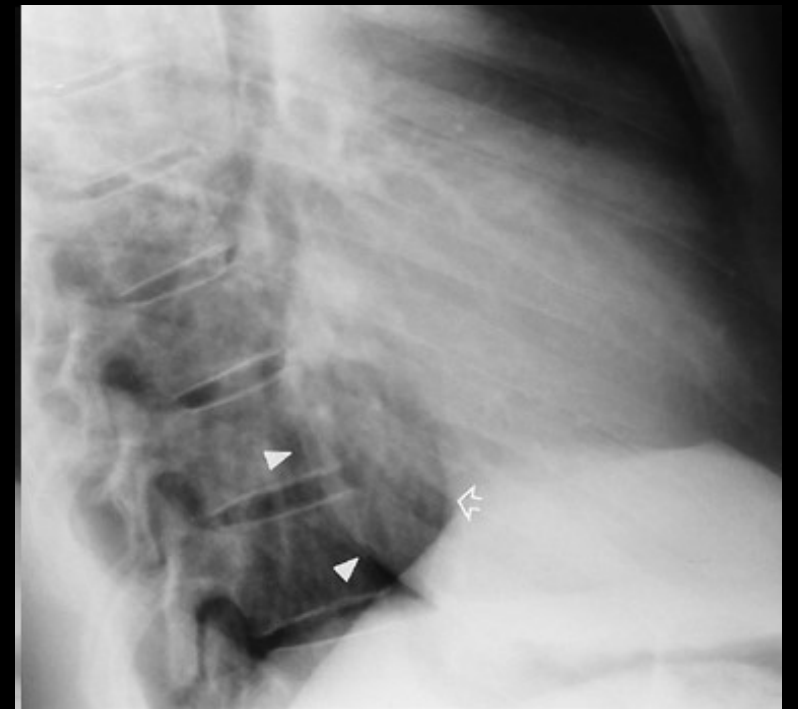
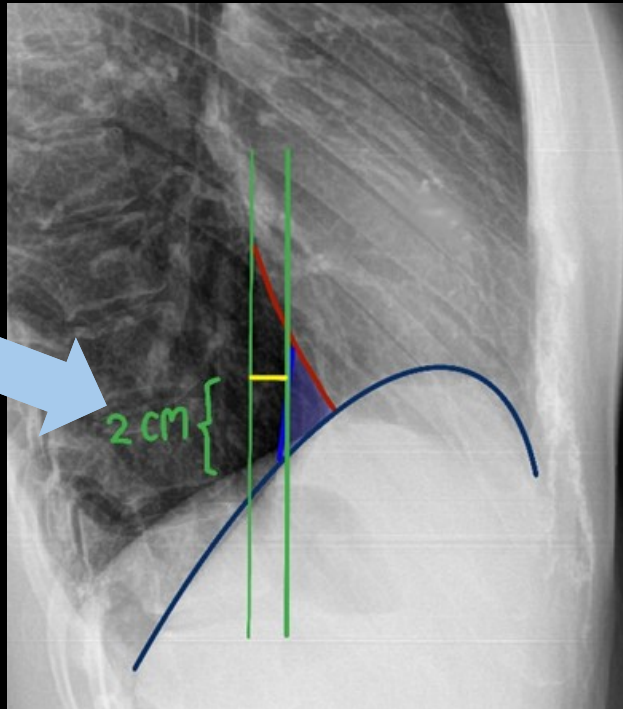
Tatco V, Left ventricular enlargement on chest radiography (illustration). Case study, Radiopaedia.org

LV Enlargement – lateral film



Hoffman-Rigler sign

- Measure 2 cm above intersection between IVC and right hemidiaphragm
- (+) if border extends > 1.8 cm posterior to IVC



Thoracic Great Vessels Evaluation

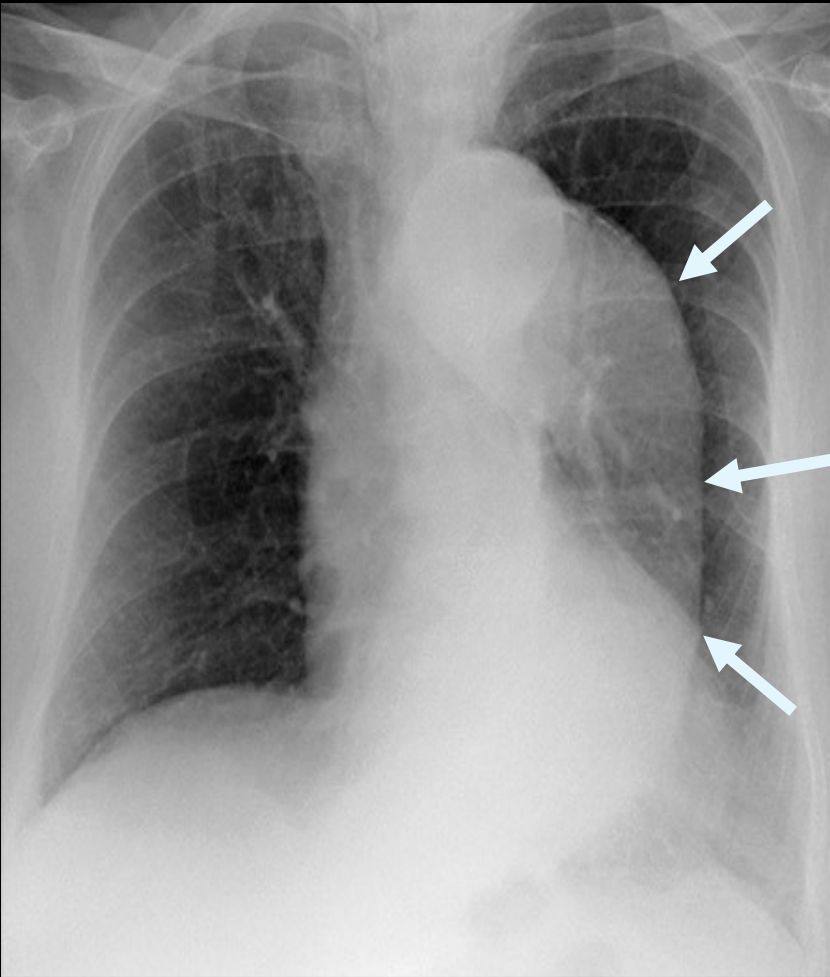
Enlarged Aortic Shadow



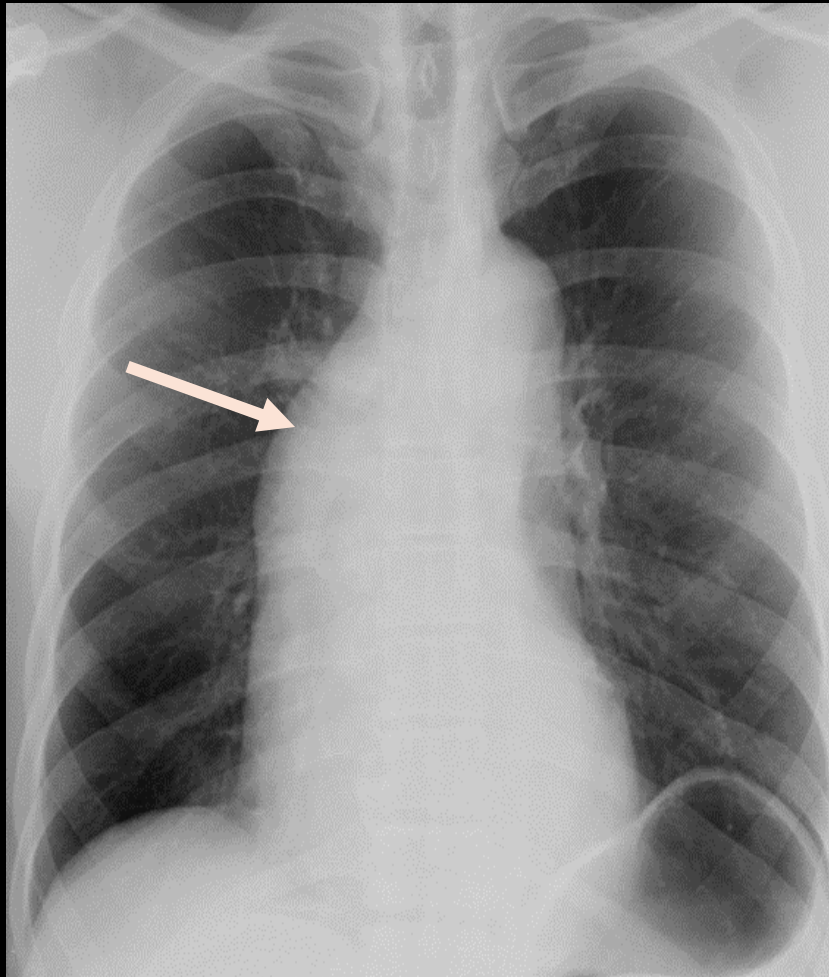
Possible causes

- Turbulence flow (AV stenosis)
- Volume overload (AV regurgitation)
- Pressure overload (hypertension)
- Aortic aneurysm/dissection
- Aortic coarctation with post-stenotic dilatation
- Connective tissue disease

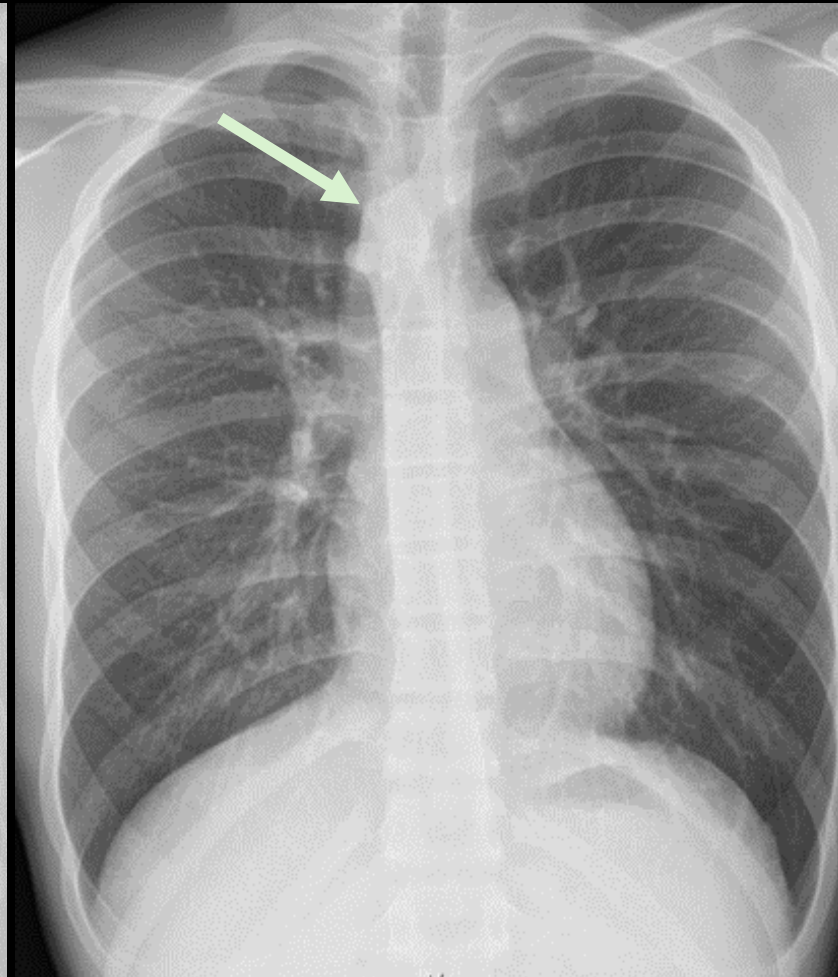
Abnormal Aortic Shadow



Descending thoracic
aortic aneurysm



Ascending aortic
aneurysm



Right-sided aortic arch

Enlarged Main pulmonary artery



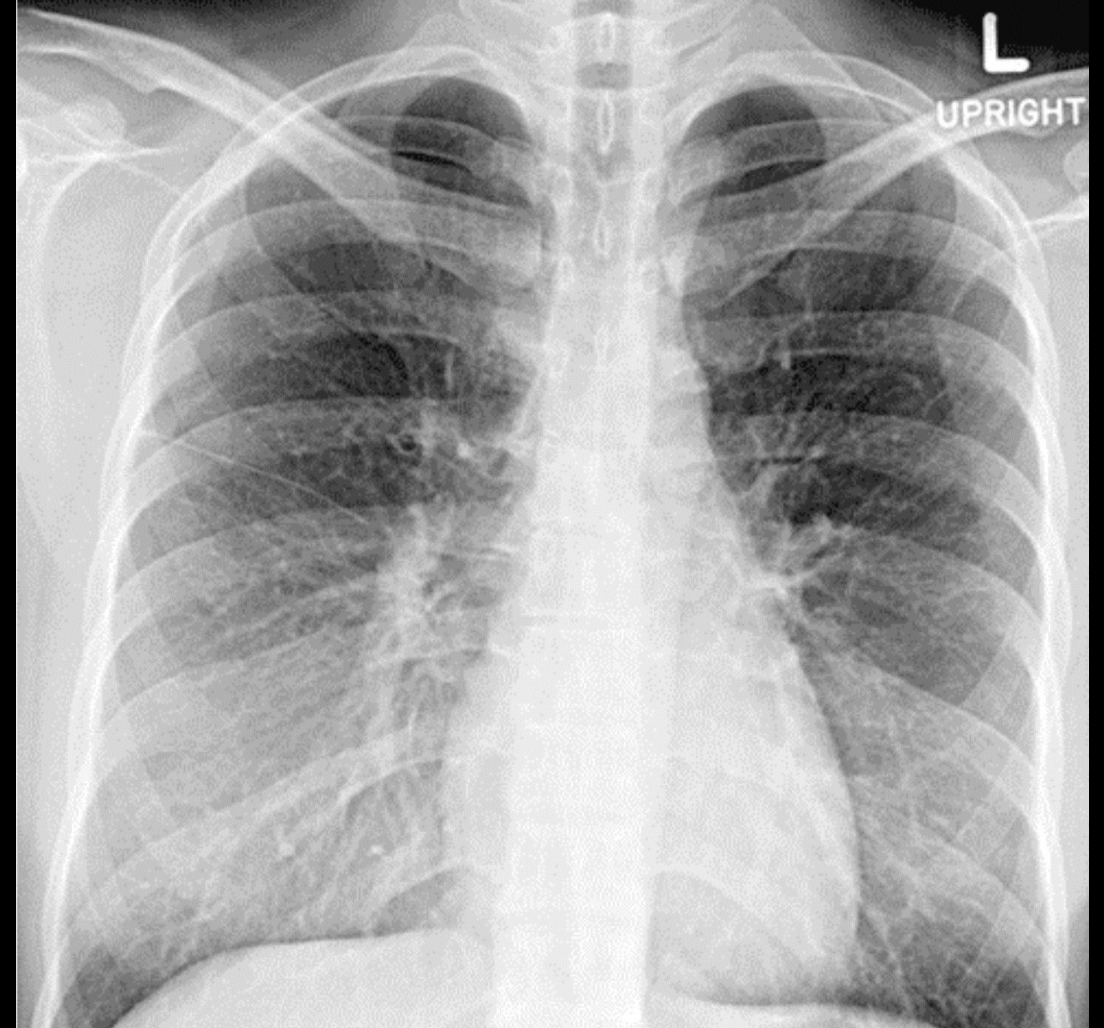
Possible causes

- Pulmonary arterial hypertension
- Excess pulmonary blood flow
- Left to right shunt
- PV stenosis/regurgitation
- Pulmonary artery aneurysm
- Idiopathic dilatation

Enlarged Main pulmonary artery



Dilated MPA

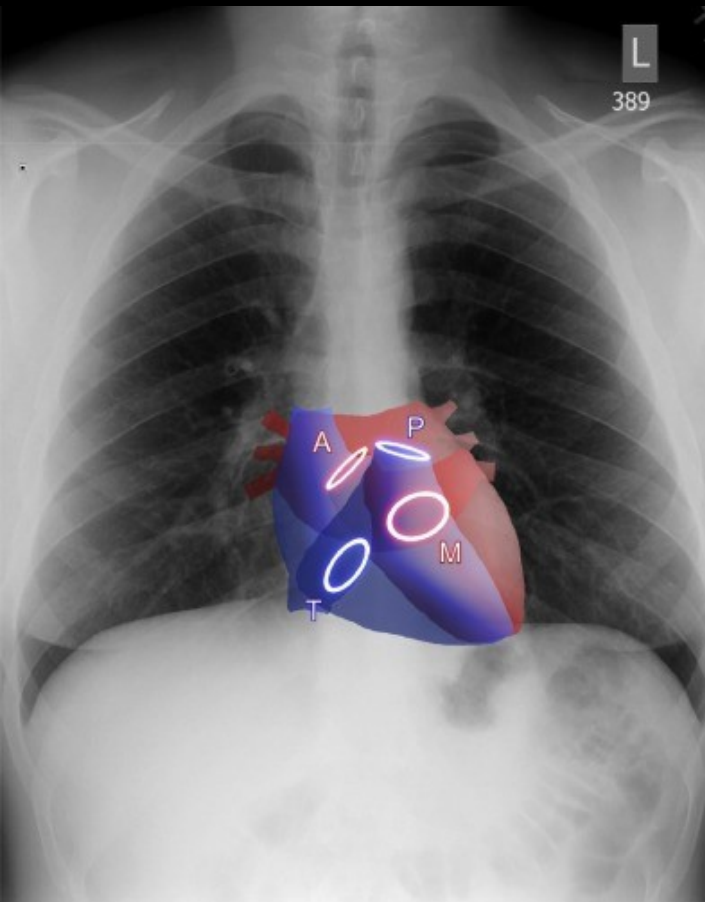


Normal-sized MPA

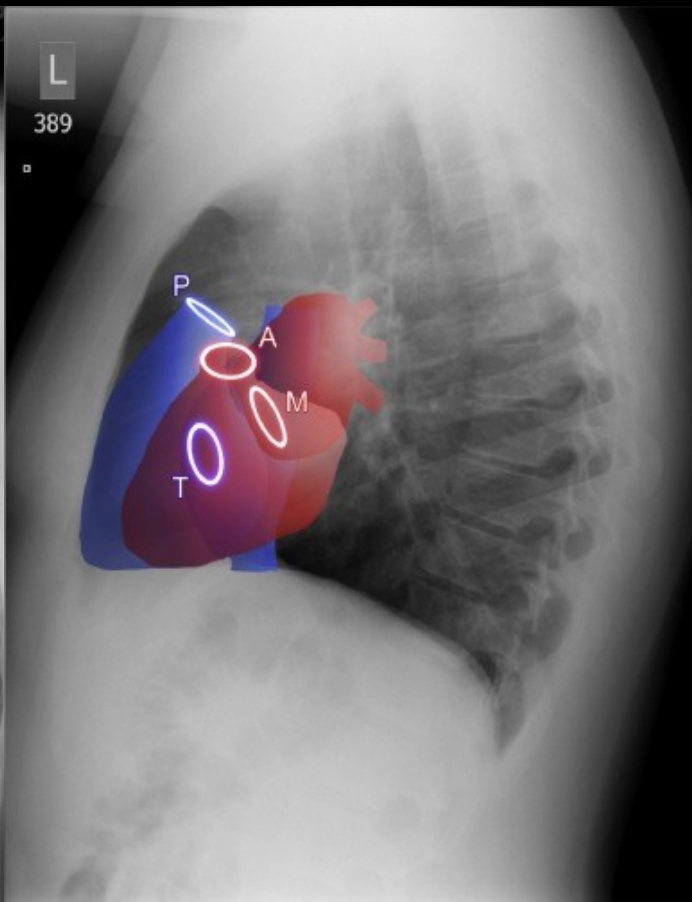


Expected Valve Position

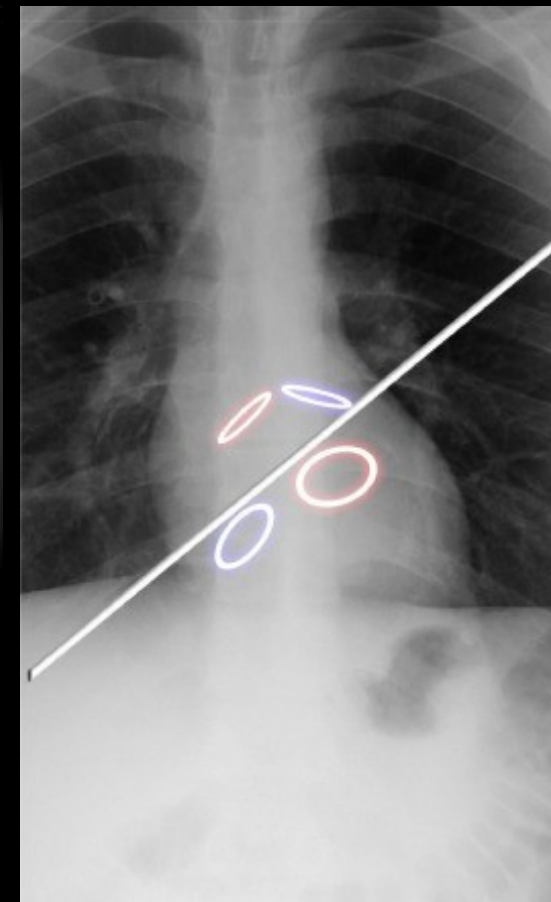
Cardiac Valves



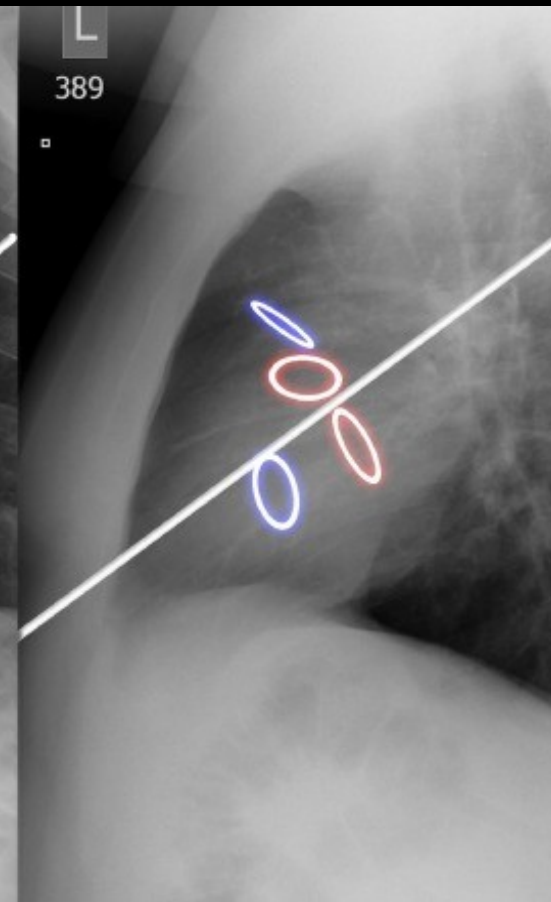
Frontal view



Lateral view



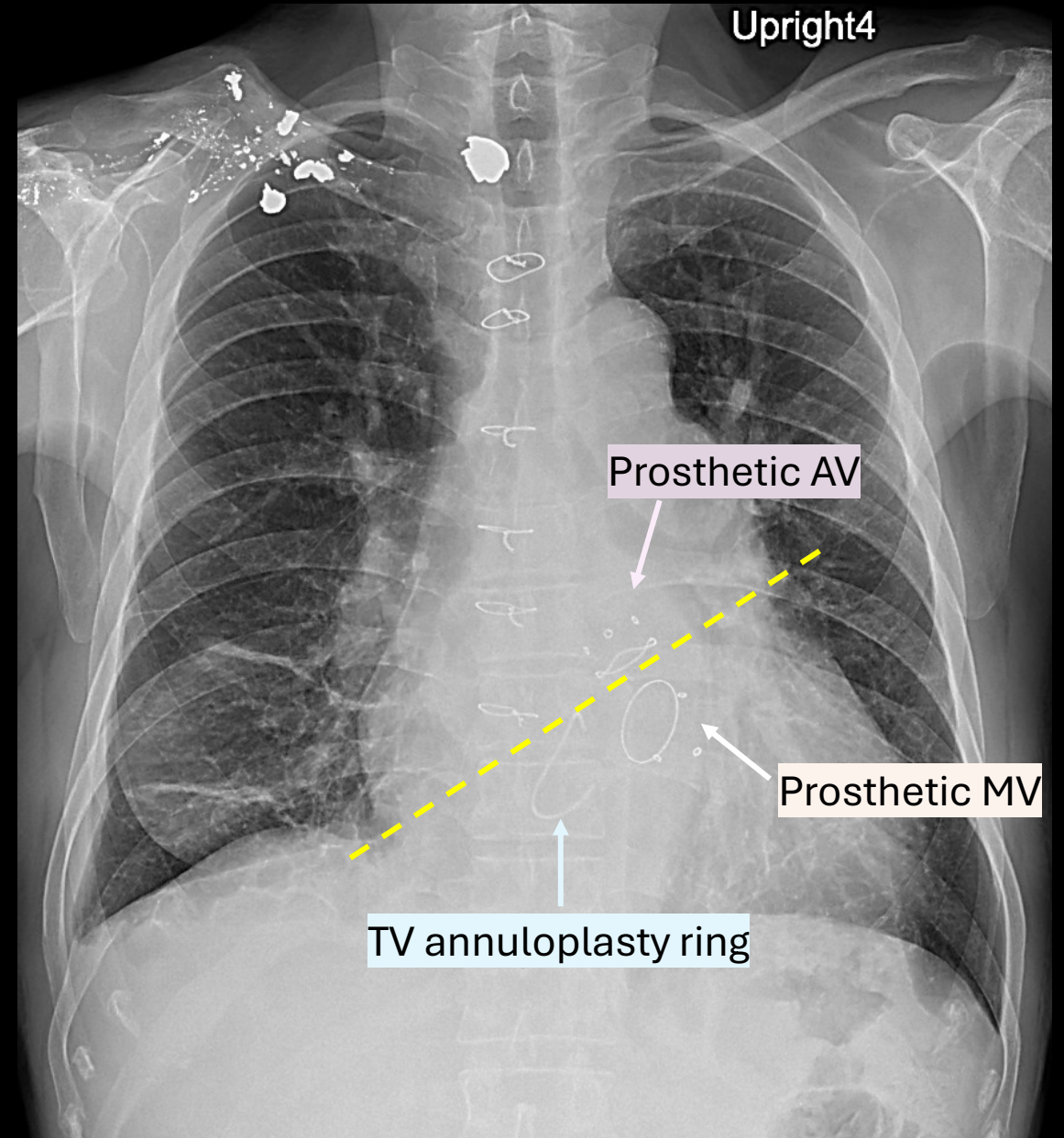
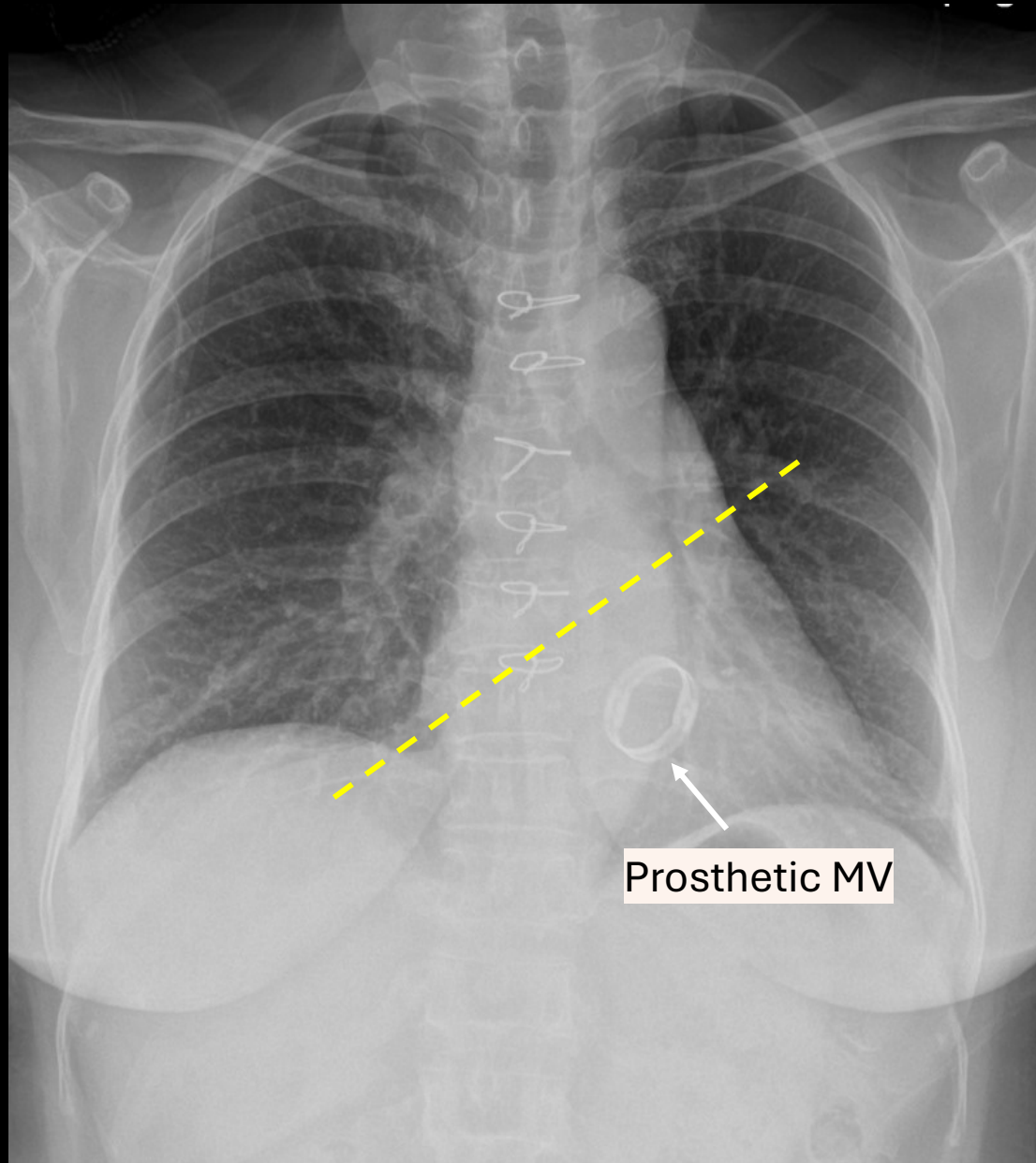
Frontal view



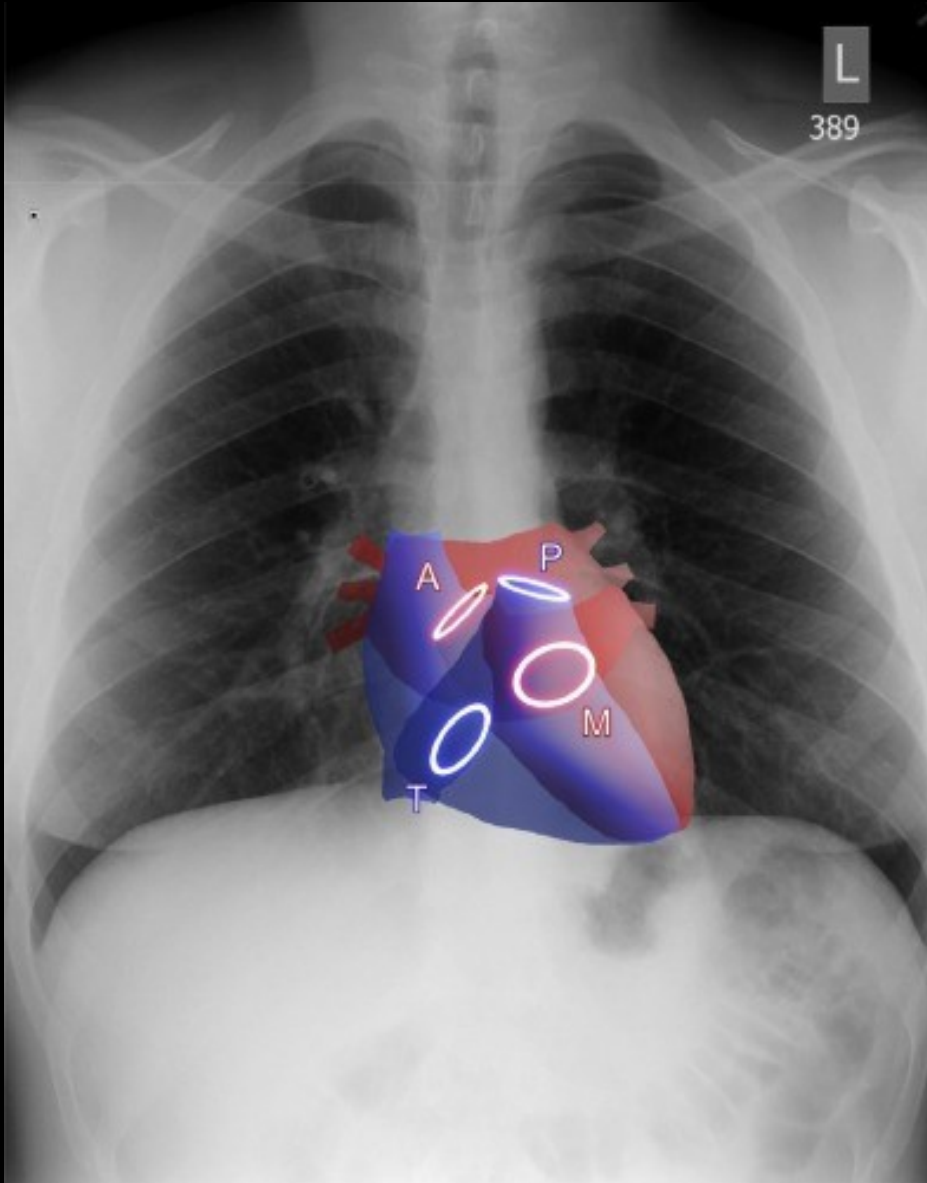
Lateral view

Imaginary line { Frontal view – line drawn from right cardiophrenic angle to distal left main bronchus
Lateral view – parallel to the long axis of the heart

Cardiac Valves: Prosthetic valve



Cardiac Valves: Helping signs for identify valve problem



LA enlargement

MV

RA enlargement

TV

Ascending aortic
dilatation

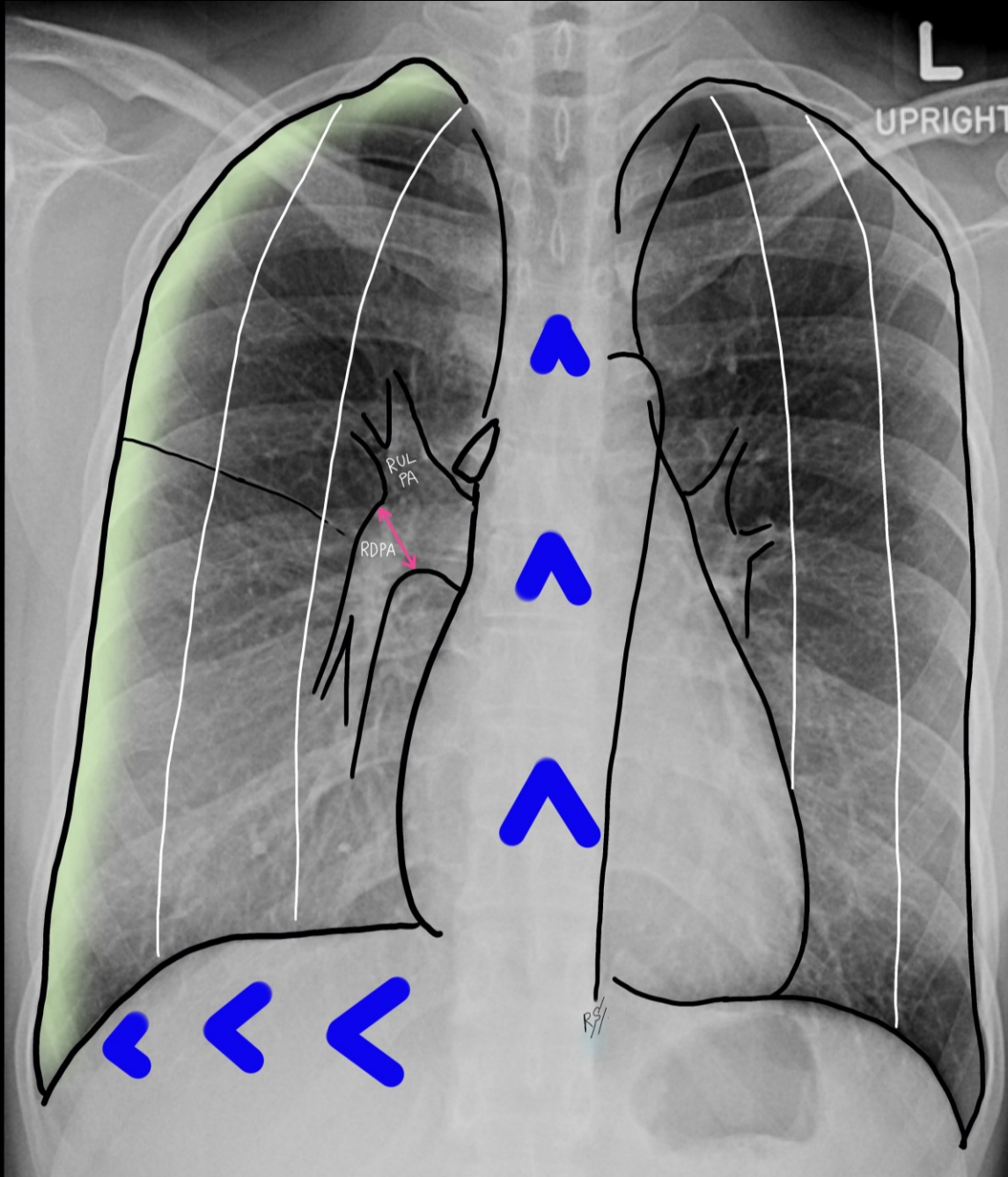
AV

Pulmonary Vascularity

Pulmonary vascularity

Normal vasculature

- Central > peripheral
- Lower > upper
- Compared within the same zones***
- RDPA (Rt interlobar PA) <17 mm
- No vessels in 1 cm from periphery



High flow or Over circulation



Possible cause

- Left to right shunt
- Cyanotic heart disease with bidirectional shunt
- High cardiac output

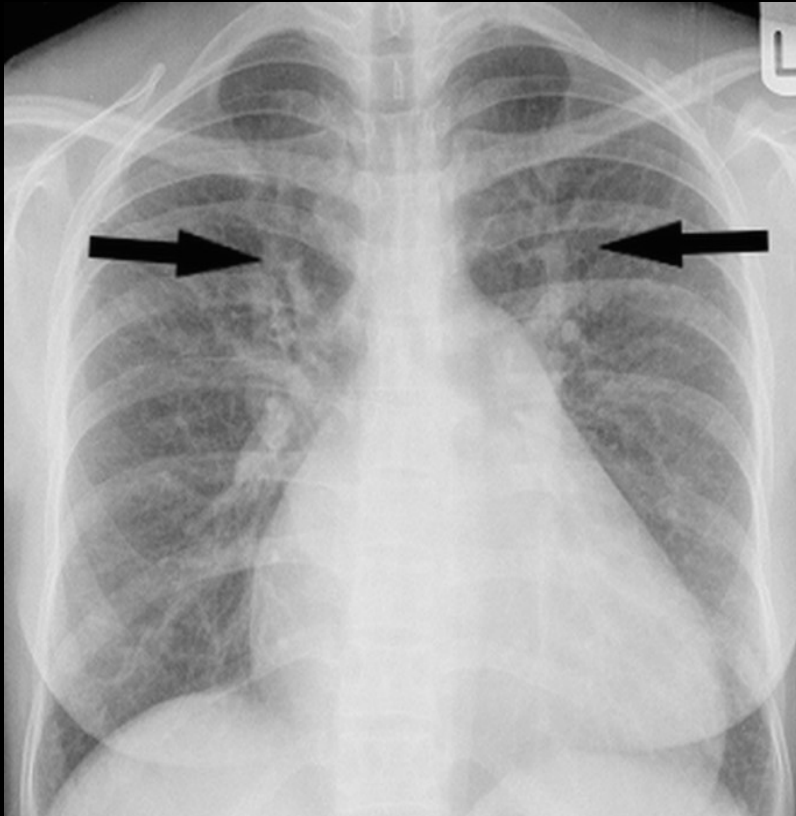
Decreased vascularity



Possible cause

- Lesions with right to left shunt
- Pulmonary atresia/stenosis
- RVOT obstruction e.g. TOF
- Right-sided heart failure
- Pericardial disease e.g. constrictive pericarditis, effusion + cardiac tamponade
- Pulmonary artery embolism

Pulmonary venous hypertension

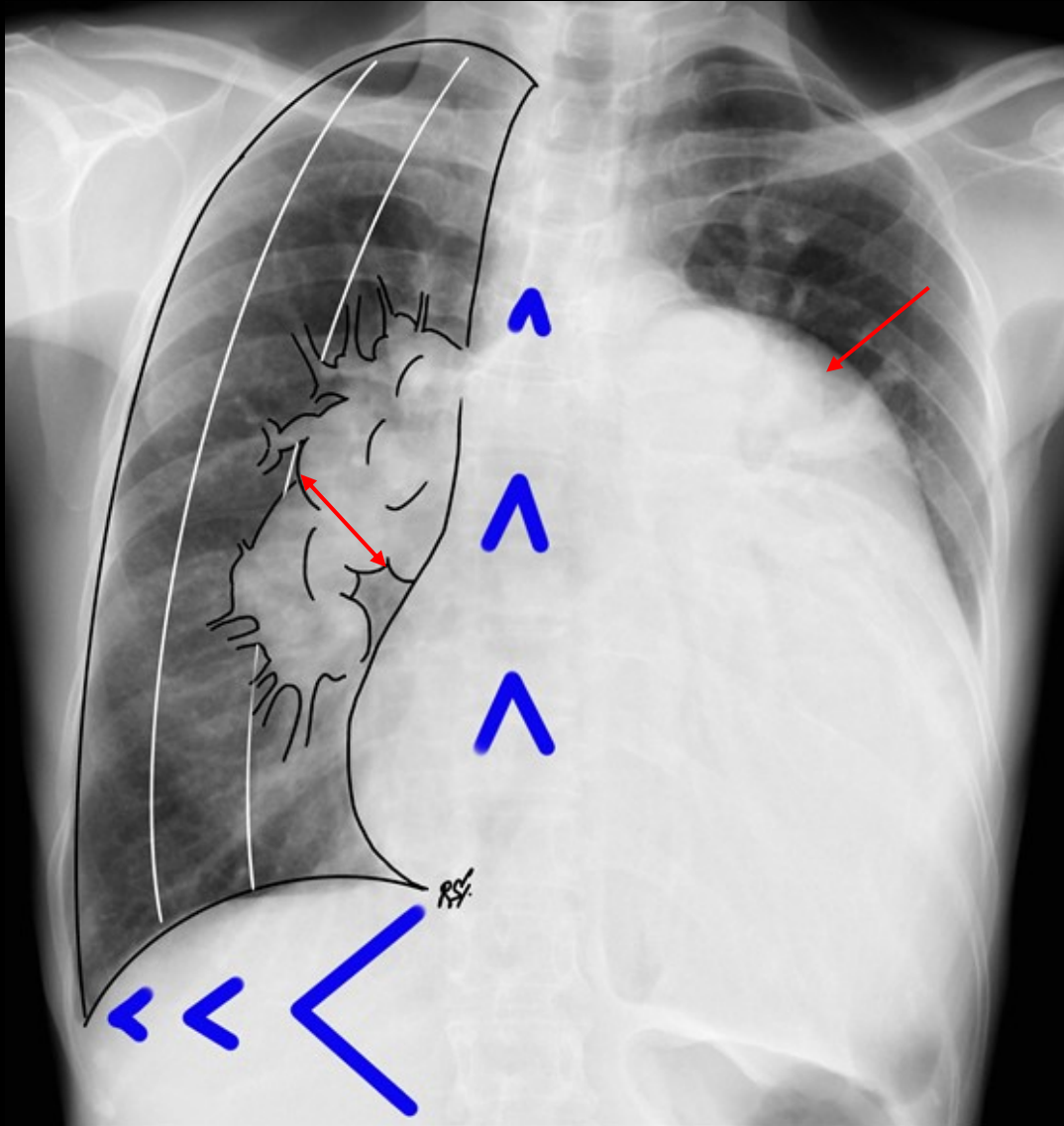


Stage 1

PCWP 12-19 mmHg

Vascular redistribution
(cephalization)

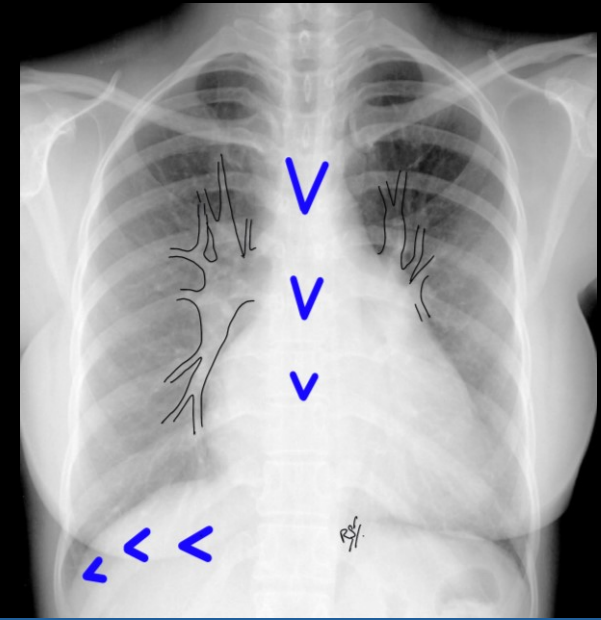
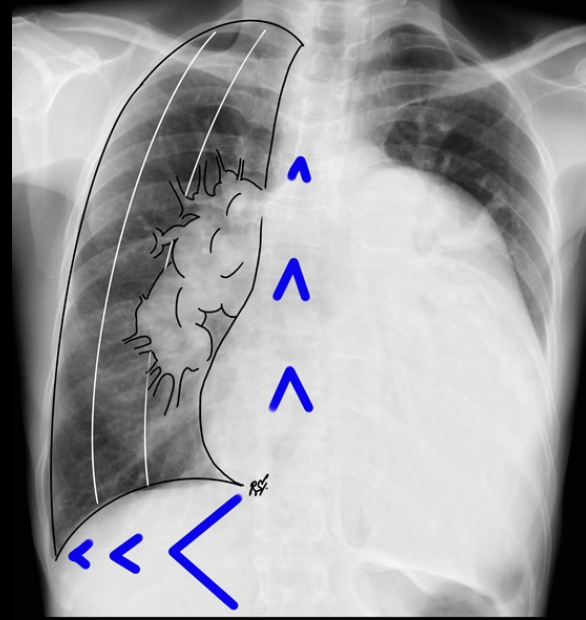
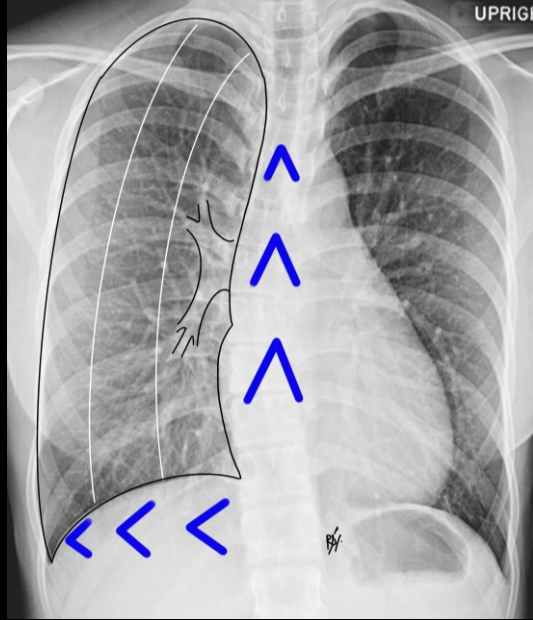
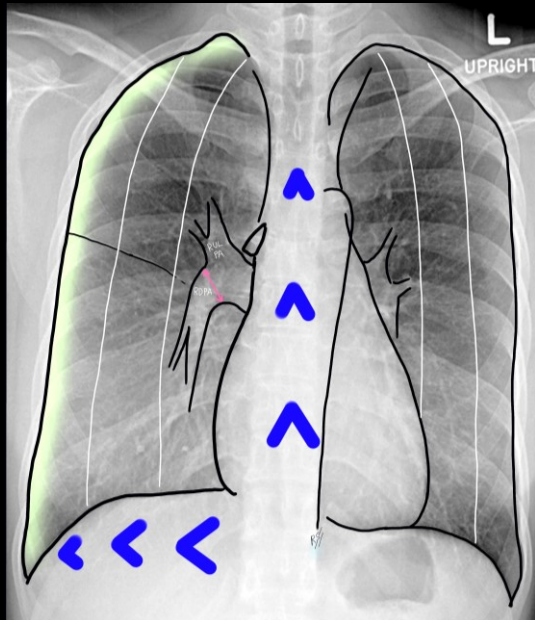
Pulmonary arterial hypertension



CXR Findings

- Increased size of MPA
- Increased size of RDPA (n 17-20 mm)
- Pruning appearance of vessels
- Centralization of vasculature
- +/- RV enlargement

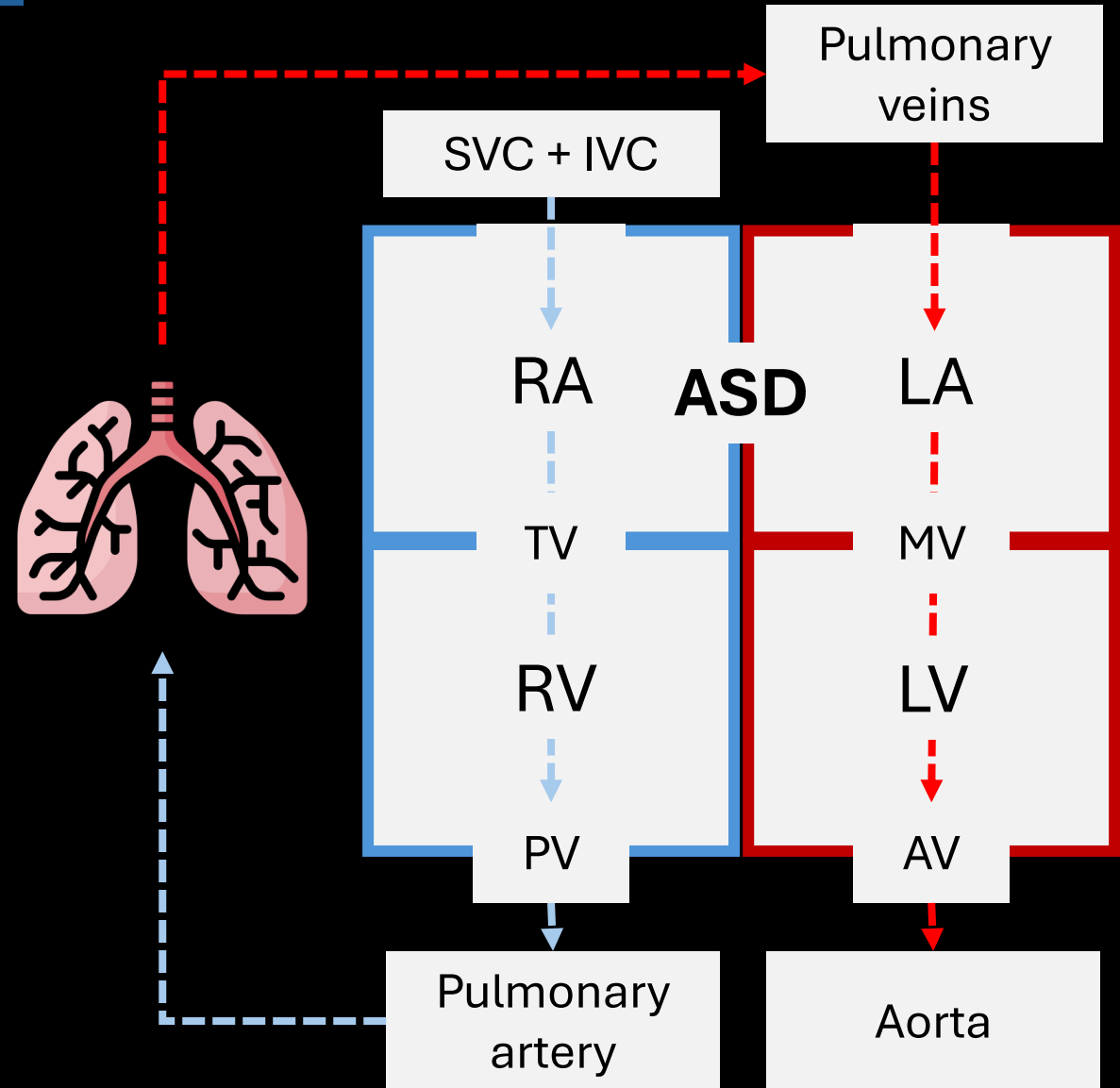
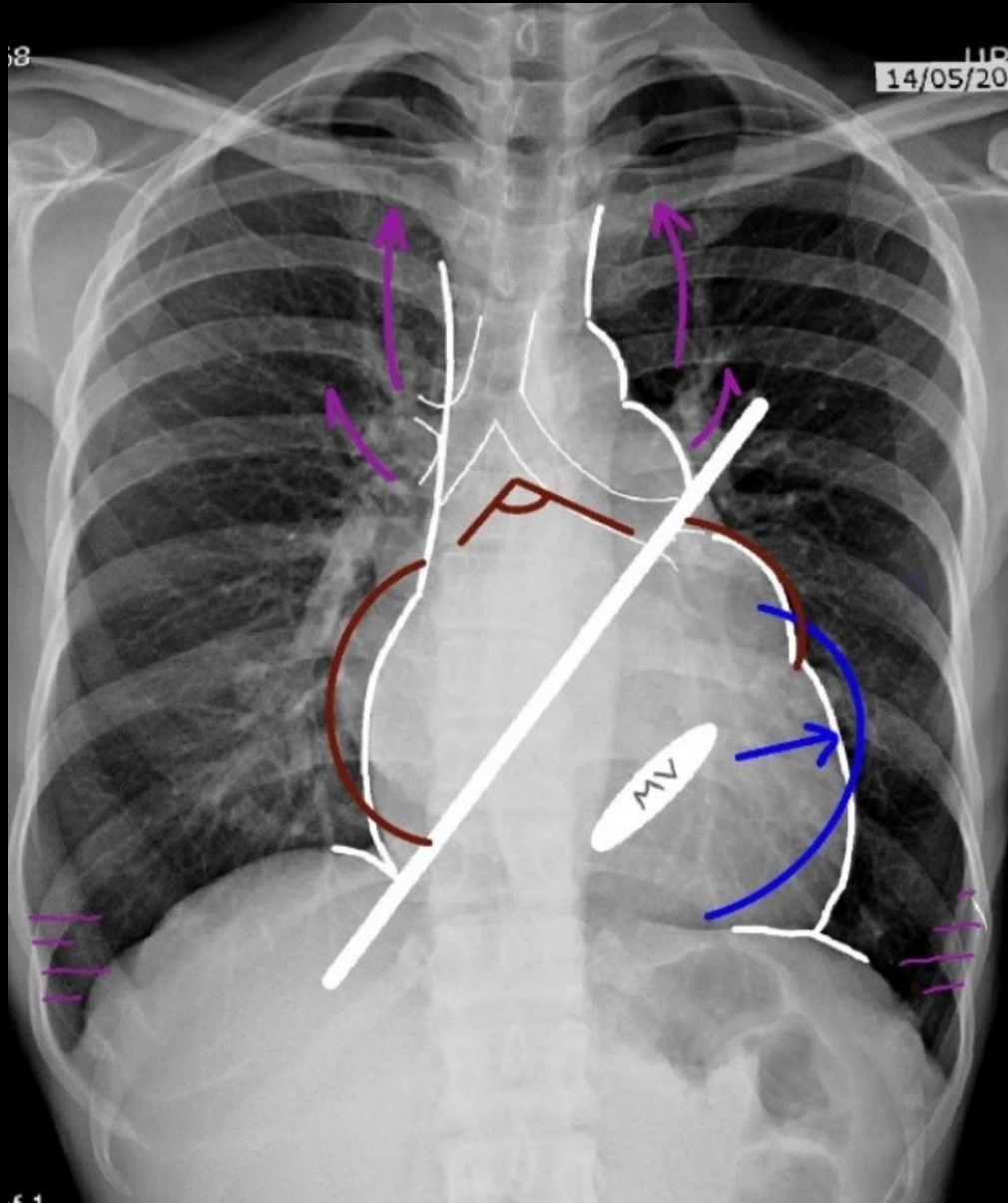
	RDPA size	Upper-lower	Central-peripheral
Normal	RDPA <17-20 mm	Lower>upper	Gradual tapering
High flow	RDPA >17-20 mm Increased size and number of vessels	Lower>upper	Gradual tapering
PAHT	RDPA >17-20 mm	Lower>upper	Central>>>peripheral Prunning apperaance
PVHT	RPDA $\pm$ >17-20 mm	Cephalization (Upper>lower)	Gradual tapering



Cardiac Morphology and Configuration

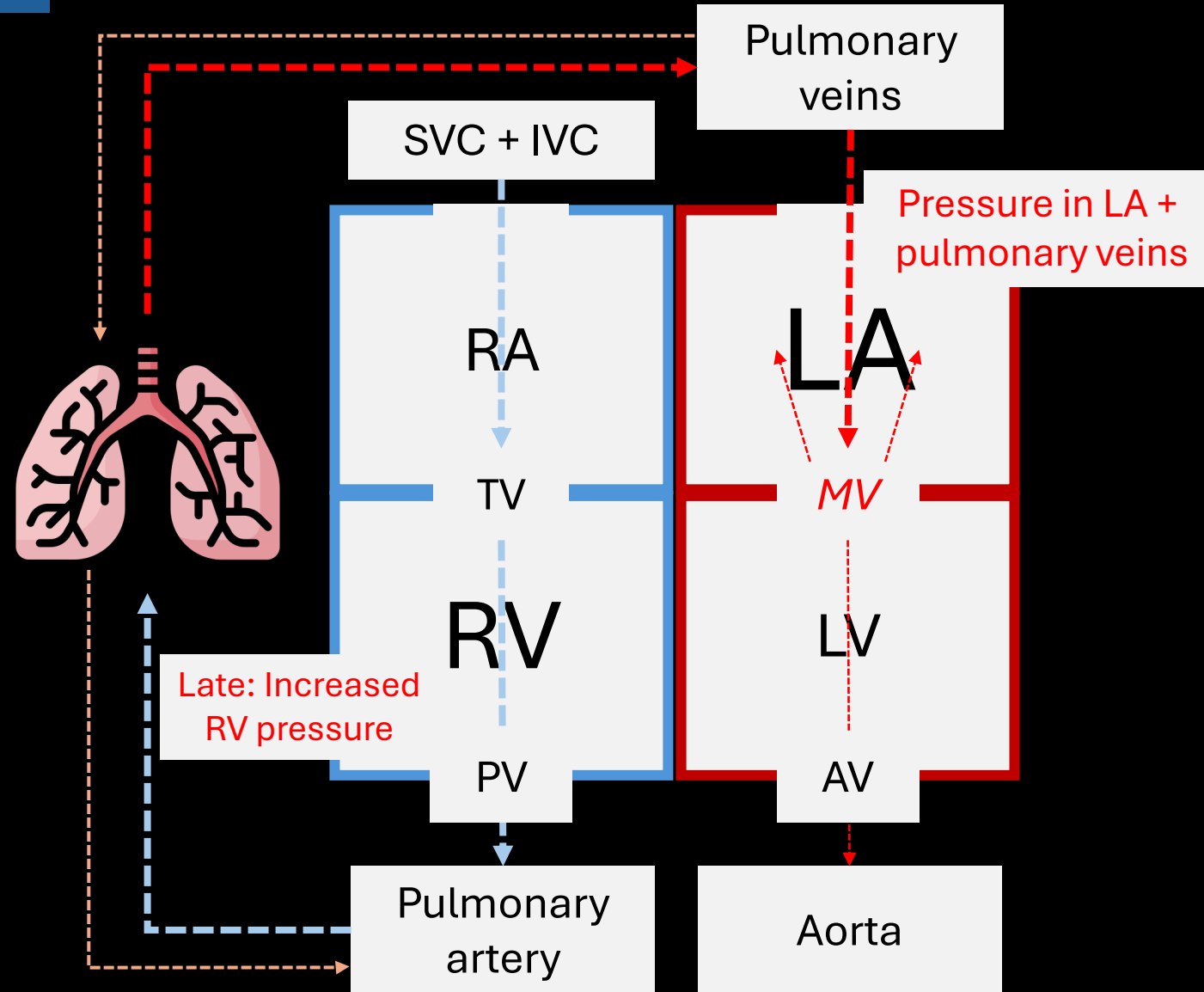
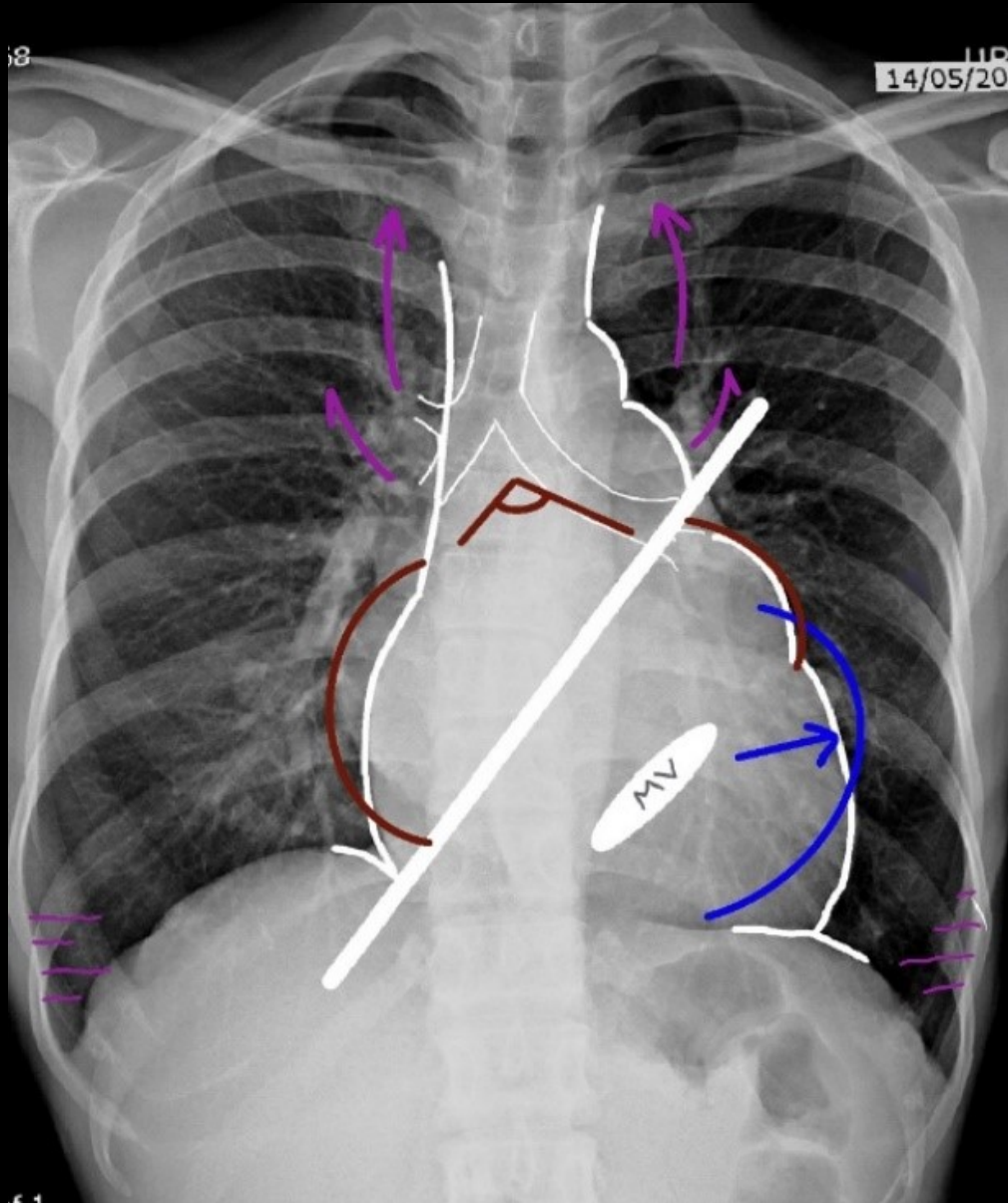


Mitral Configuration



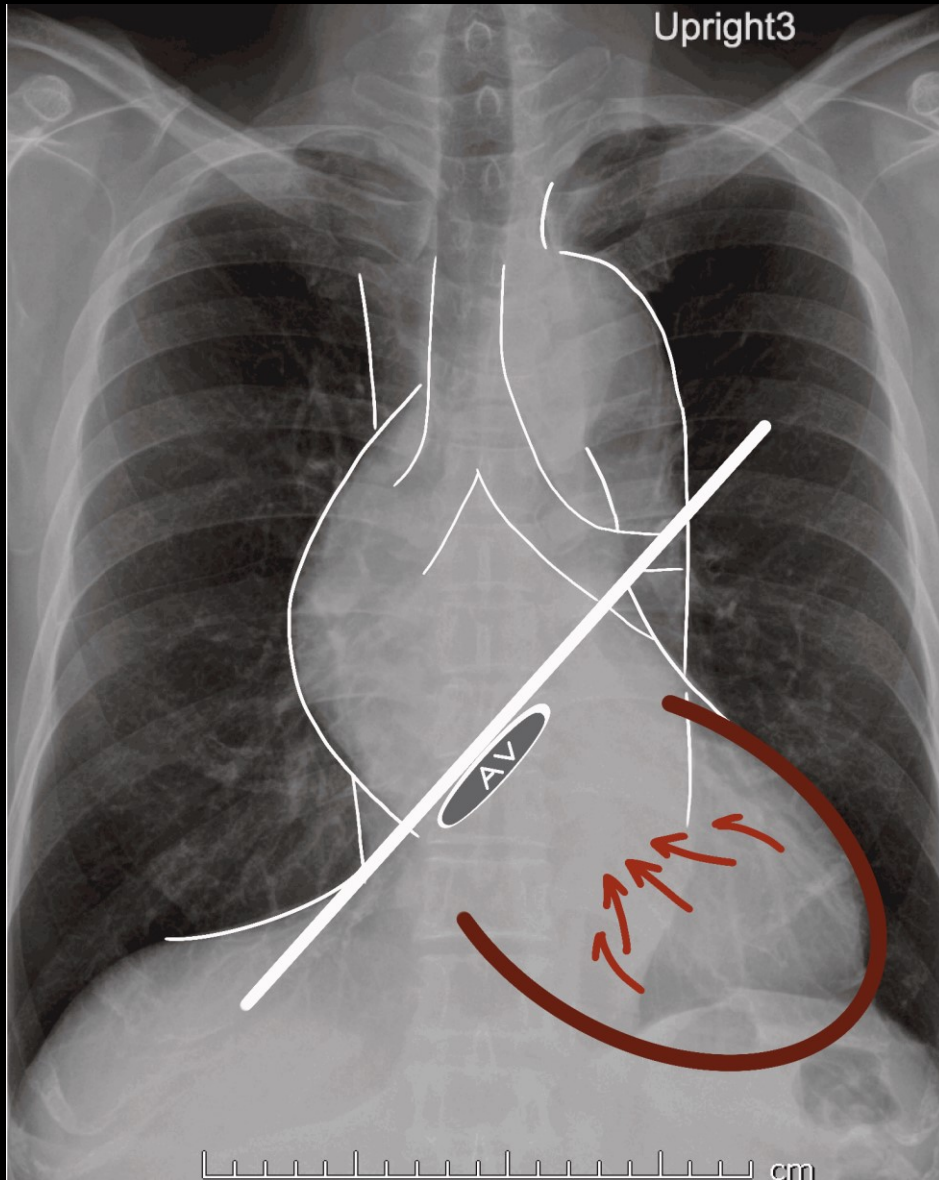
Mitral Configuration

LAE + PVHT + RVH = mitral configuration



Aortic Configuration

LVE + Dilated A Ao = Aortic configuration



AV stenosis (or regurgitation)

Pressure/volume load in LV

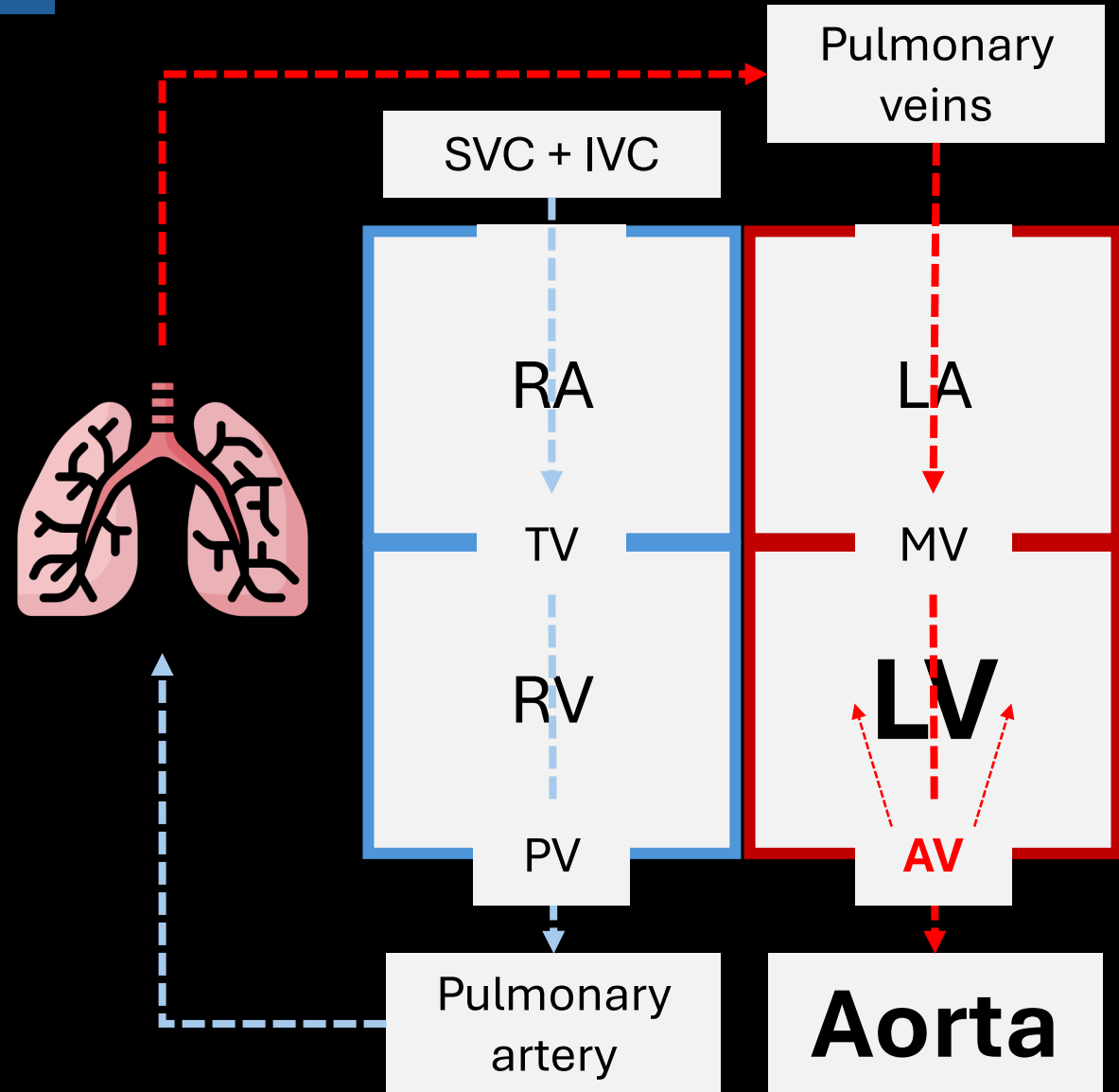
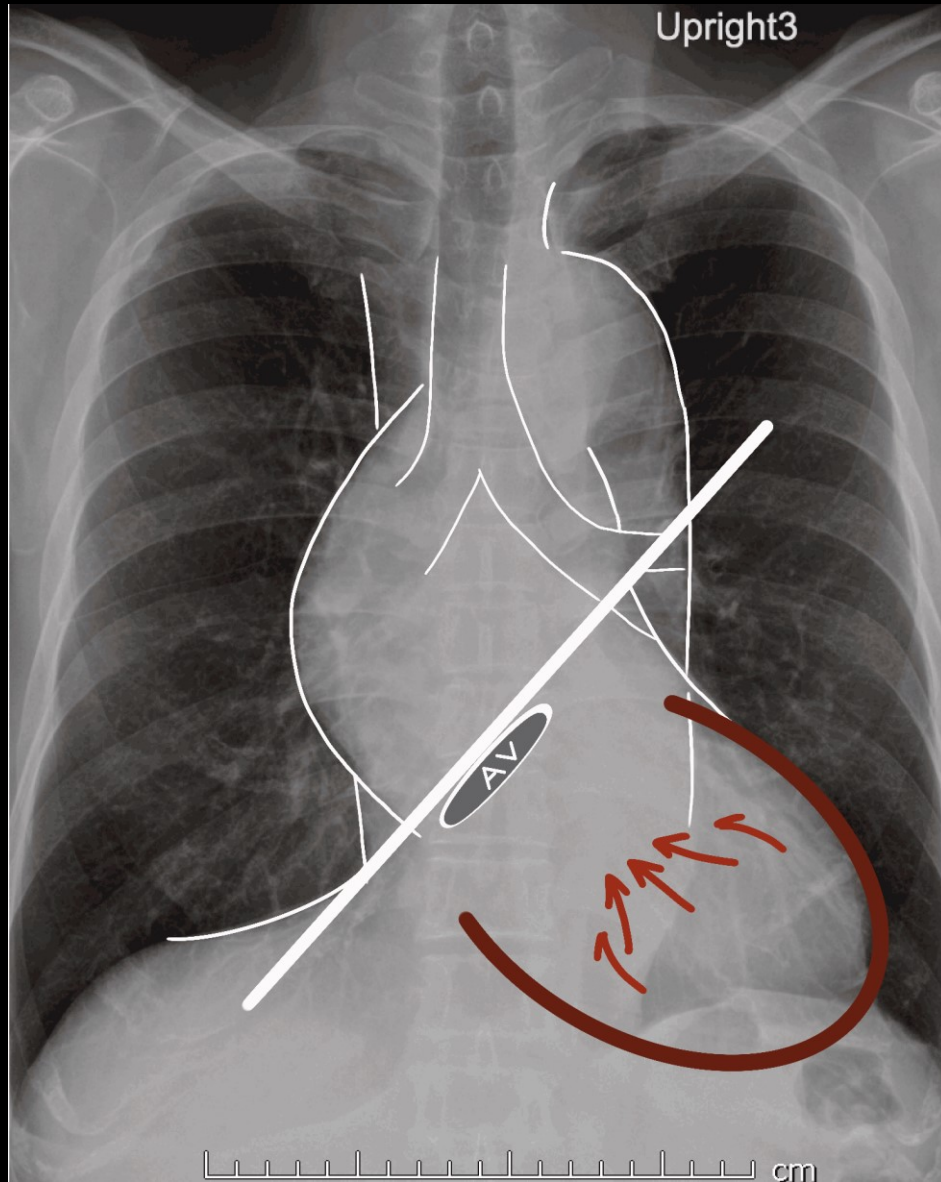
LV enlargement

Flow on ascending aorta

Dilated ascending aorta

Aortic Configuration

LVE + Dilated A Ao = Aortic configuration

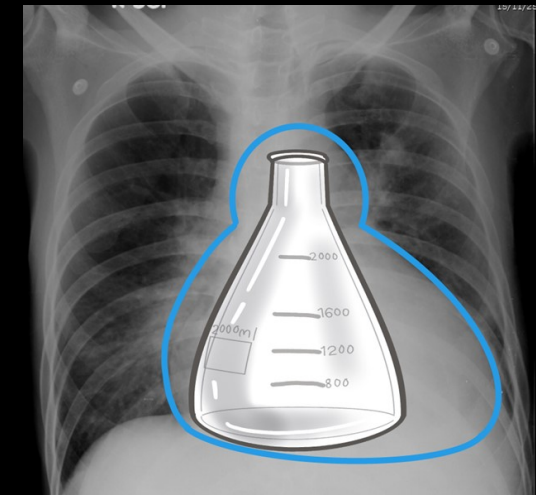
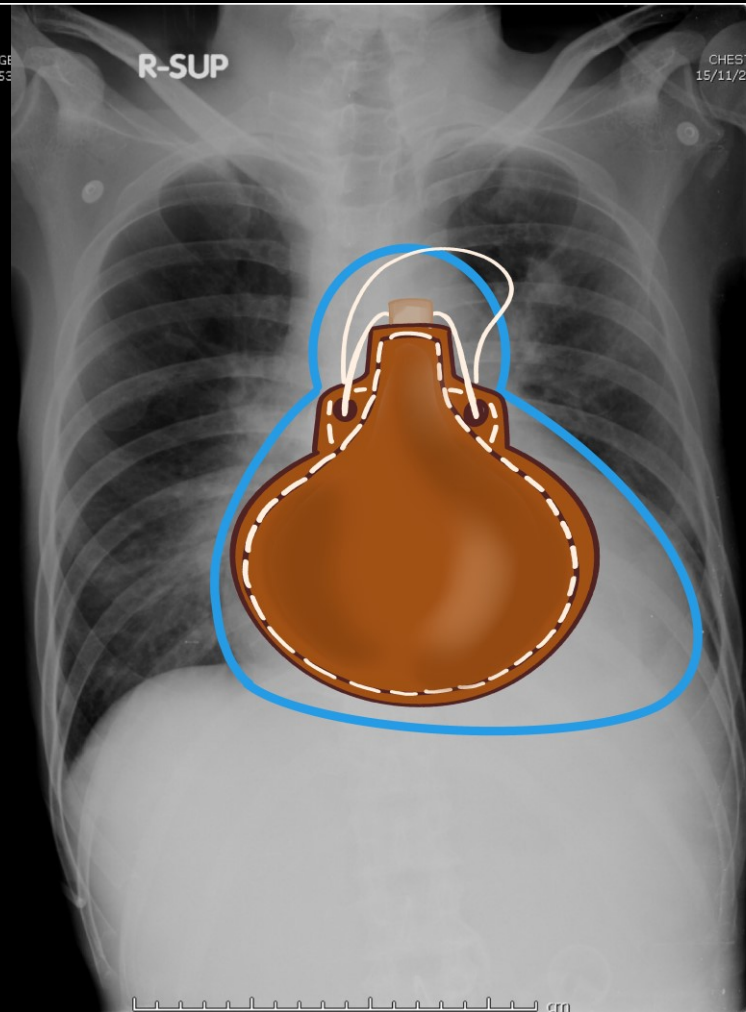
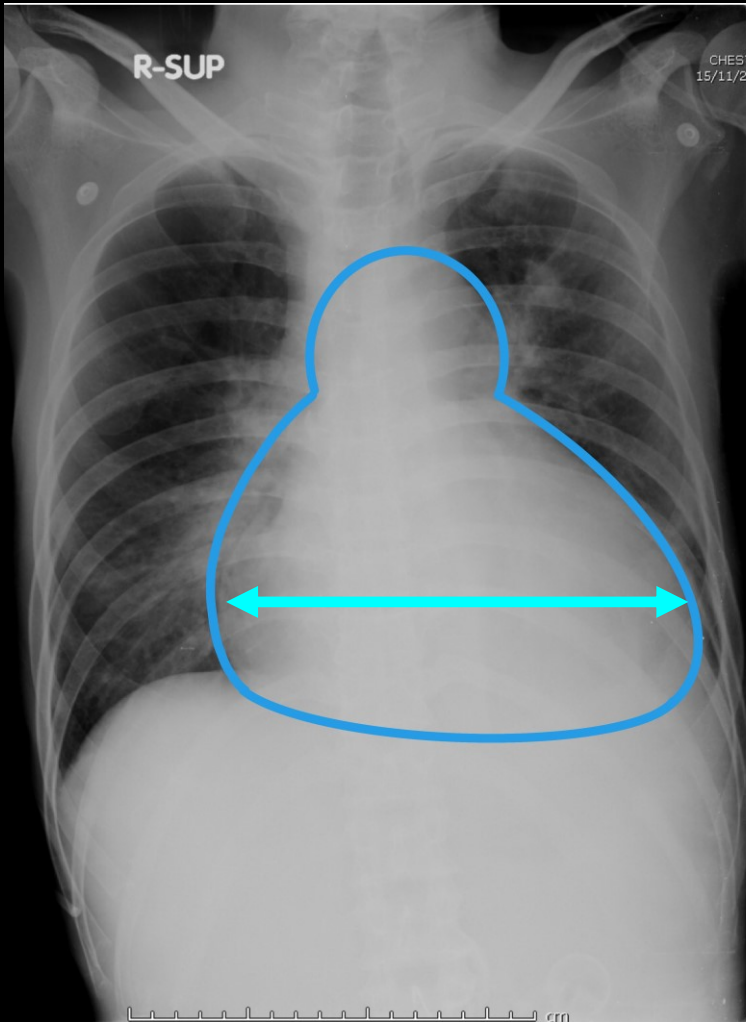


Globular-shape Heart

= Flask shape heart
= Water bottle shape heart

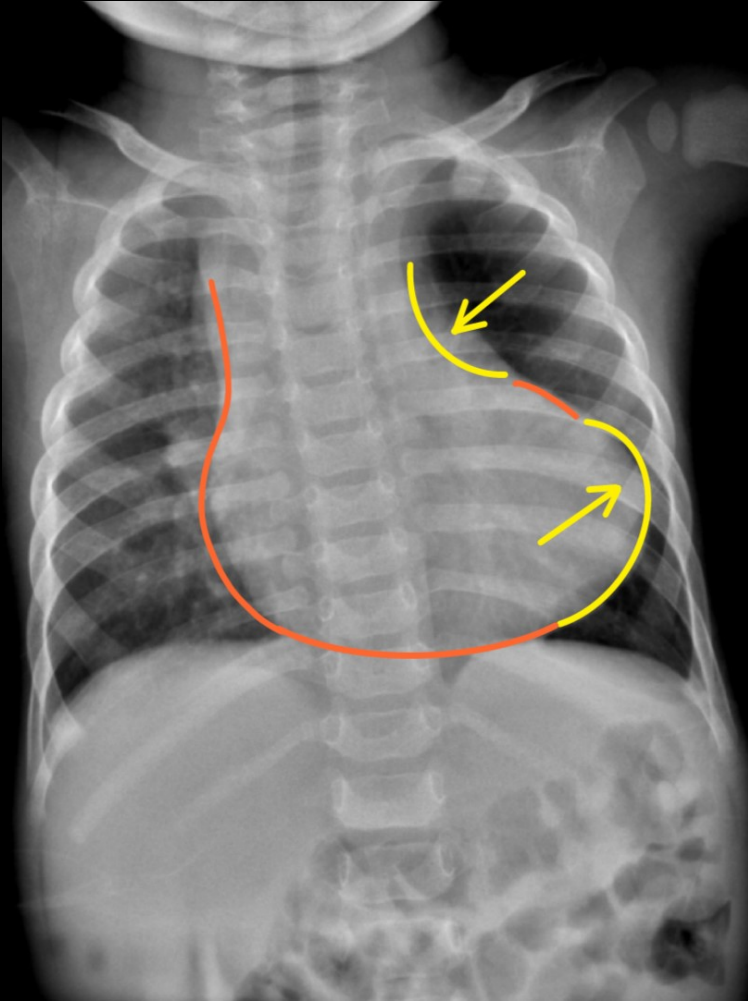
Possible causes

- Pericardial effusion
- Dilated cardiomyopathy
- Multivalvular heart disease



Boot-shaped Heart

Seen in cases of
Tetralogy of Fallot (TOF)

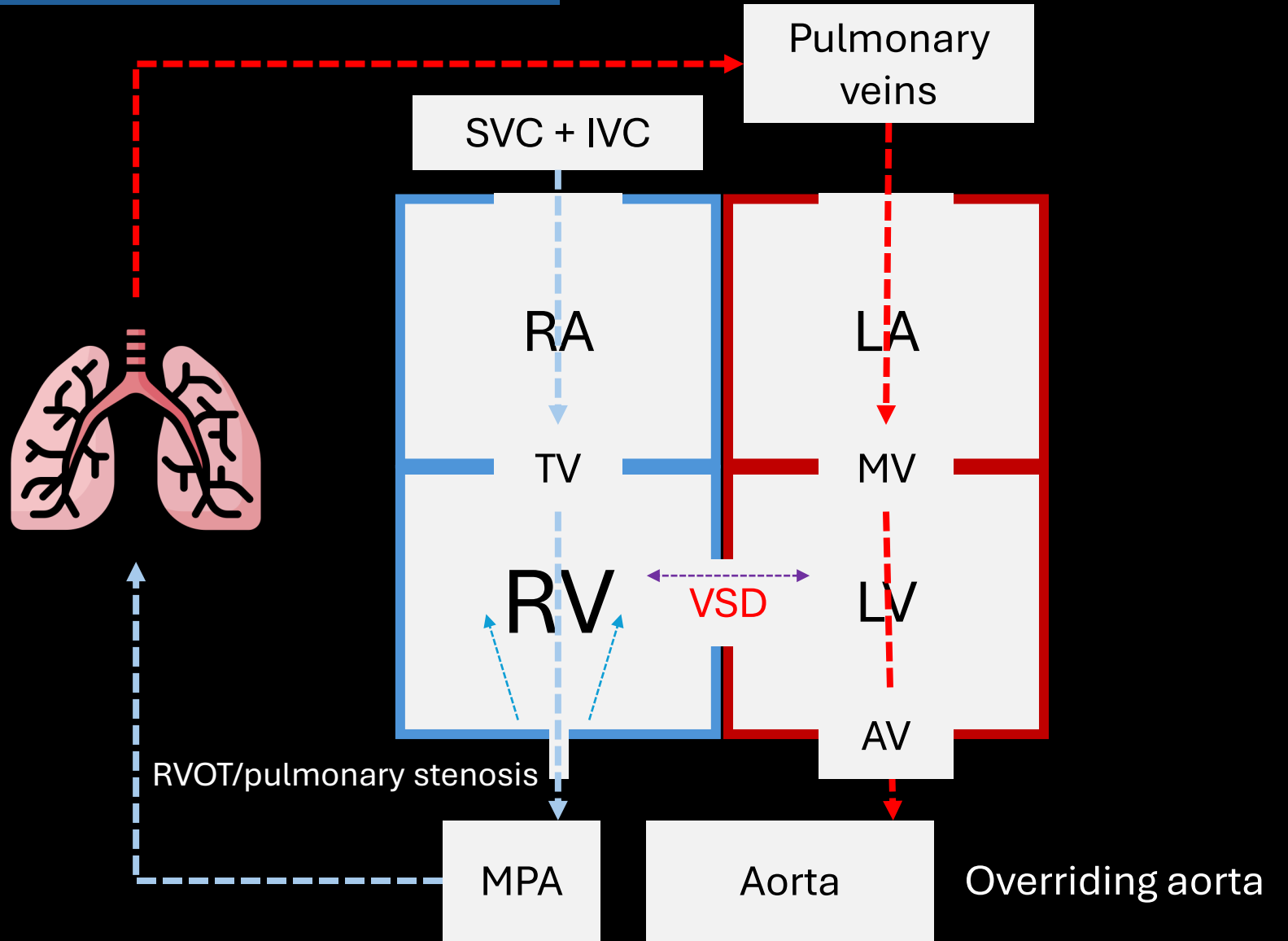
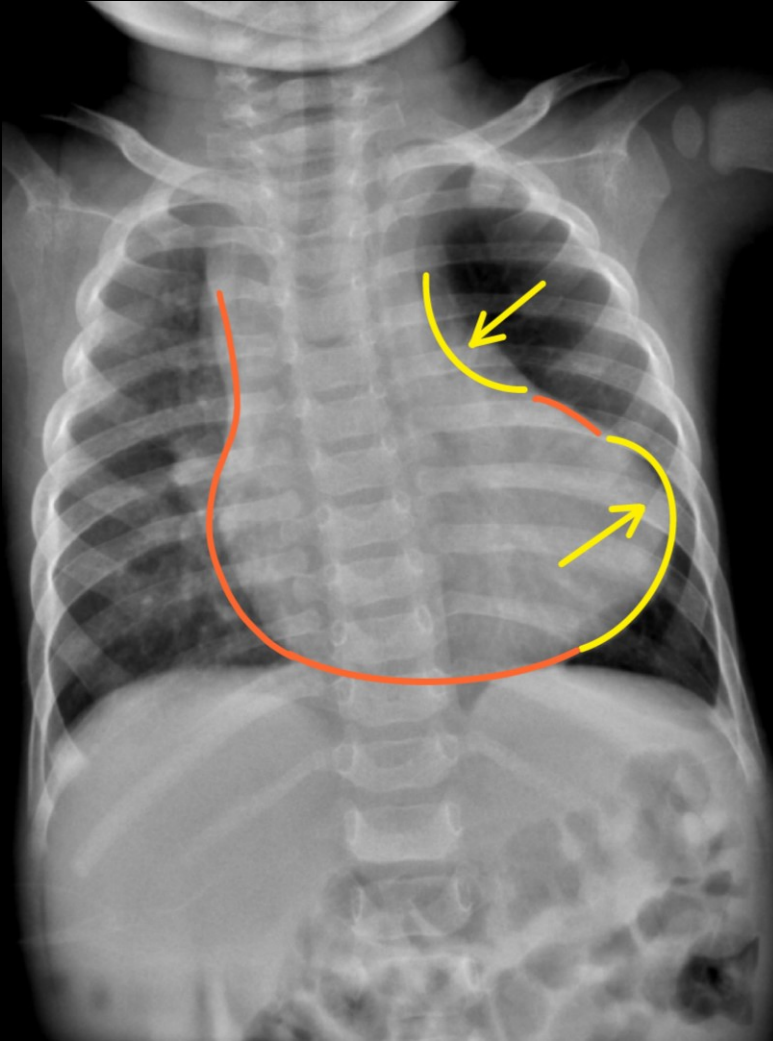


Features

- | | |
|----|-------------------------------------|
| 1. | Overriding aorta |
| 2. | Ventricular septal defect (VSD) |
| 3. | RVOT obstruction (concave MPA) |
| 4. | RV hypertrophy (upward lifted apex) |

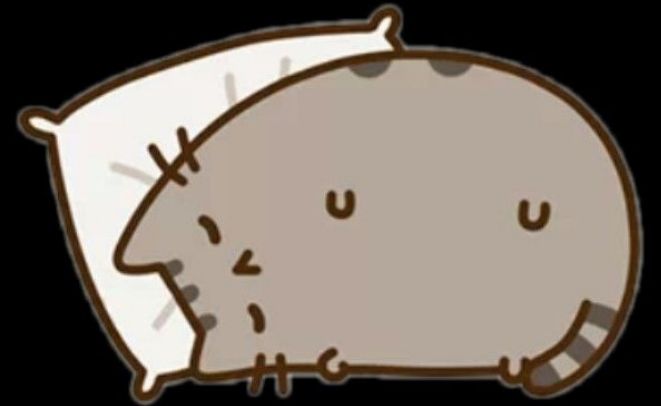
Pulmonary vascularity may vary decrease according to the severity of RVOT obstruction and VSD size

Boot-shape Heart



Non-cyanotic Congenital Heart Disease

Increased pulmonary vasculature: ASD, VSD, PDA



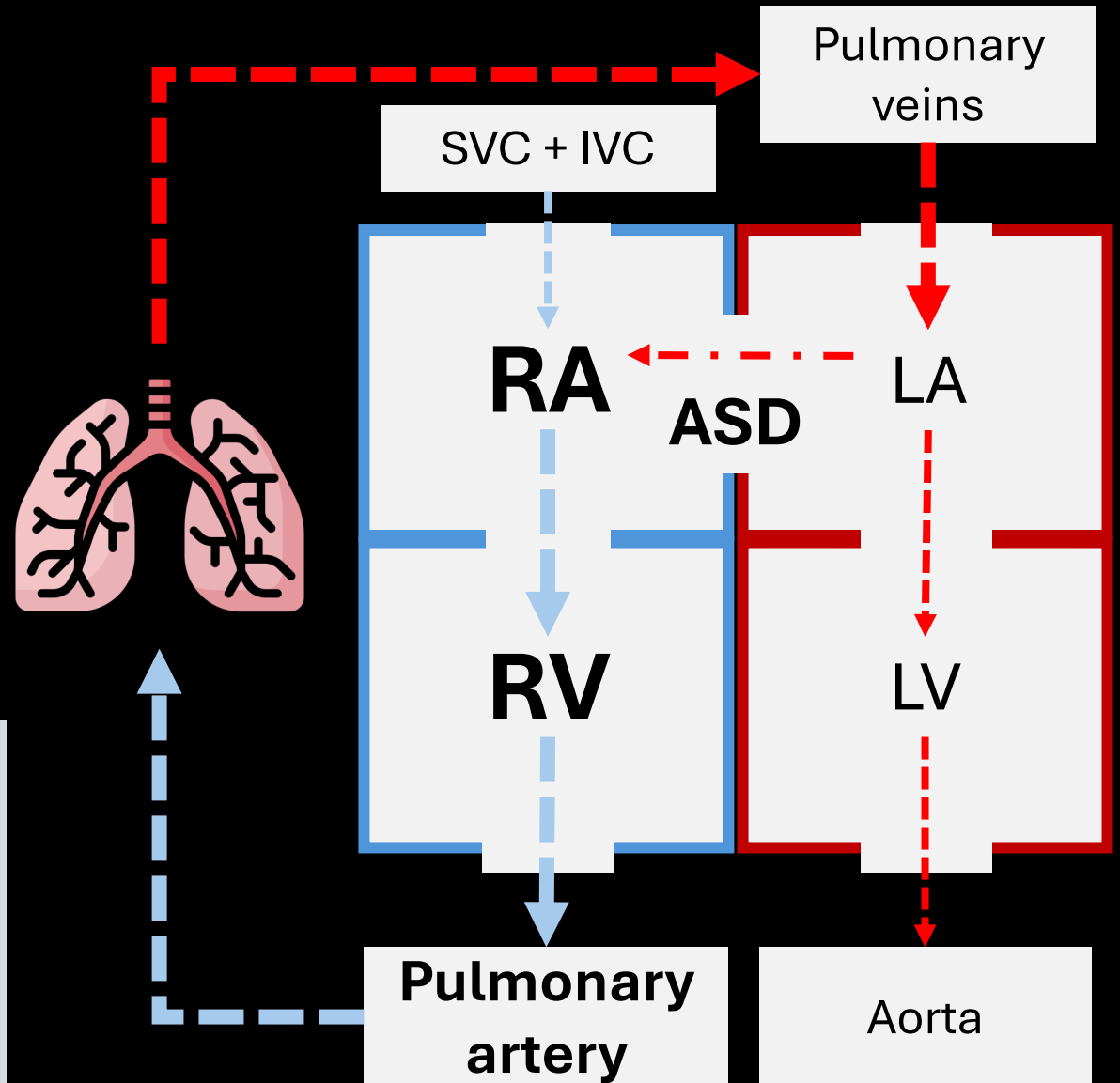


CXR

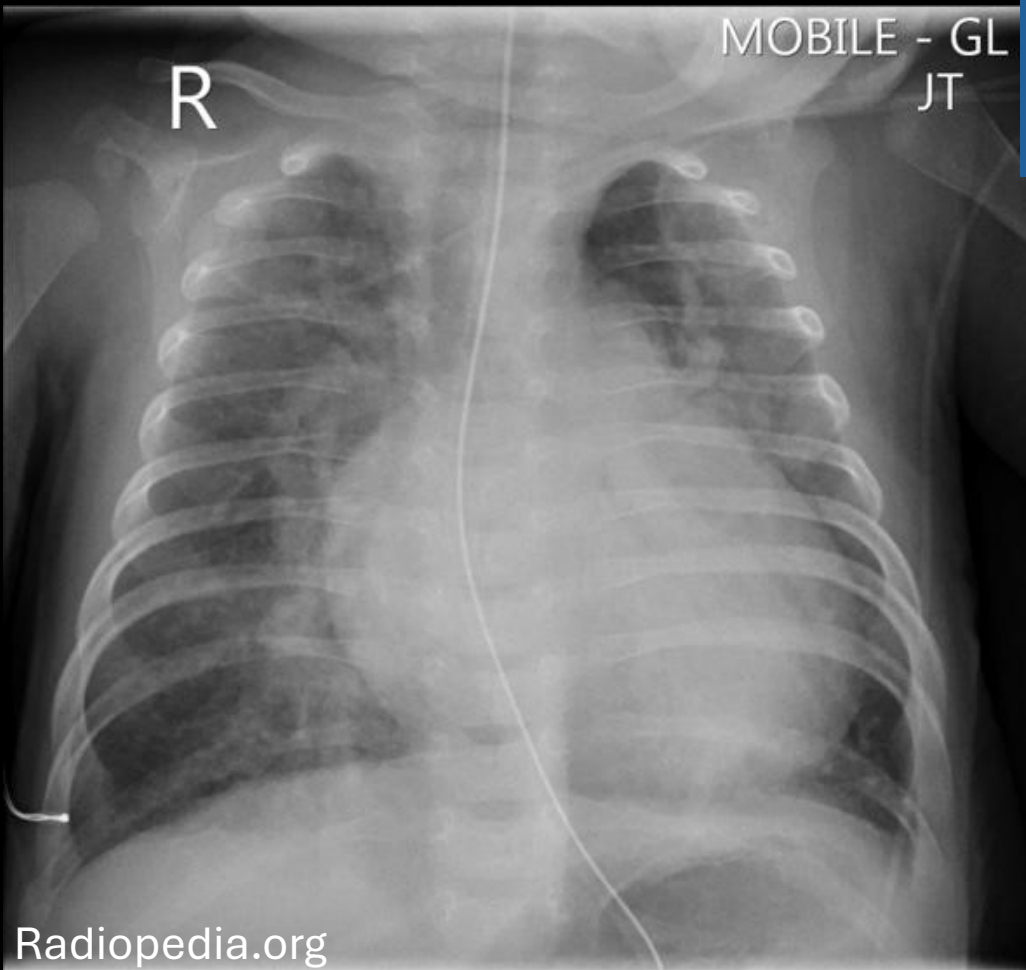
Severity depend on ASD size

1. RA and RV enlargement
2. Pulmonary artery enlargement
3. Increased pulmonary vasculature (High flow or overcirculation)

Atrial Septal Defect (ASD)



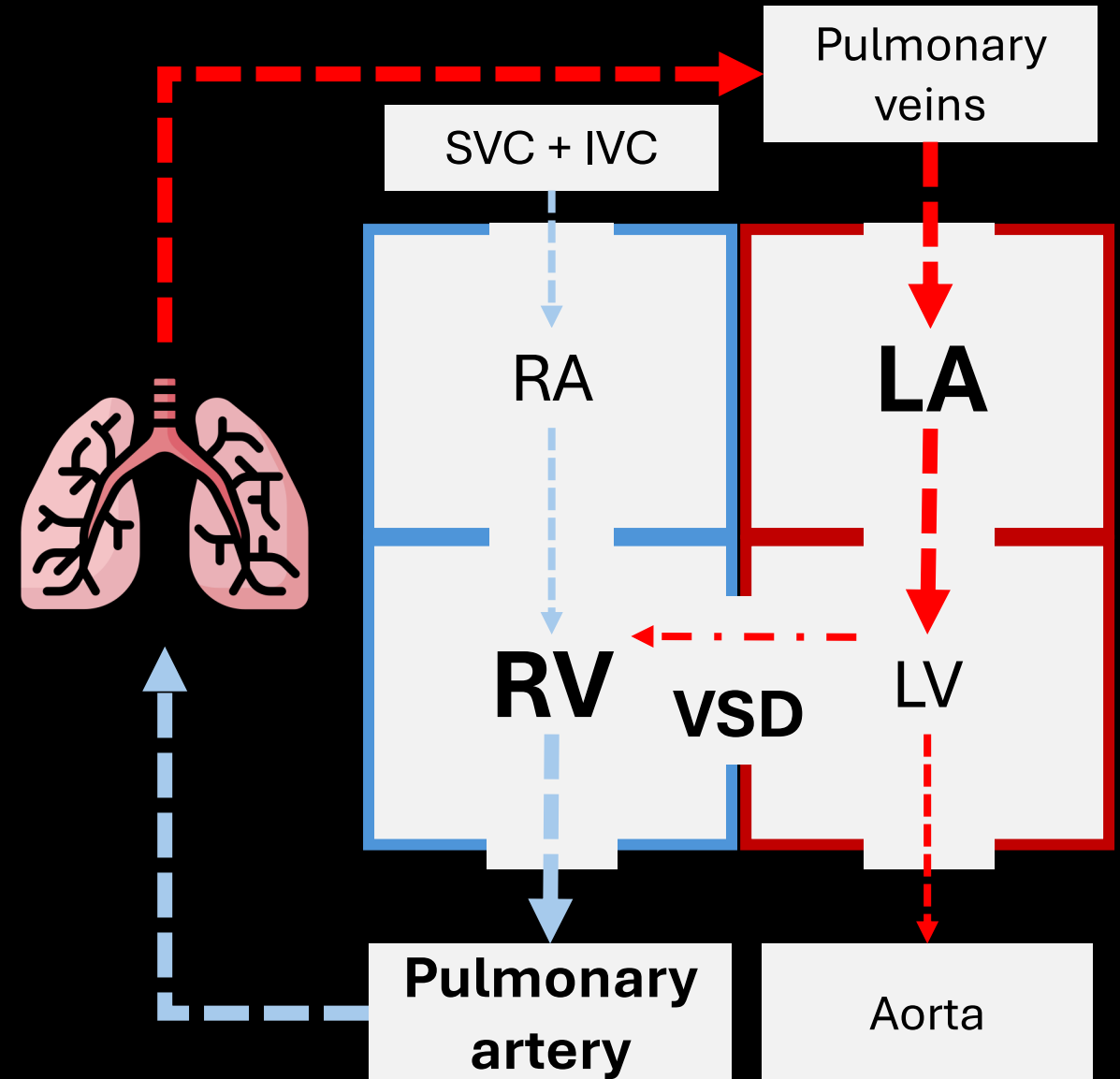
Ventricular Septal Defect (VSD)



CXR

Severity depend on VSD size

1. LA and RV enlargement
2. Pulmonary artery enlargement
3. Increased pulmonary vasculature (High flow or overcirculation)



Patent ductus arteriosus (PDA)

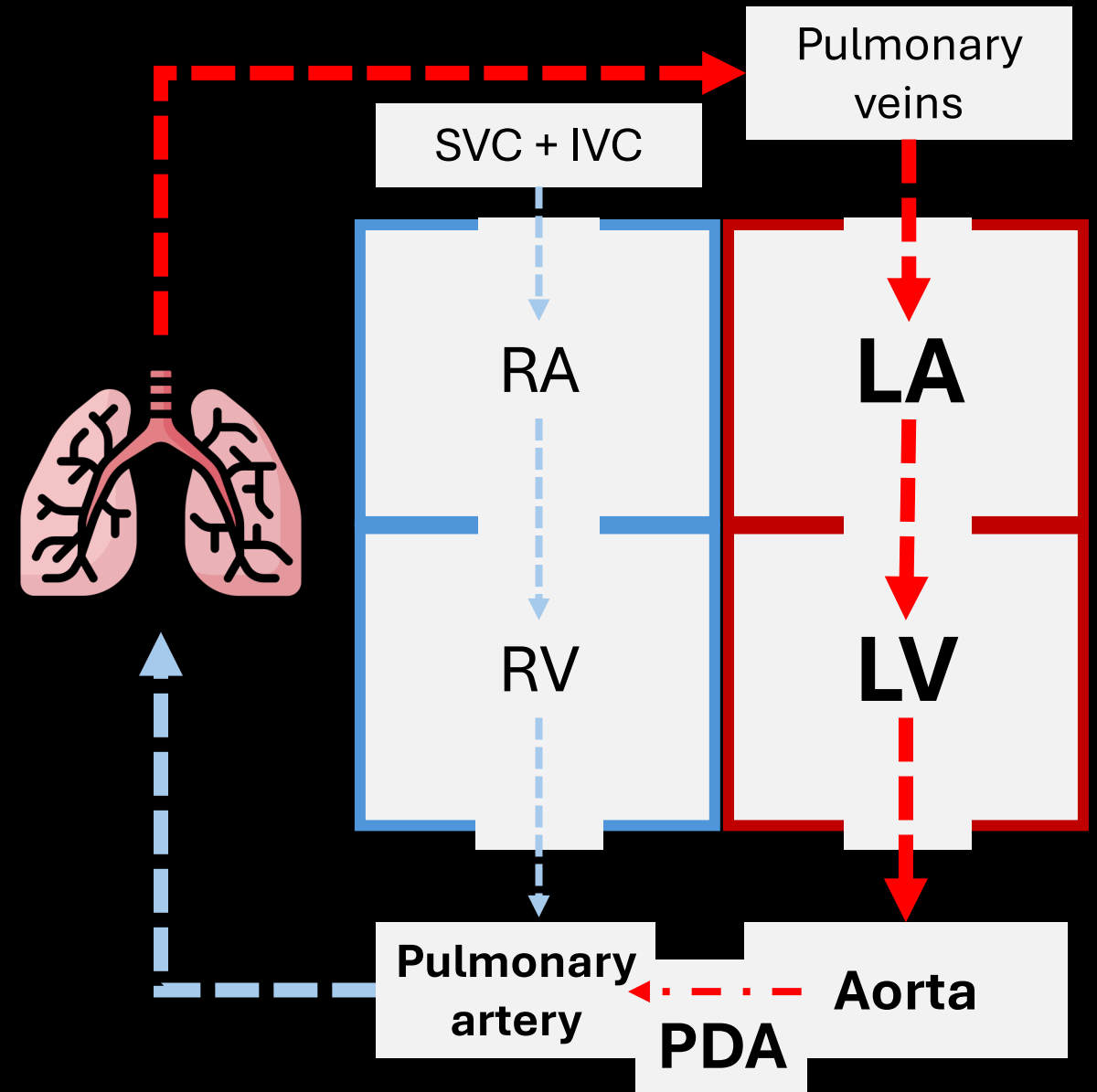


Radiopedia.org

CXR

Severity depend on PDA size

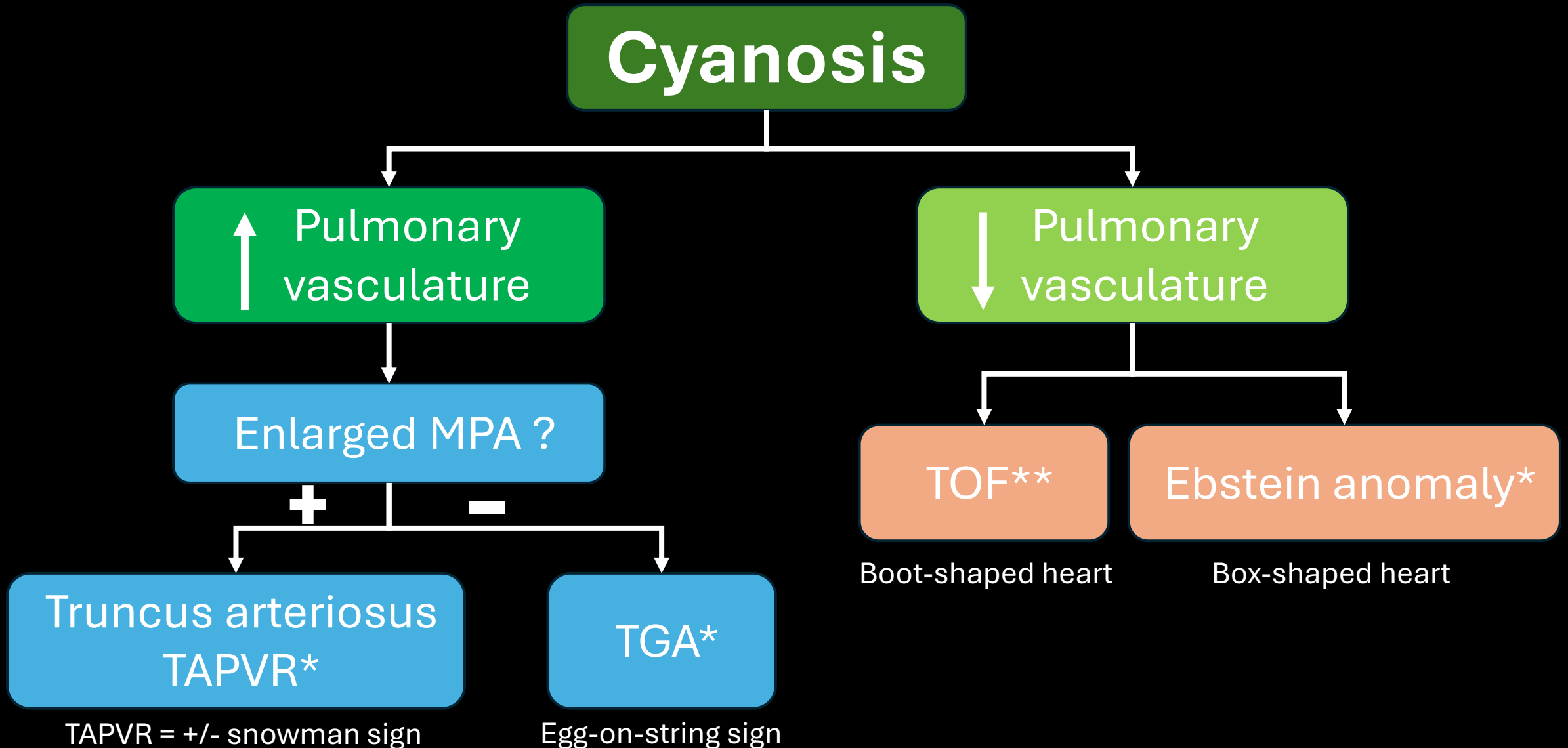
1. LA and LV (+/- RV) enlargement
2. Pulmonary artery enlargement
3. Increased pulmonary vasculature (High flow or overcirculation)
4. Aortic knob enlargement



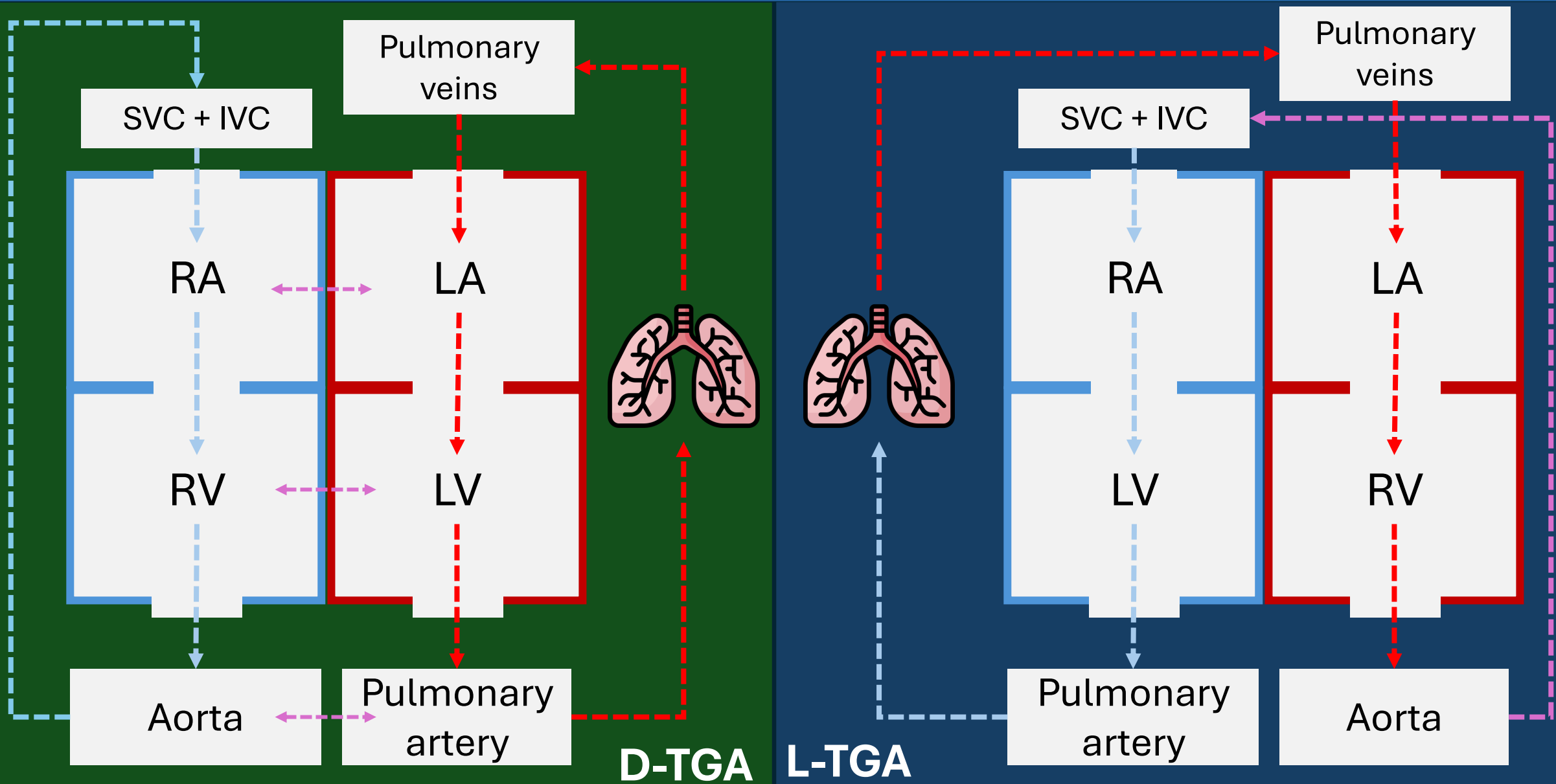
Cyanotic Congenital Heart Disease



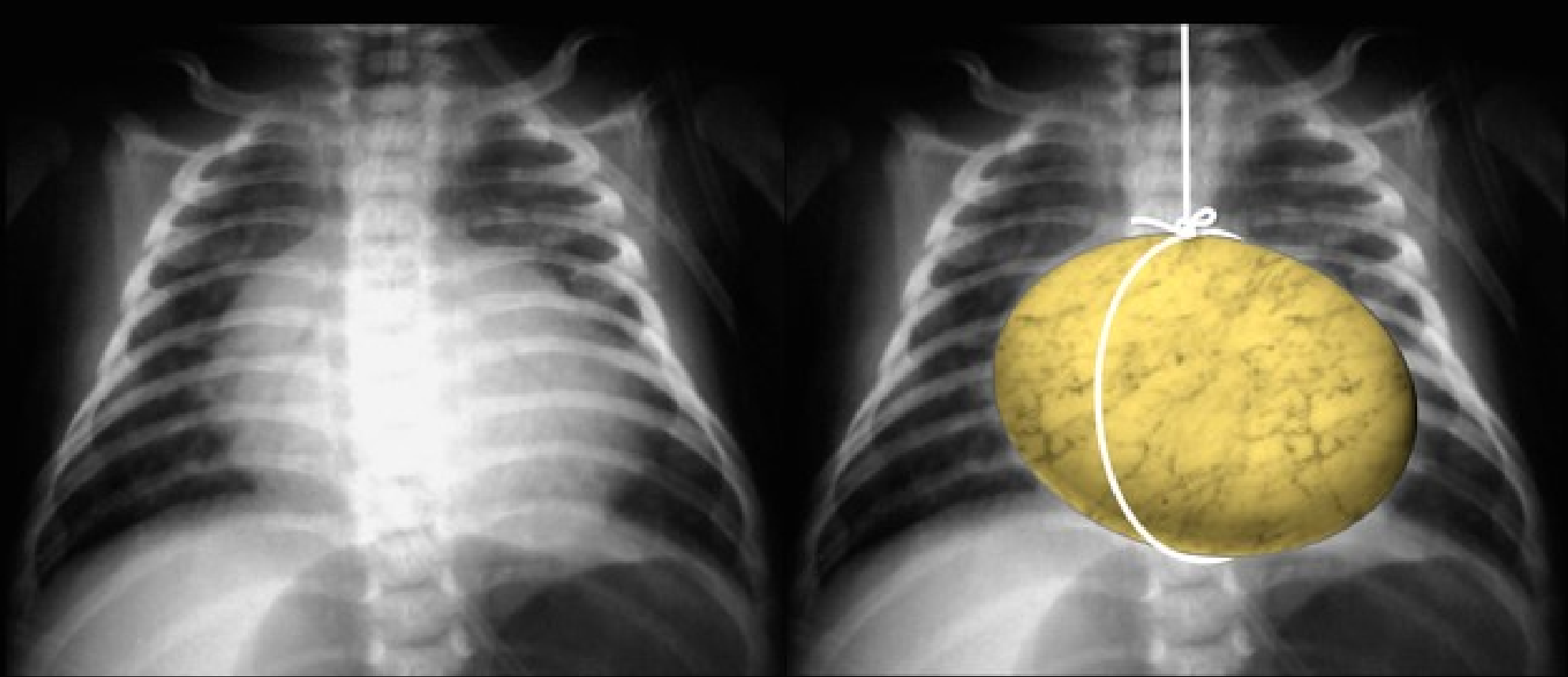
Radiographic approach to Cyanotic CHD



Transposition of great arteries (TGA)



D-TGA: Egg-on-a-string Sign



Transposition of the Great Arteries (TGA)

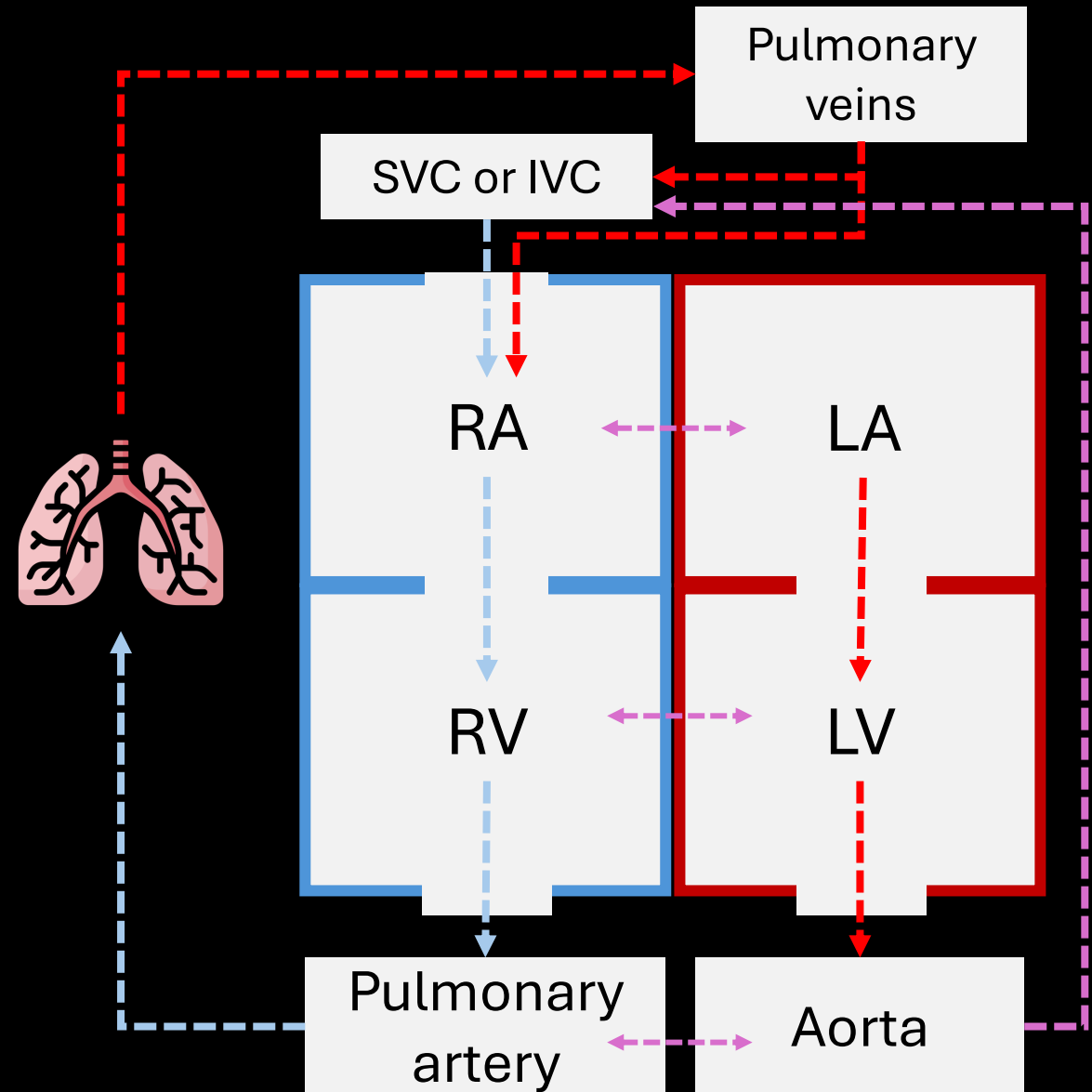
- Egg = Heart (convexity of the RA border with LA enlargement)
- String = narrowed superior mediastinum with stress-induced thymic atrophy and hyperinflated lungs

Total Anomalous Pulmonary Venous Return (TAPVR)

- Normal pulmonary veins drain to LA
- In this condition, pulmonary veins drain to RA or other veins (then into RA)

Types of TAPVR

- Supracardiac type:
 - Drain to systemic veins above heart
 - *CXR – snowman sign*
- Cardiac type: To RA or coronary sinus
- Infracardiac type:
 - Drain to veins below heart
 - Such as IVC or hepatic veins.
- Mixed types



Supracardiac TAPVR: Snowman (Figure of 8) sign

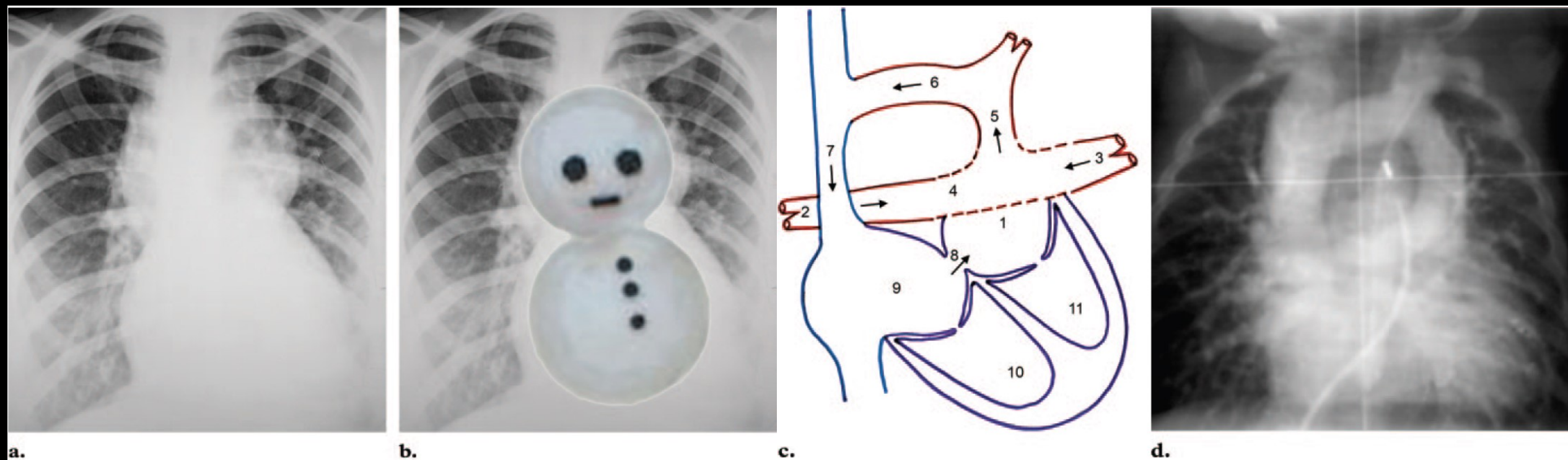
Pulmonary veins

Ascending vertical vein

Brachiocephalic vein

SVC

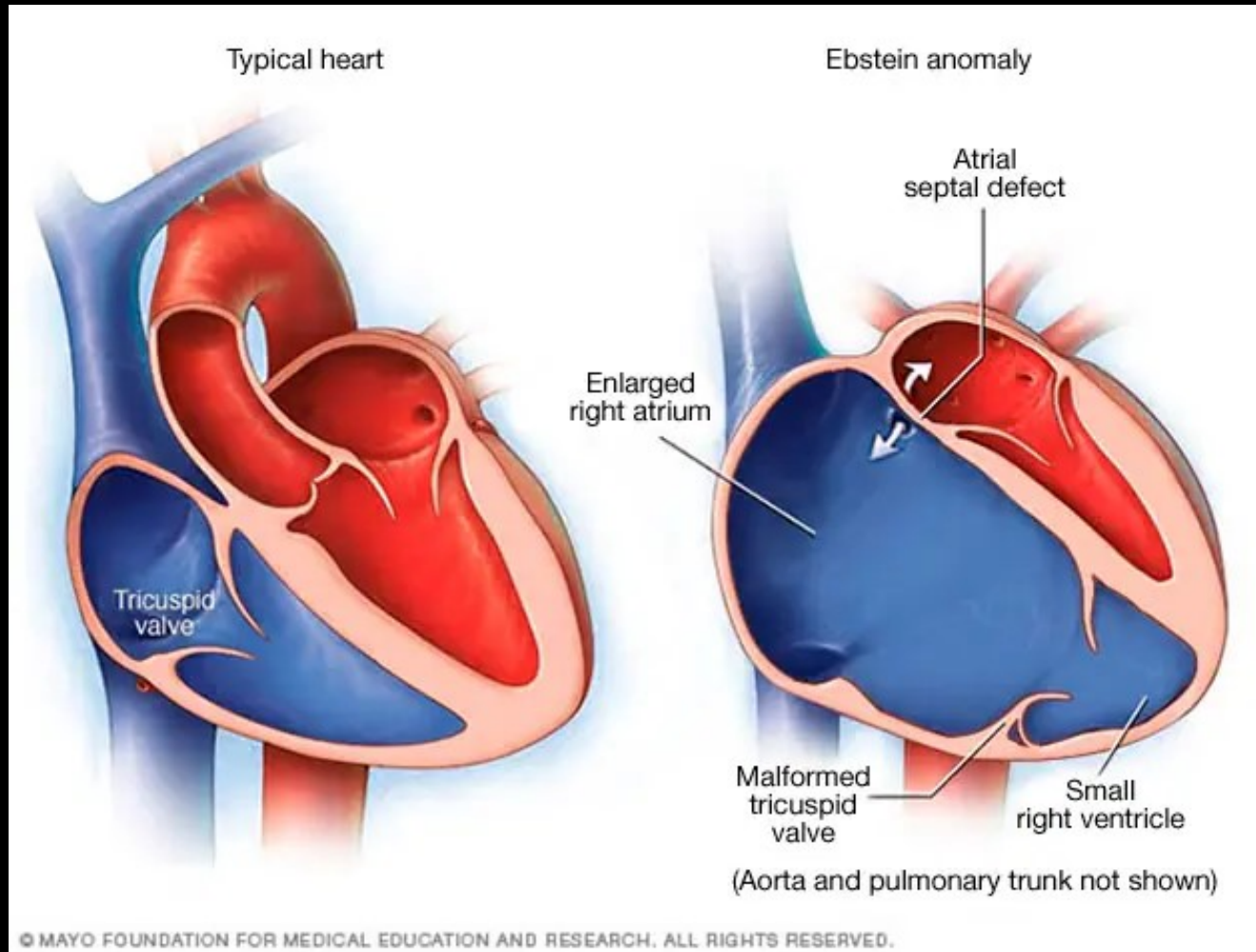
RA



Snowman head = Dilated ascending vertical vein (and SVC)

Snowman body = Cardiomegaly with enlarged RA

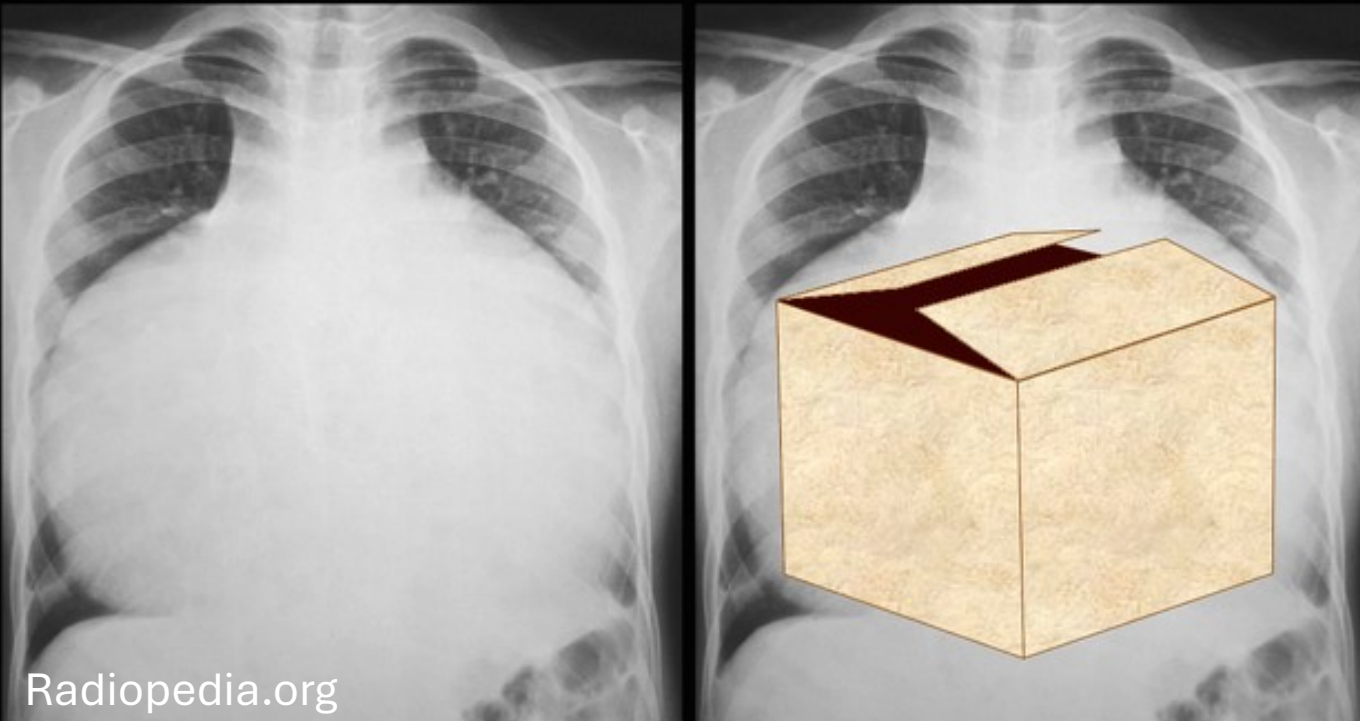
Ebstein's anomaly



Characteristics

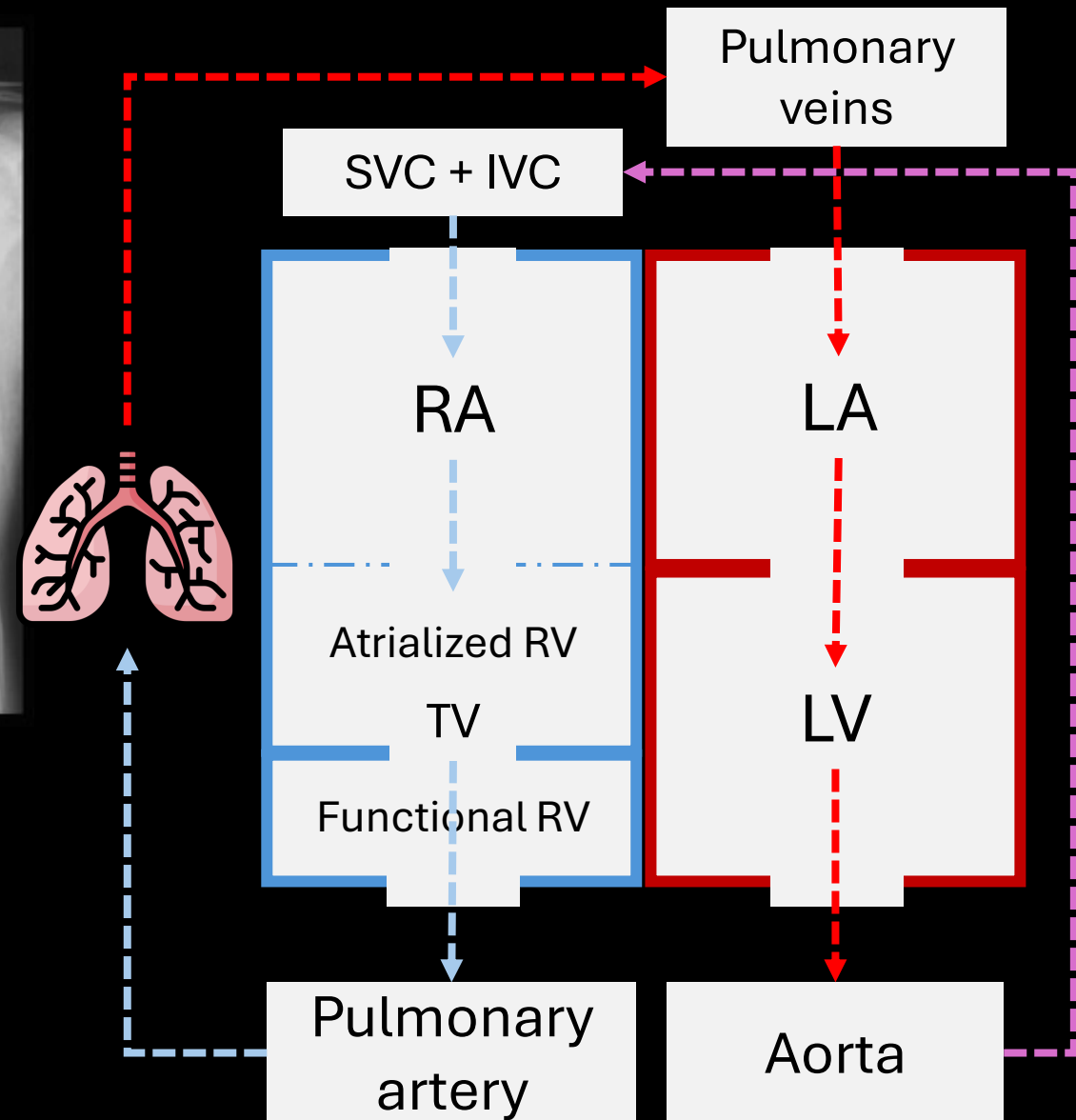
- Downward displacement of TV (posterior and septal leaflets)
- Enlarged RA and small sized RV
- Decreased pulmonary vasculature

Ebstein's anomaly: Box-shaped heart



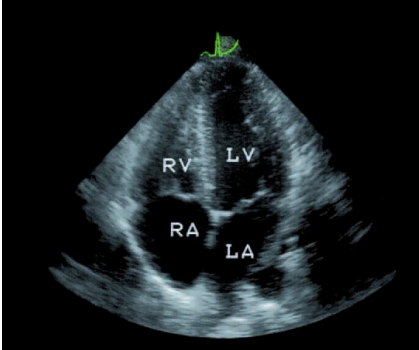
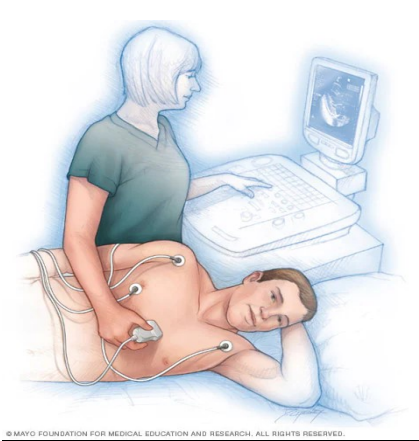
- Markedly enlarged RA (right heart border)
- Displacement of RVOT (to the left side)
- Decreased pulmonary vasculature

Not all EA cases show these CXR findings
Depended on disease severity



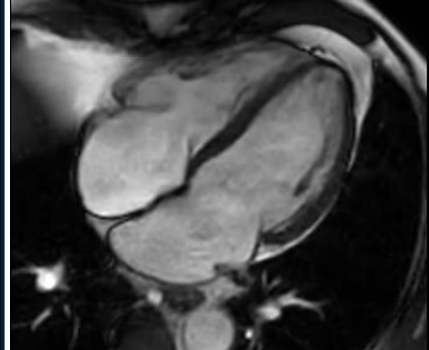
Imaging Modality for Cardiac Evaluation

Echocardiogram



- Transthoracic (TTE)
- Transesophageal (TEE)
- Stress-ECHO

Cardiac MRI



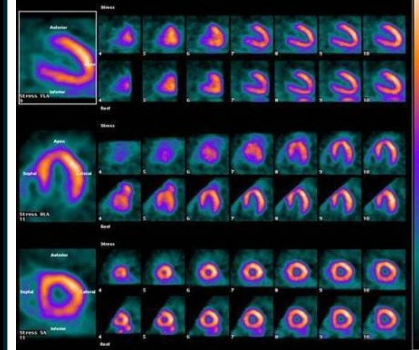
- Cardiomyopathy
- Stress test
- Evaluate cardiac mass

Cardiac CT/ Coronary CTA



- Structure and function
- Coronary artery
- Coronary calcium score

Nuclear Medicine



- MUGA - EF
- MIBI – stress test
- PET and PET/CT

Invasive Angiogram



- Diagnosis and treatment
- Coronary: IVUS, FFR, OCT

Question ?

